QIQY5

Whisky3.0 (Y400)

LA-8691P Rev0.3 Schematic

Intel IVY Bridge Processor with DDRIII + Panther Point PCH
nVIDIA N13P GT1-A2 + 2nd VGA N13P GT1-A2

2012-05-10 Rev0.3

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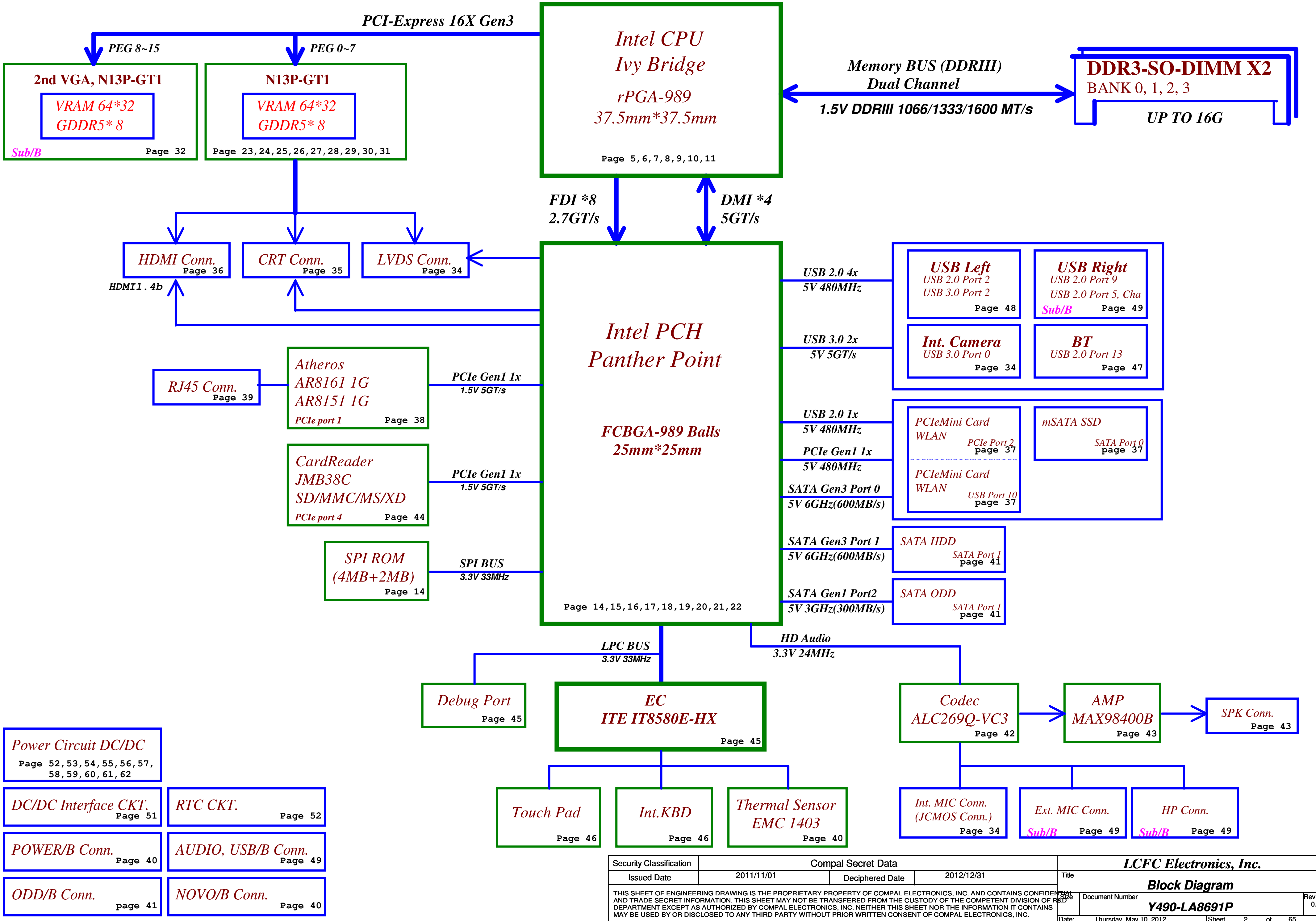

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Title: Block Diagram

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Voltage Rails (O --> Means ON , X --> Means OFF)

State \ Power Plane	B+	+3VALW +5VALW	+1.5V	+5VS +3VS +1.5VS +VCCSA +V1.5S_VCCP +CPU_CORE +VGA_CORE +GFX_CORE +1.8VS +1.05VS +0.75VS +3.3VS_VGA +1.5VS_VGA +1.05VS_VGA
S0	O	O	O	O
S3	O	O	O	X
S5 S4/AC Only	O	O	X	X
S5 S4 Battery only	O	X	X	X
S5 S4 AC & Battery don't exist	X	X	X	X

STATE \ SIGNAL	SLP_S1#	SLP_S3#	SLP_S4#	SLP_S5#	+VALW	+V	+VS	Clock
Full ON	HIGH	HIGH	HIGH	HIGH	ON	ON	ON	ON
S1(Power On Suspend)	LOW	HIGH	HIGH	HIGH	ON	ON	ON	LOW
S3 (Suspend to RAM)	LOW	LOW	HIGH	HIGH	ON	ON	OFF	OFF
S4 (Suspend to Disk)	LOW	LOW	LOW	HIGH	ON	OFF	OFF	OFF
S5 (Soft OFF)	LOW	LOW	LOW	LOW	ON	OFF	OFF	OFF

USB Port Table

USB 2.0	USB 3.0	Port	4 External USB Port
EHCI1	XHCI 1	0	Camera
	2	1	
	3	2	USB Port (Left Side)
	4	3	
		4	
		5	USB Port (Right Side)
		6	
		7	
EHCI2		8	
		9	USB Port (Right Side)
		10	Mini Card(WLAN)
		11	
		12	
		13	Blue Tooth

BOM Structure Table

BOM Structure	BTO Item
HDMI@	HDMI part
CHG@	USB charger part
NOCHG@	No USB charger part
CMOS@	CMOS Camera part
8161@	AR8161 LAN part
8151@	AR8151 LAN part
8161S@	AR8161 LAN surge part
8151S@	AR8151 LAN surge part
SURGE@	AR8151&8161 LAN surge part
61@	X76 P/N for AR8161
51@	X76 P/N for AR8151
X76@	X76 Level part for VRAM
GC6@	NV CG6 support part
NOGC6@	NV no CG6 support part
AOAC@	AOAC support part
KBL@	K/B Light part
ME@	ME part
OPT@	For optimus function part
SLI@	For SLI function part
DS3@	Deep S3 support part
S3@	For S3 function part
GT@	NV chip part
@	Unpop

SMBUS Control Table

	SOURCE	Main VGA	2nd VGA	BATT	IT8580E	SODIMM	WLAN WiMAX	Thermal Sensor	PCH	TP Module
EC_SMB_CK1 EC_SMB_DA1	IT8580E +3VALW	X	X	V +3VALW	X	X	X	X	X	X
EC_SMB_CK2 EC_SMB_DA2	IT8580E +3VS	V +3VS	V +3VS	X	X	X	X	V +3VS	V +3V_PCH	X
SMB_CLK_S3 SMB_DATA_S3	PCH +3VS	X	X	X	X	V +3VS	V +3VS	X	V +3V_PCH	V +3VS

PCIE PORT LIST

Port	Device
1	LAN
2	WLAN
3	
4	Card Reader
5	
6	
7	
8	

EC SM Bus1 address

Device	Address
Smart Battery	0001 011X b

EC SM Bus2 address

Device	Address
Thermal Sensor EMC1403-2	1001_101xb
Master VGA	0x9E
Slave VGA	0x9C

PCH SM Bus address

Device	Address
DDR DIMM0	1001 000Xb
DDR DIMM2	1001 010Xb

ZZZ1

DA80000T10J

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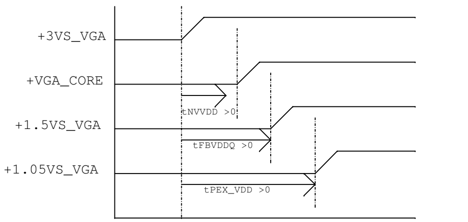
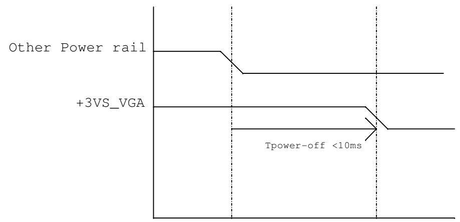
Hot plug detect for IFP link E

VGA and GDDR5 Voltage Rails (N13Px GPIO)

GPIO	I/O	ACTIVE	Function Description
GPIO0	OUT	-	GPU VID4
GPIO1	OUT	-	GPU VID3
GPIO2	OUT	-	VGA_BL_PWM
GPIO3	OUT	-	VGA_ENVDD
GPIO4	OUT	-	VGA_ENBKL
GPIO5	OUT	-	GPU VID1
GPIO6	OUT	-	GPU VID2
GPIO7	OUT	-	DPRSLPVR_VGA
GPIO8	I/O	-	Thermal Catastrophic Over Temperature
GPIO9	OUT	-	GPIO9
GPIO10	OUT	-	Memory VREF Control
GPIO11	OUT	-	GPU VID0
GPIO12	IN		AC Power Detect Input (10K pull High)
GPIO13	OUT	-	GPU VID5
GPIO14	OUT	-	FB_CLAMP_TOGGLE_REQ#
GPIO15	IN	N/A	(100K pull low)
GPIO16	OUT	-	FRMLCK#
GPIO17	IN	N/A	
GPIO18	IN	-	dGPU_HDMI_HPD
GPIO19	IN	-	HPD_IRQ



1. all power rail ramp up time should be larger than 40us



1.all GPU power rails should be turned off within 10ms
2. Optimus system VDD33 avoids drop down earlier than NVDD and FBVDDQ

Performance Mode P0 TDP at Tj = 102 C* (GDDR5)

Products	GPU (4)	Mem (1,5)	NVCLK /MCLK	NVVDD			FBVDD (1.35V)		FBVDDQ (GPU+Mem) (1.35V)		PCI Express (1.05V) (6)		I/O and PLLVDD (1.8V)		I/O and PLLVDD (1.05V)		Other (3.3V)	
	(W)	(W)	(MHz)	(V)	(A)	(W)	(A)	(W)	(A)	(W)	(mA)	(W)	(mA)	(W)	(mA)	(W)	(mA)	(W)
N13X 128bit 1GB GDDR5	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Physical Strapping pin	Power Rail	Logical Strapping Bit3	Logical Strapping Bit2	Logical Strapping Bit1	Logical Strapping Bit0
ROM_SCLK	+3VS_VGA	PCI_DEVID[4]	SUB_VENDOR	SLOT_CLK_CFG	PEX_PLL_EN_TERM
ROM_SI	+3VS_VGA	RAM_CFG[3]	RAM_CFG[2]	RAM_CFG[1]	RAM_CFG[0]
ROM_SO	+3VS_VGA	FB[1]	FB[0]	SMB_ALT_ADDR	VGA_DEVICE
STRAP0	+3VS_VGA	USER[3]	USER[2]	USER[1]	USER[0]
STRAP1	+3VS_VGA	3GIO_PAD_CFG_ADR[3]	3GIO_PAD_CFG_ADR[2]	3GIO_PAD_CFG_ADR[1]	3GIO_PAD_CFG_ADR[0]
STRAP2	+3VS_VGA	PCI_DEVID[3]	PCI_DEVID[2]	PCI_DEVID[1]	PCI_DEVID[0]
STRAP3	+3VS_VGA	SOR3_EXPOSED	SOR2_EXPOSED	SOR1_EXPOSED	SOR0_EXPOSED
STRAP4	+3VS_VGA	RESERVED	PCIE_SPEED_ CHANGE_GEN3	PCIE_MAX_SPEED	DP_PLL_VDD33V

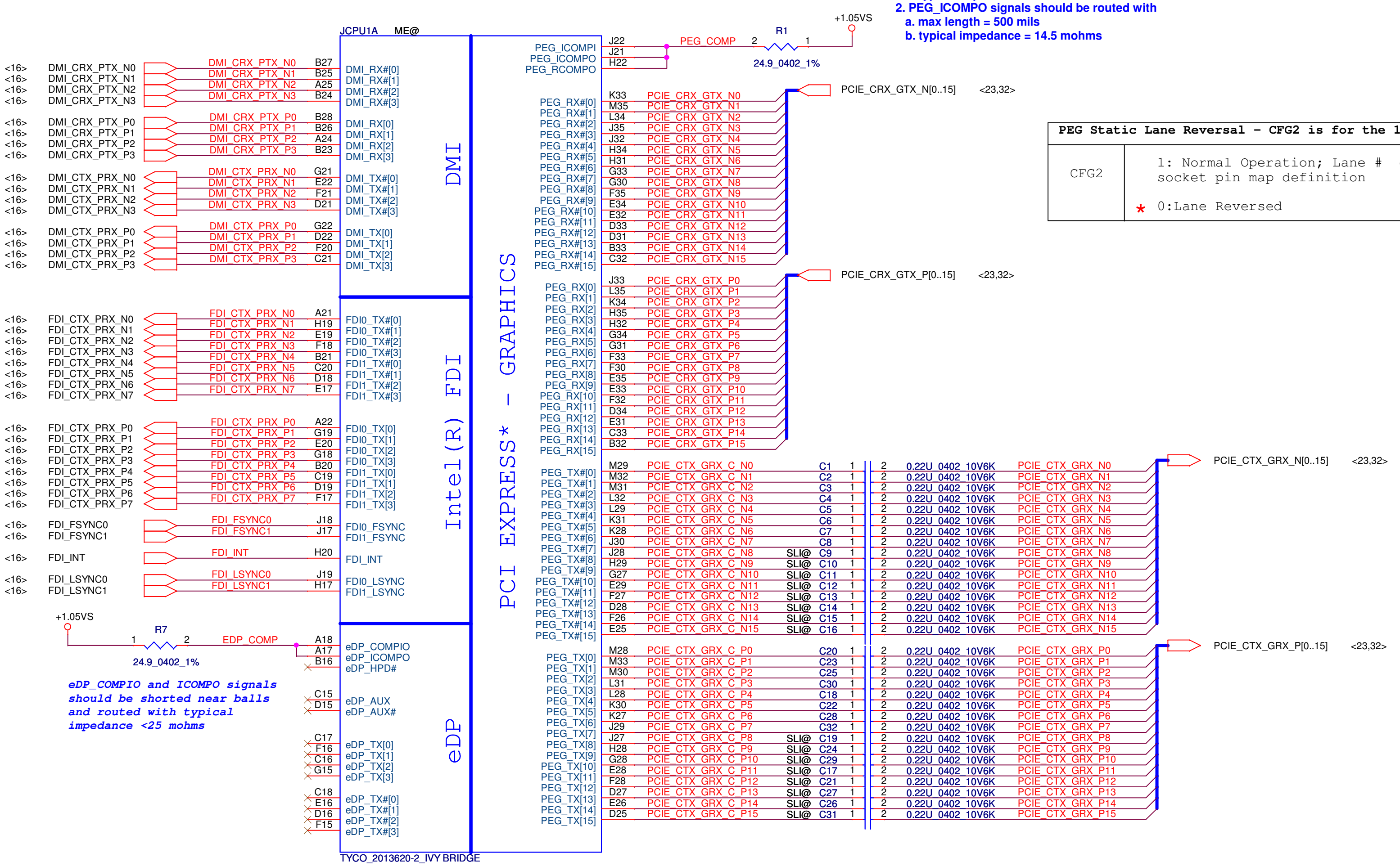
	Device ID
N13P-GT (28nm)	0x0FDB

	setting	I2C Slave addrees ID
SMB_ALT_ADDR (ROM_SO Bit 1)	0	0x9E
	1	0x9C

GPU	ROM_SO	ROM_SCLK	STRAP0	STRAP1	STRAP2	STRAP3	STRAP4	
N13P-GT1 28nm	PU 10K	PU 25K	PU 45K	PD 35K	PD 10K	PU 5K	PD 10K	Master
	PU 20K	PU 25K	PU 45K	PD 35K	PD 10K	PD 5K	PD 10K	Slave

GPU		N13P-GT		
FB Memory (GDDR5)		ROM_SI		
Samsung 2500MHz	K4G10325FG-HC04			
	32Mx32	PD 45K		
Hynix 2500MHz	H5GQ1H24BFR-T2C			
	32Mx32	PD 35K		
Samsung 2500MHz	K4G20325FD-FC04			
	64Mx32	PD 30K		
Hynix 2500MHz	H5GQ2H24MFR-T2C			
	64Mx32	PD 25K		

1. PEG_ICOMPI and RCOMPO signals should be shorted and routed with
   a. max length = 500 mils
   b. typical impedance = 43 mohms
2. PEG_ICOMPO signals should be routed with
   a. max length = 500 mils
   b. typical impedance = 14.5 mohms



eDP_COMPIO and ICOMPO signals should be shorted near balls and routed with typical impedance <25 mohms

PEG Static Lane Reversal - CFG2 is for the 16x	
CFG2	1: Normal Operation; Lane # definition matches socket pin map definition * 0:Lane Reversed

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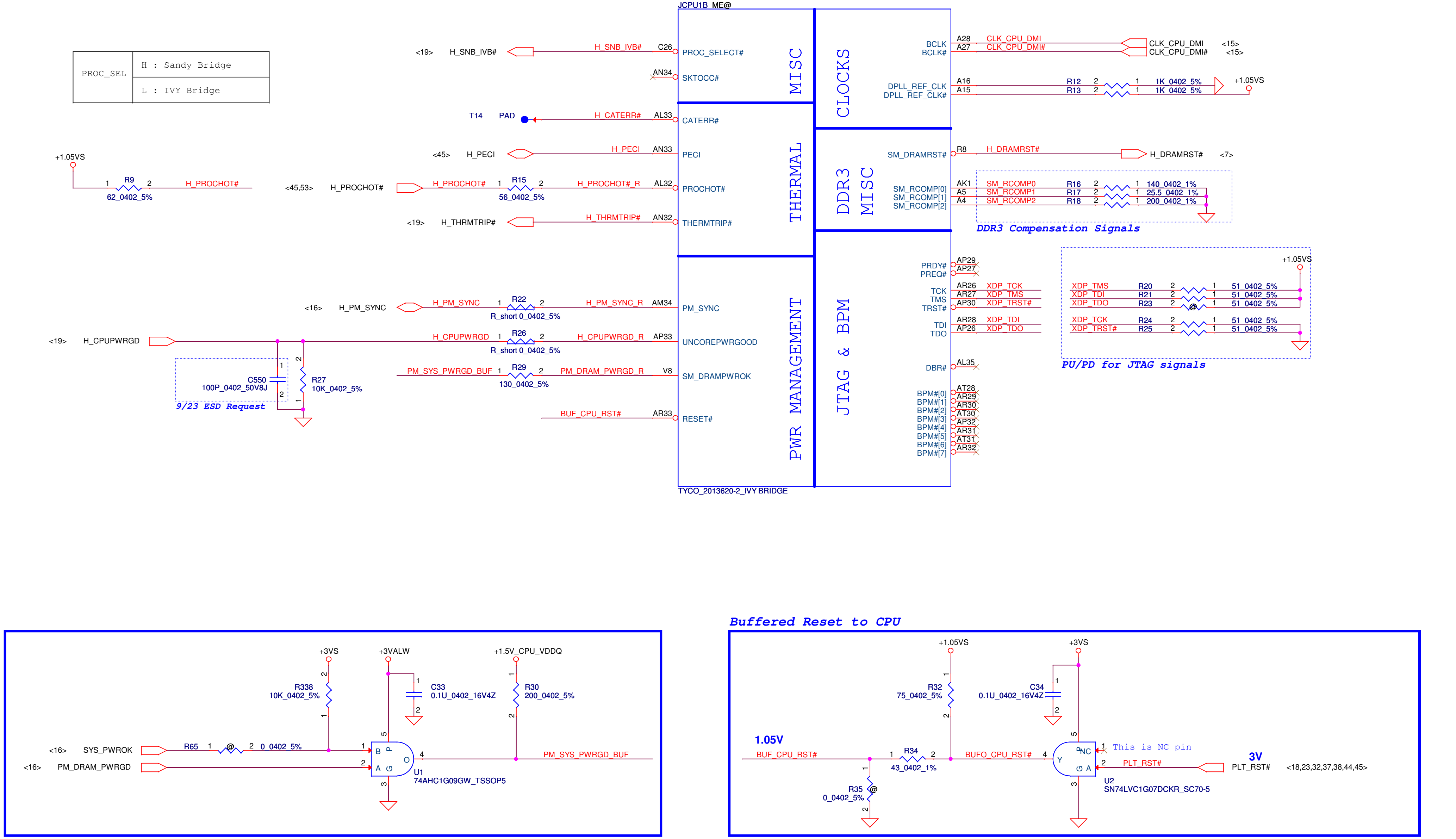
				Date: Thursday, May 10, 2012; Sheet 5 of 65



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Issued Date	2011/11/01	Deciphered Date	2012/12/31

Compal Electronics, Inc.

Title: **PROCESSOR(2/7) PM,XDP,CLK**

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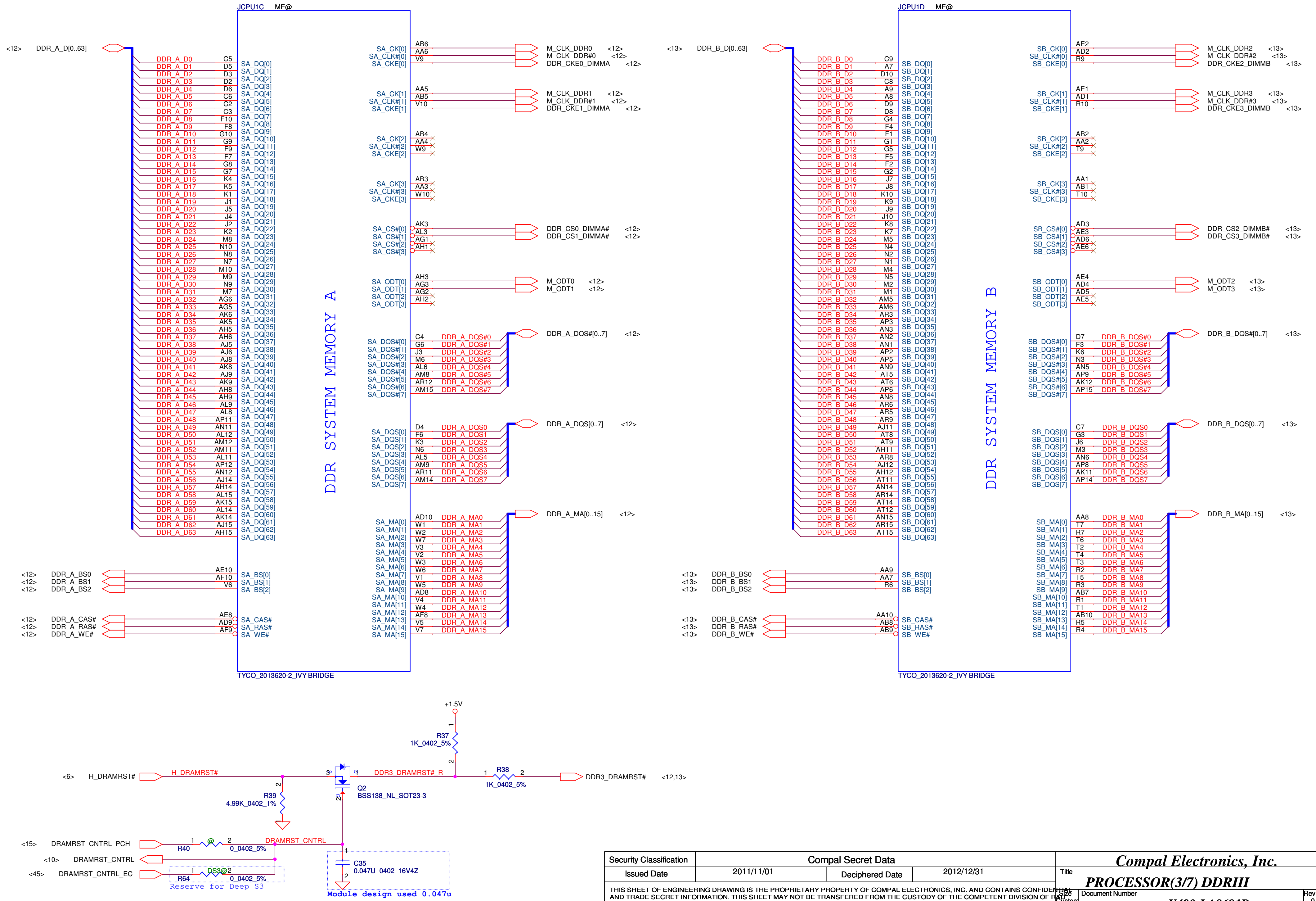

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Title: **PROCESSOR(3/7) DDRIII**

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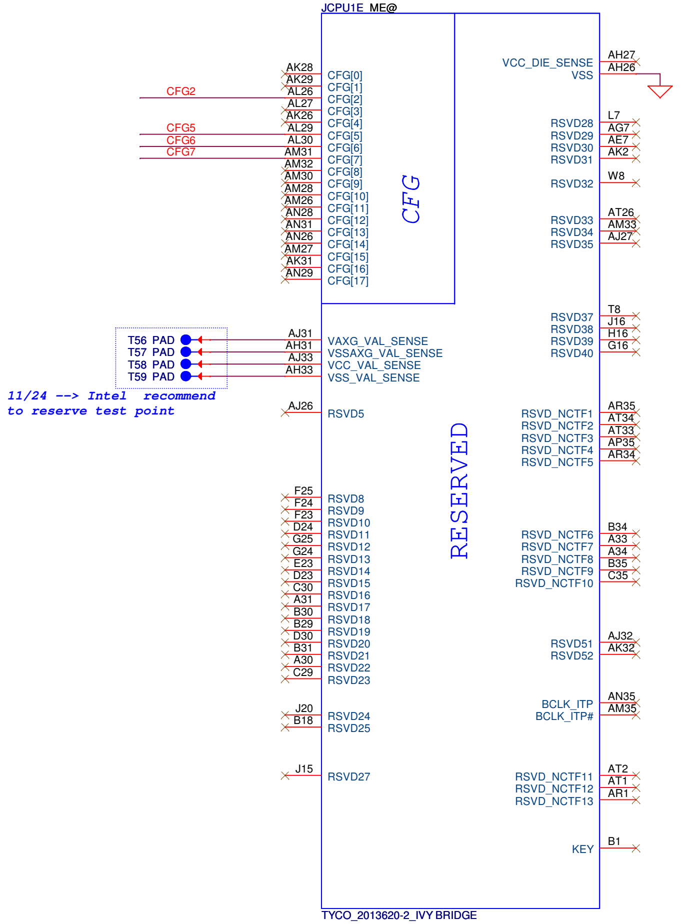
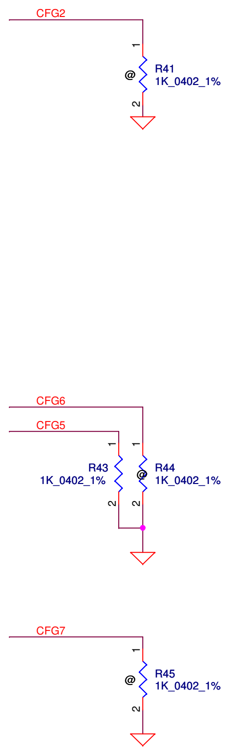


CFG Straps for Processor



PEG Static Lane Reversal - CFG2 is for the 16x	
CFG2	* 1: Normal Operation; Lane # definition matches socket pin map definition 0:Lane Reversed

Display Port Presence Strap	
CFG4	* 1 : Disabled; No Physical Display Port attached to Embedded Display Port 0 : Enabled; An external Display Port device is connected to the Embedded Display Port

PCIE Port Bifurcation Straps	
CFG[6:5]	11: (Default) x16 - Device 1 functions 1 and 2 disabled *10: x8, x8 - Device 1 function 1 enabled ; function 2 disabled 01: Reserved - (Device 1 function 1 disabled ; function 2 enabled) 00: x8,x4,x4 - Device 1 functions 1 and 2 enabled

PEG DEFER TRAINING	
CFG7	1: (Default) PEG Train immediately following xxRESETB de assertion 0: PEG Wait for BIOS for training

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Title	PROCESSOR(4/7) RSVD,CFG	
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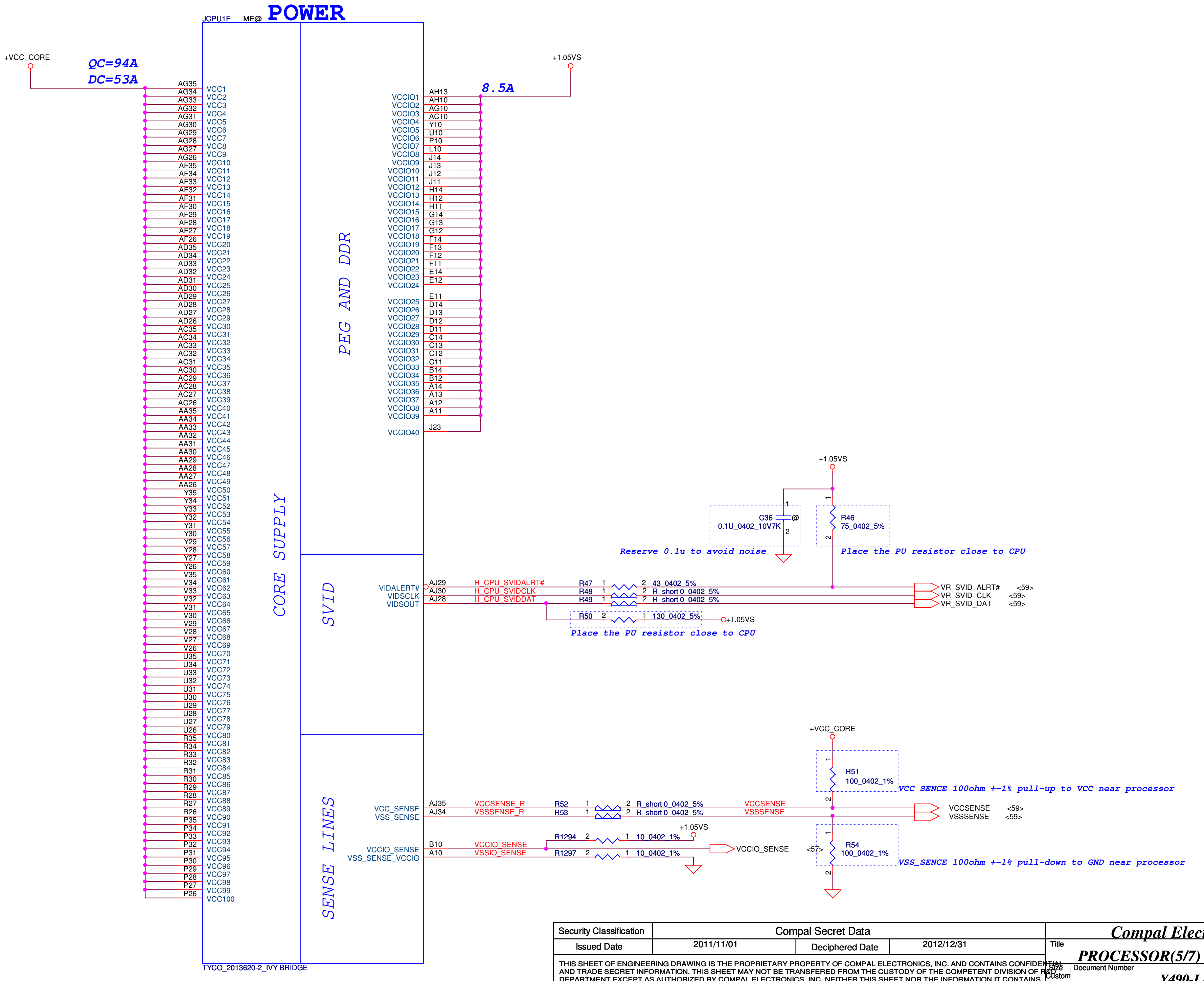

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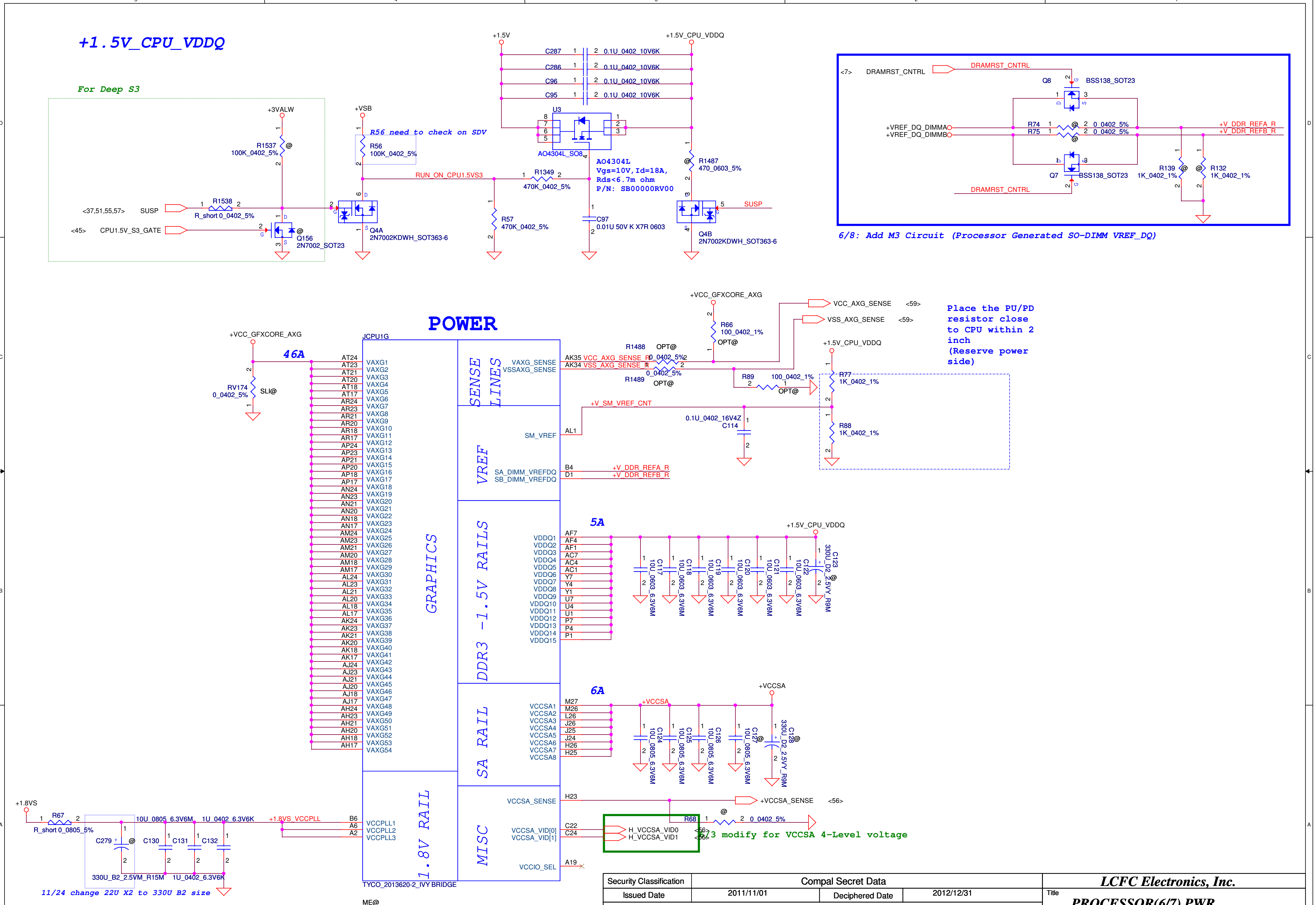

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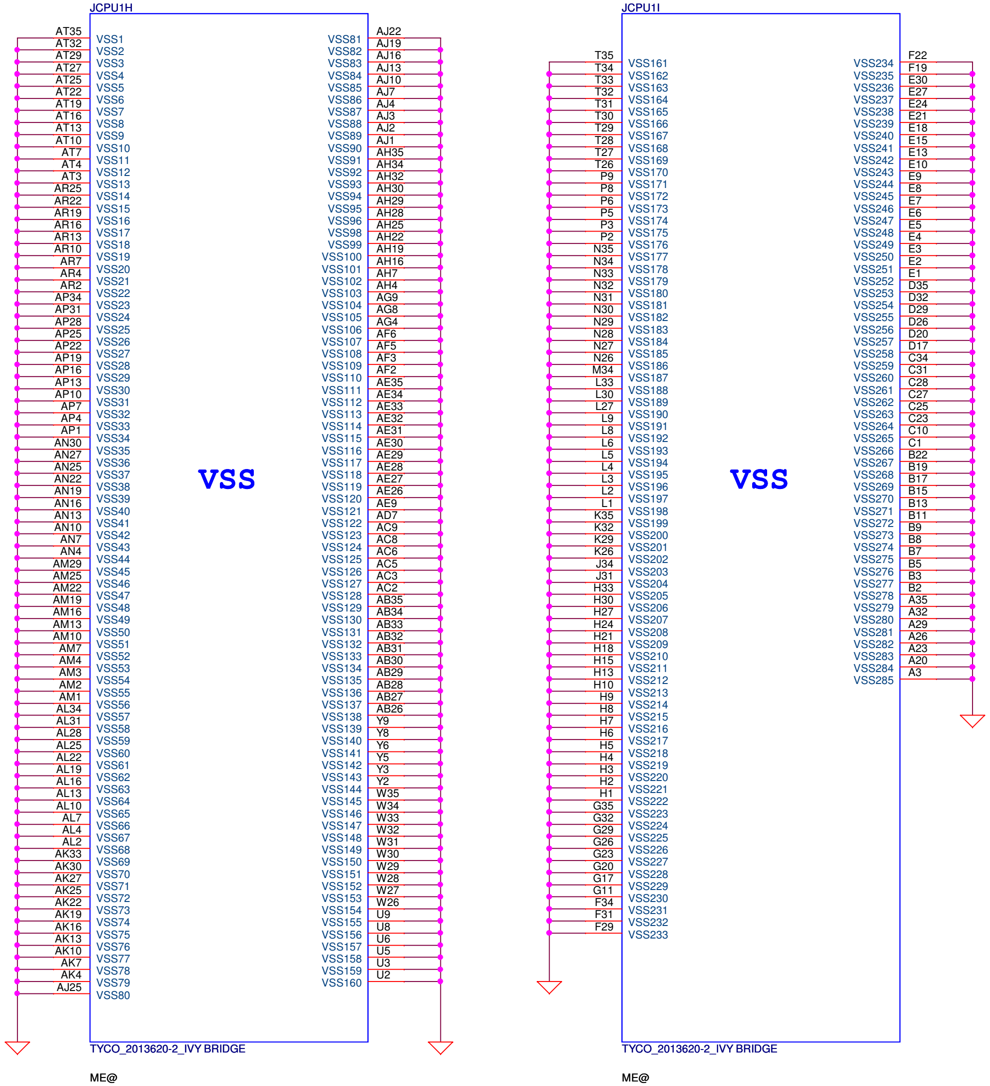

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Title	PROCESSOR(7/7) VSS	
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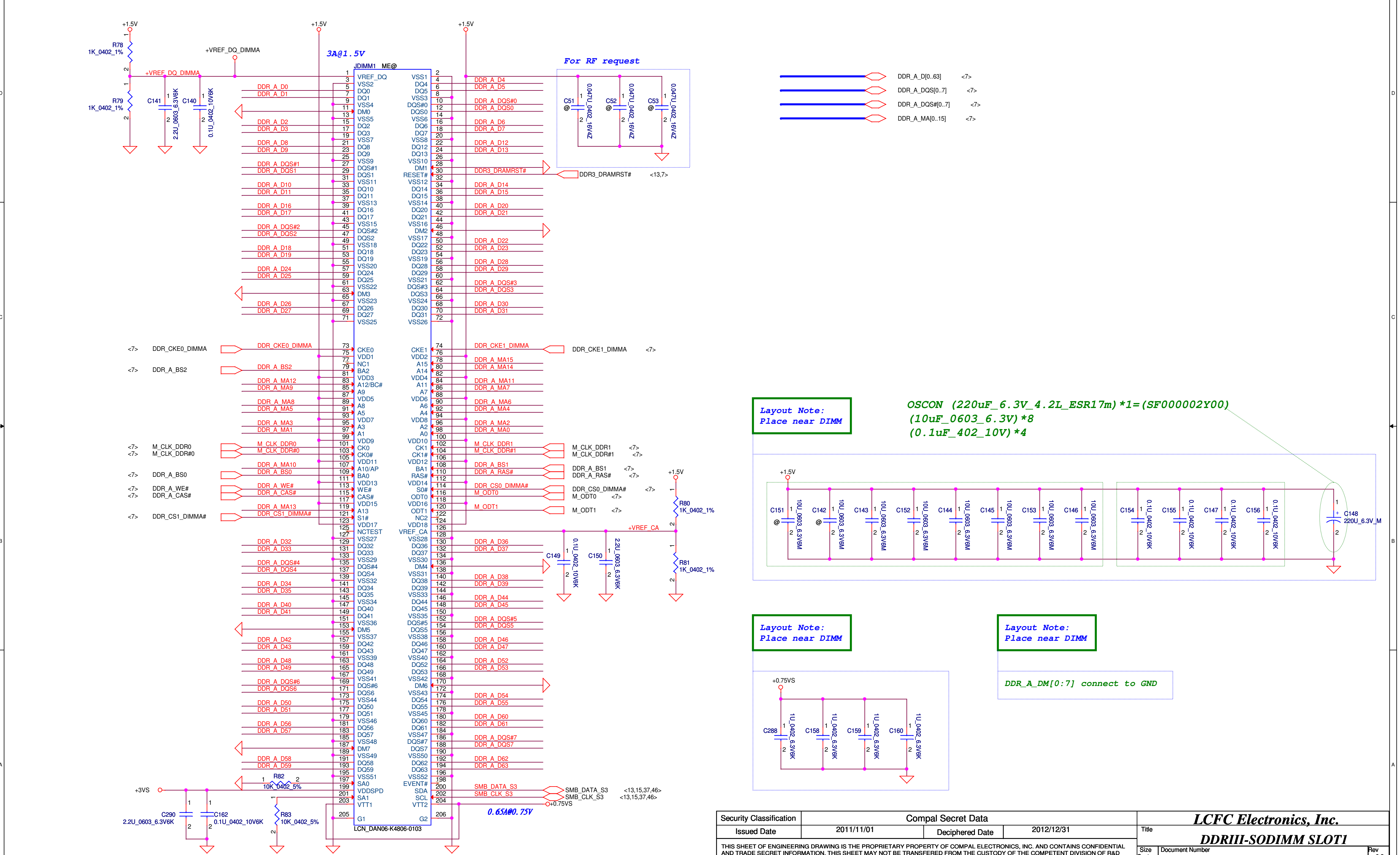
DDR3 SO-DIMM A



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Title: **DDRIII-SODIMM SLOT1**

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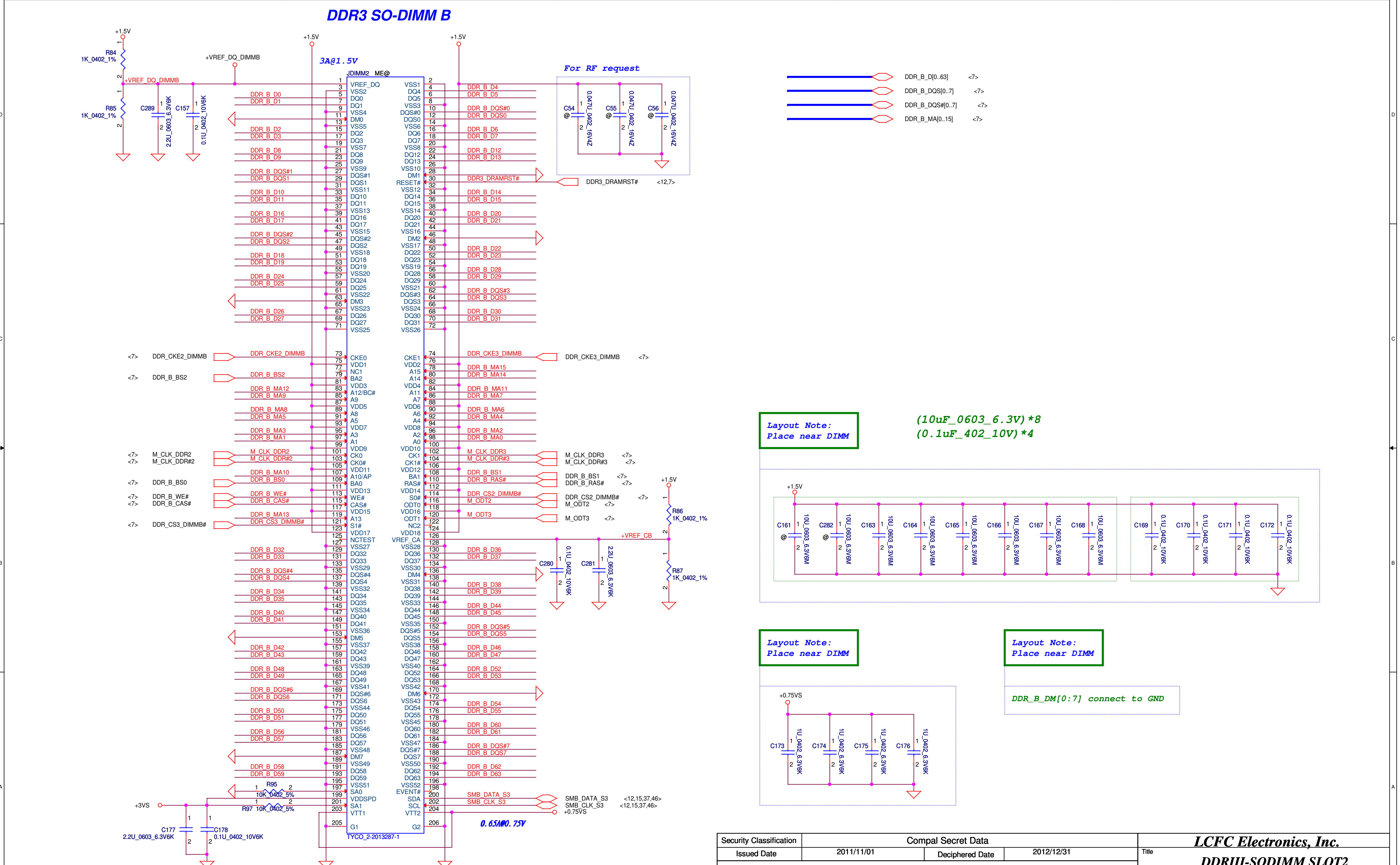

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Title: DDRIII-SODIMM SLOT2

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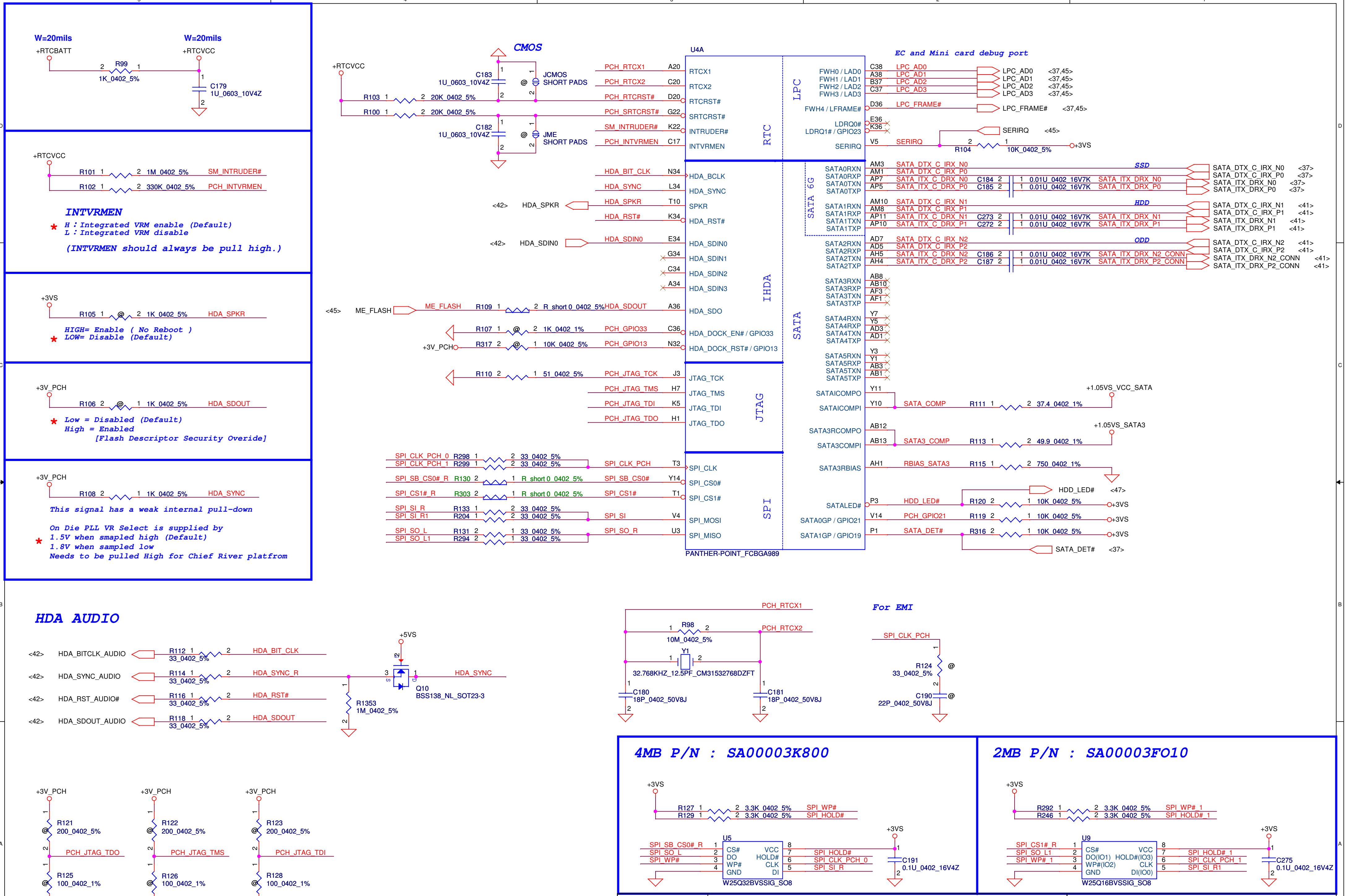

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Issued Date	2011/11/01	Deciphered Date	2012/12/31	Title: PCH (1/9) SATA,HDA,SPI, LPC, XDP

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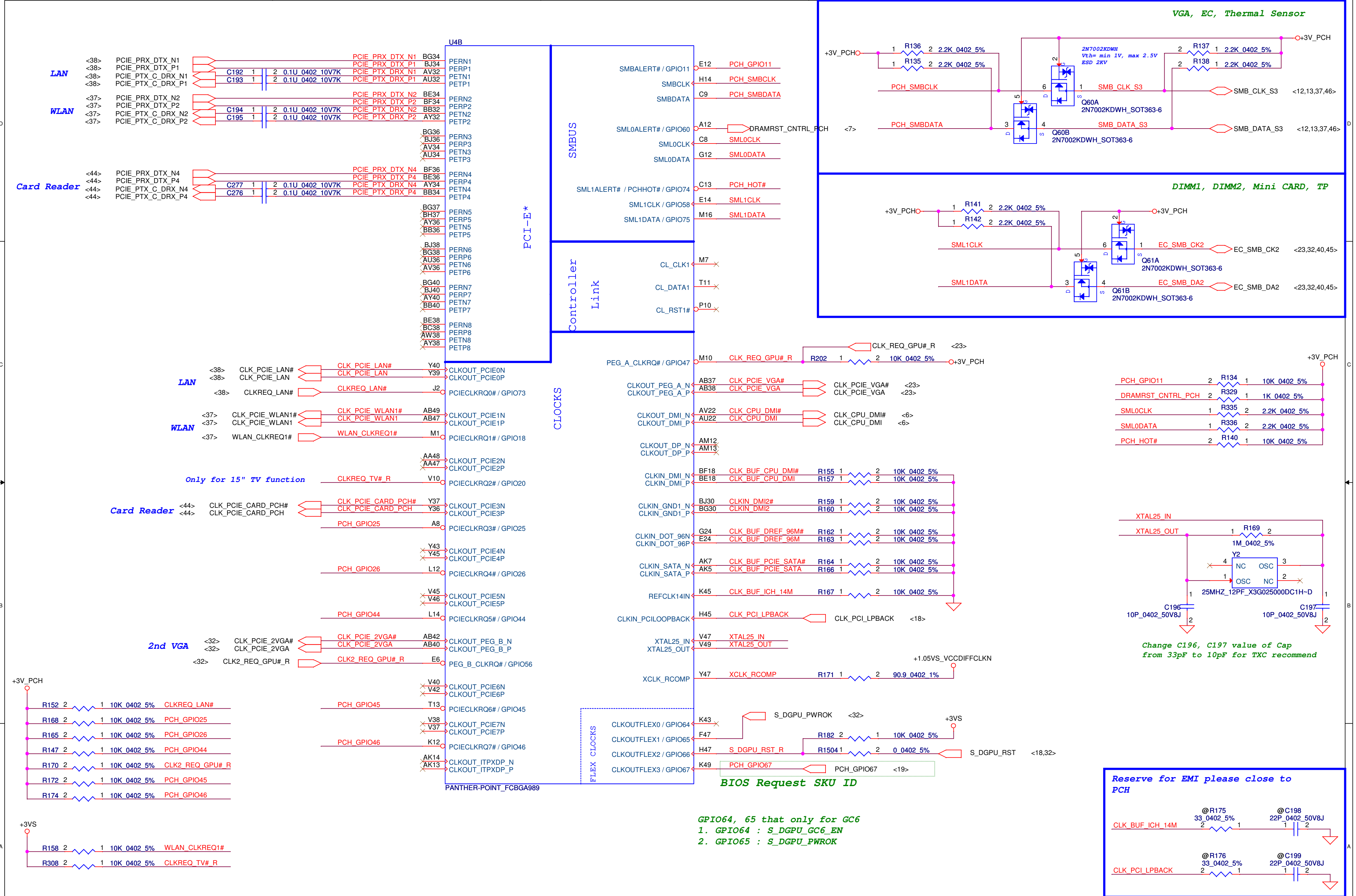

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Title: **PCH (2/9) PCIE, SMBUS, CLK**

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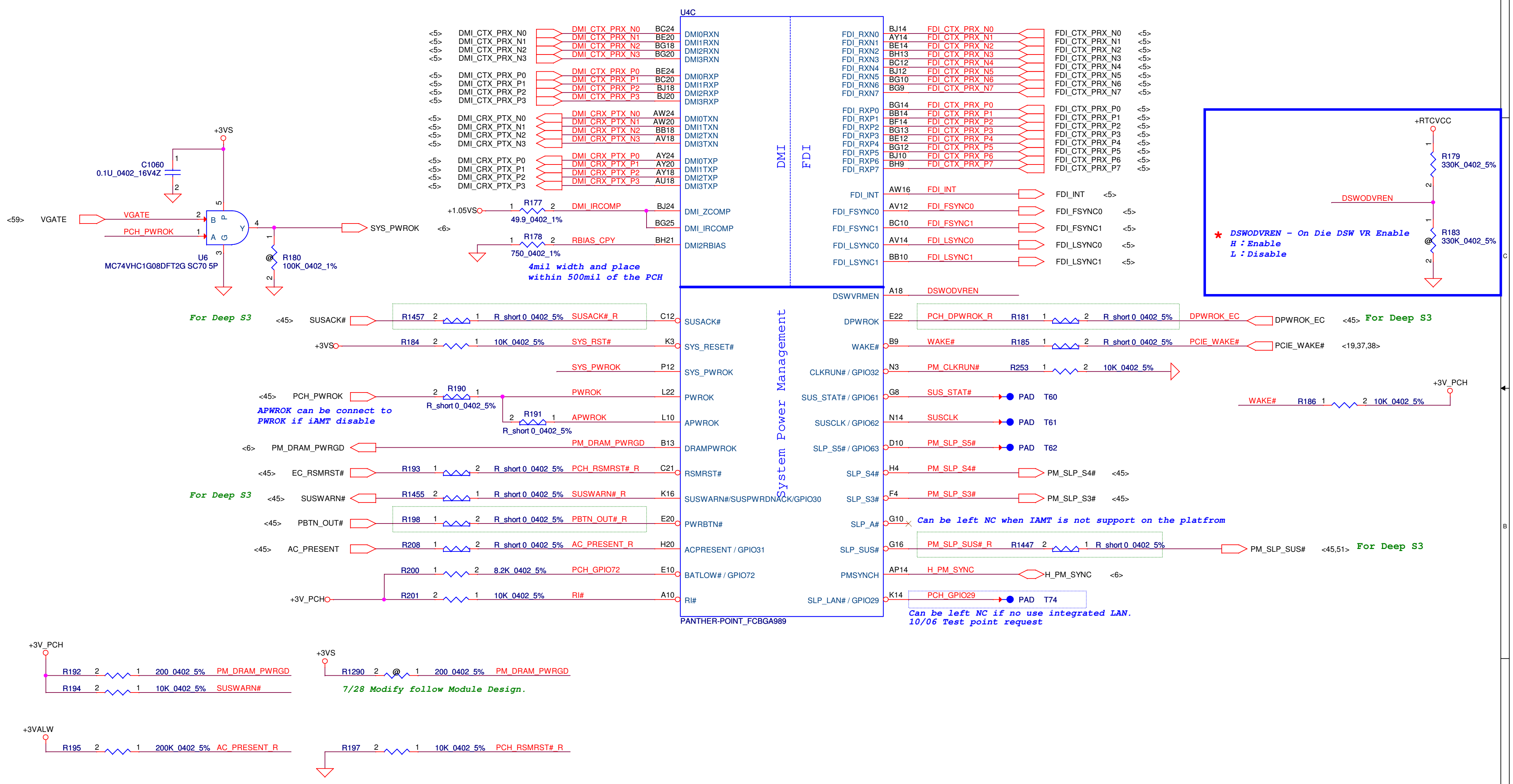

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Title: PCH (3/9) DMI,FDI,PM,

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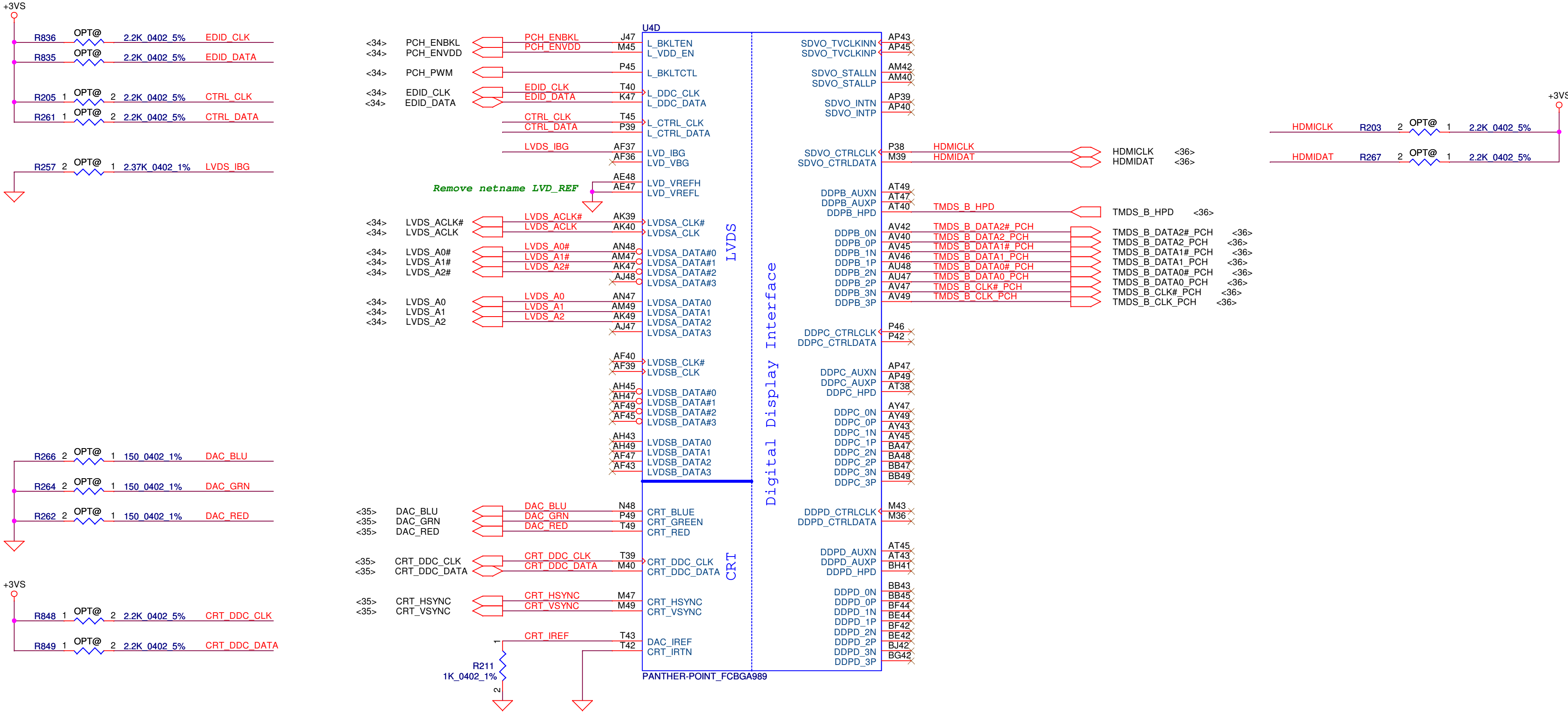

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Issued Date	2011/11/01	Deciphered Date	2012/12/31

Title: **PCH (4/9) LVDS,CRT,DP,HDMI**

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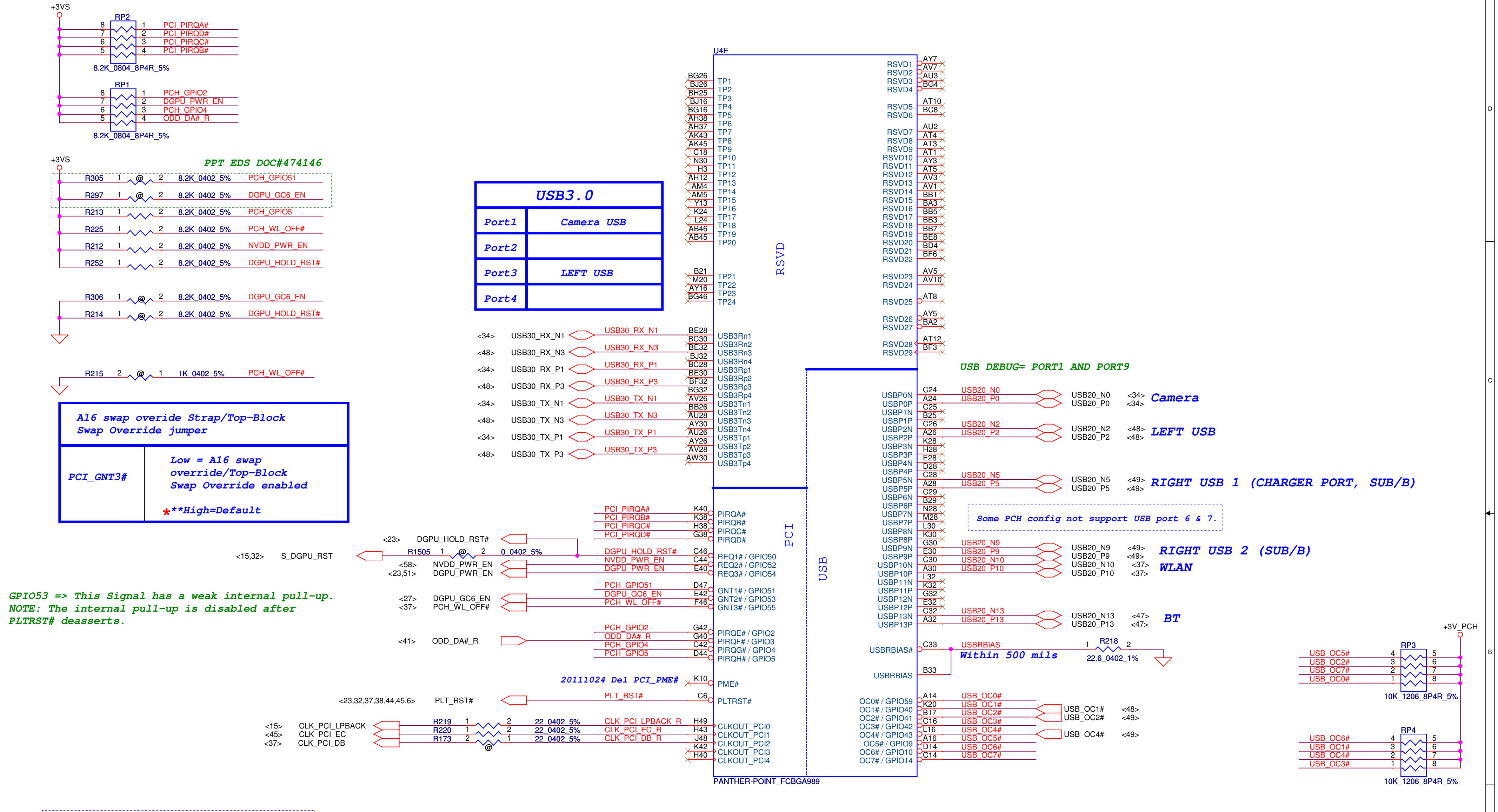

USB3.0	
Port1	Camera USB
Port2	
Port3	LEFT USB
Port4	

A16 swap overide Strap/Top-Block Swap Override jumper	
PCI_GNT3#	Low = A16 swap override/Top-Block Swap Override enabled **High=Default

GPIO53 => This Signal has a weak internal pull-up.
NOTE: The internal pull-up is disabled after PLTRST# deasserts.

PCH_GPIO51 R221 1K_0402_5%

Boot BIOS Strap bit1 BBS1			
	Bit11	Bit10	Boot BIOS Destination
GNT1#/ GPIO51	0	1	Reserved
	1	0	Reserved
	1	1	SPI (Default)
	0	0	LPC

Security Classification	Compal Secret Data		
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Title: PCH (5/9) PCI, USB

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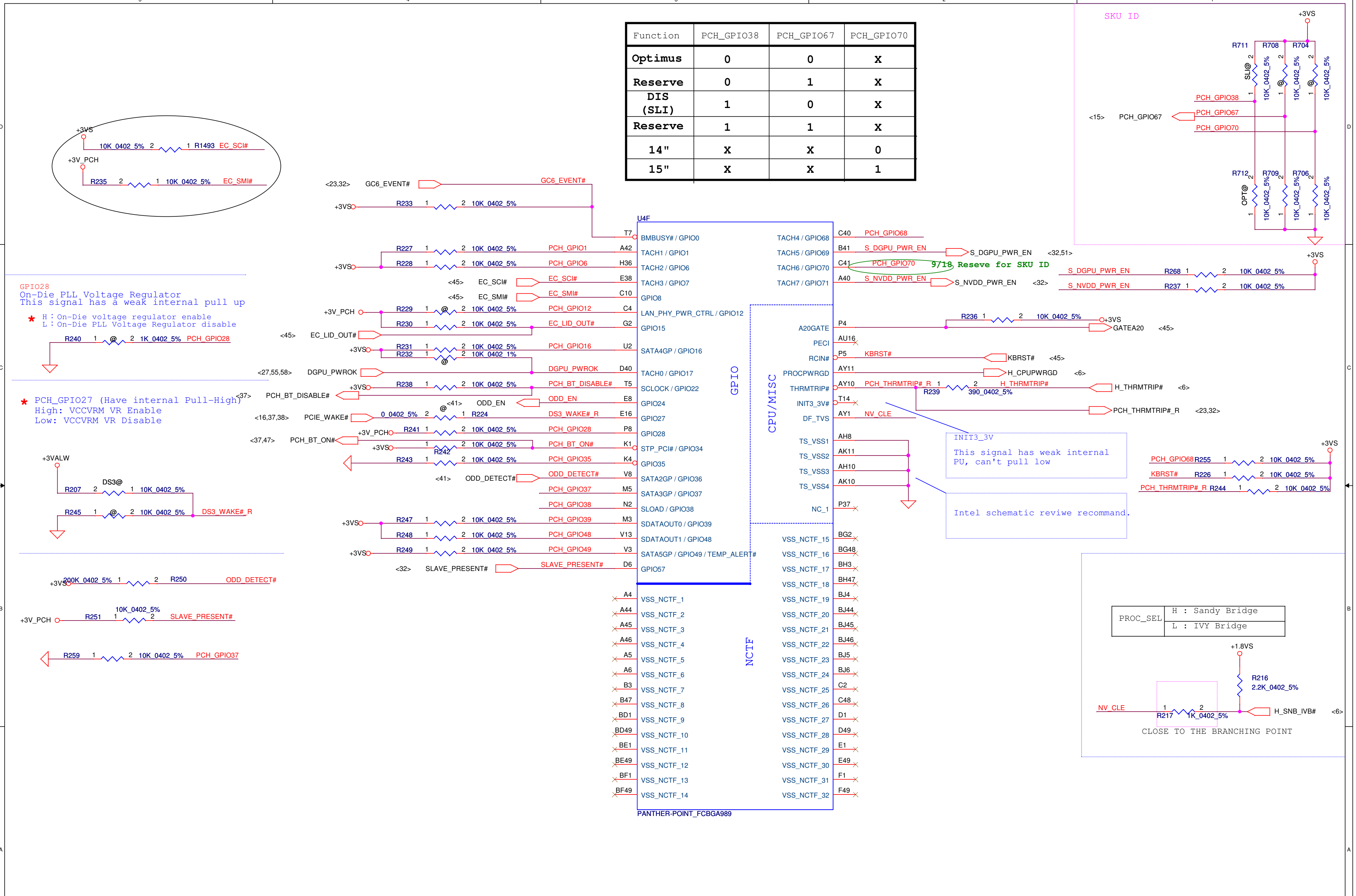

Function	PCH_GPIO38	PCH_GPIO67	PCH_GPIO70
Optimus	0	0	X
Reserve	0	1	X
DIS (SLI)	1	0	X
Reserve	1	1	X
14"	X	X	0
15"	X	X	1

PROC_SEL	
	H : Sandy Bridge
	L : IVY Bridge

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Title: **PCH (6/9) GPIO, CPU, MISC**

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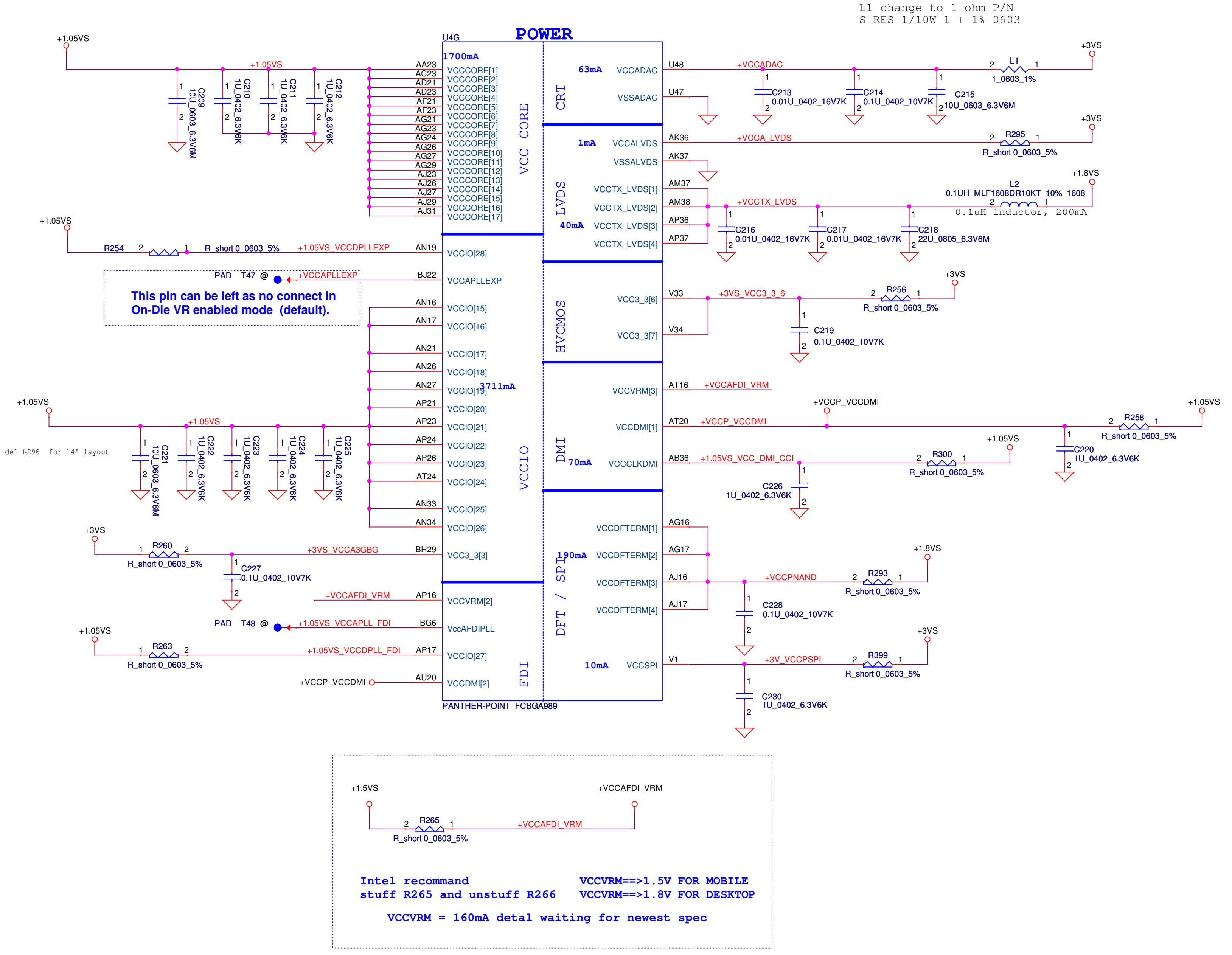
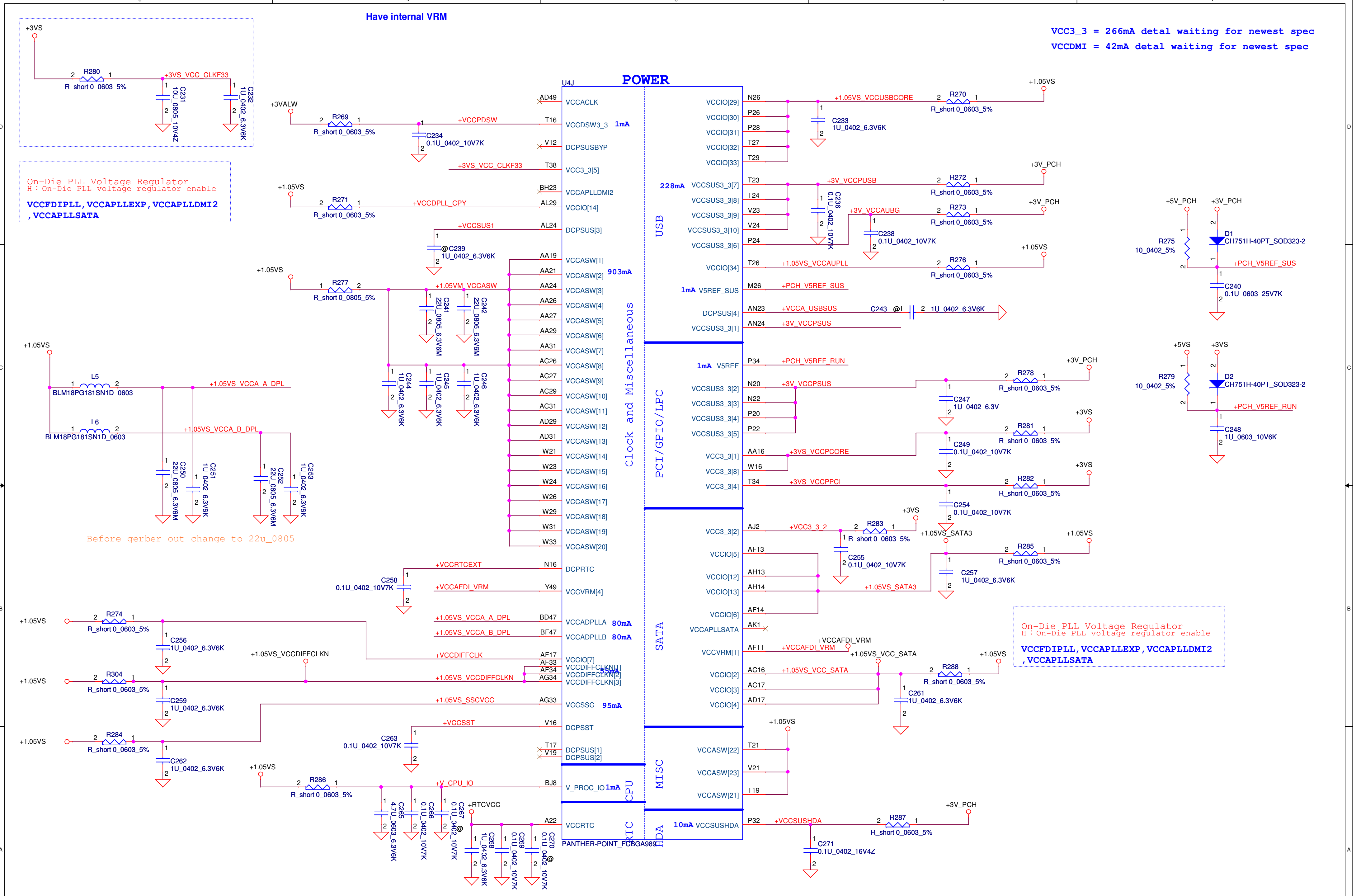


PCH Power Rail Table
Refer to CPU EDS R1.5

Voltage Rail	Voltage	S0 Iccmax Current (A)
V_PROC_IO	1.05	0.001
V5REF	5	0.001
V5REF_Sus	5	0.001
Vcc3_3	3.3	0.228
VccADAC	3.3	0.063
VccADPLLA	1.05	0.08
VccADPLLB	1.05	0.08
VccCore	1.05	1.7
VccDMI	1.05	0.047
VccIO	1.05	3.711
VccASW	1.05	0.903
VccSPI	3.3	0.01
VccDSW	3.3	0.001
VccDFTERM	1.8	0.002
VccRTC	3.3	6 uA
VccSus3_3	3.3	0.095
VccSusHDA	3.3 / 1.5	0.01
VccVRM	1.8 / 1.5	0.167
VccCLKDMI	1.05	0.07
VccSSC	1.05	0.095
VccDIFFCLKN	1.05	0.055
VccALVDS	3.3	0.001
VccTX_LVDS	1.8	0.04



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Title: **PCH (8/9) PWR**

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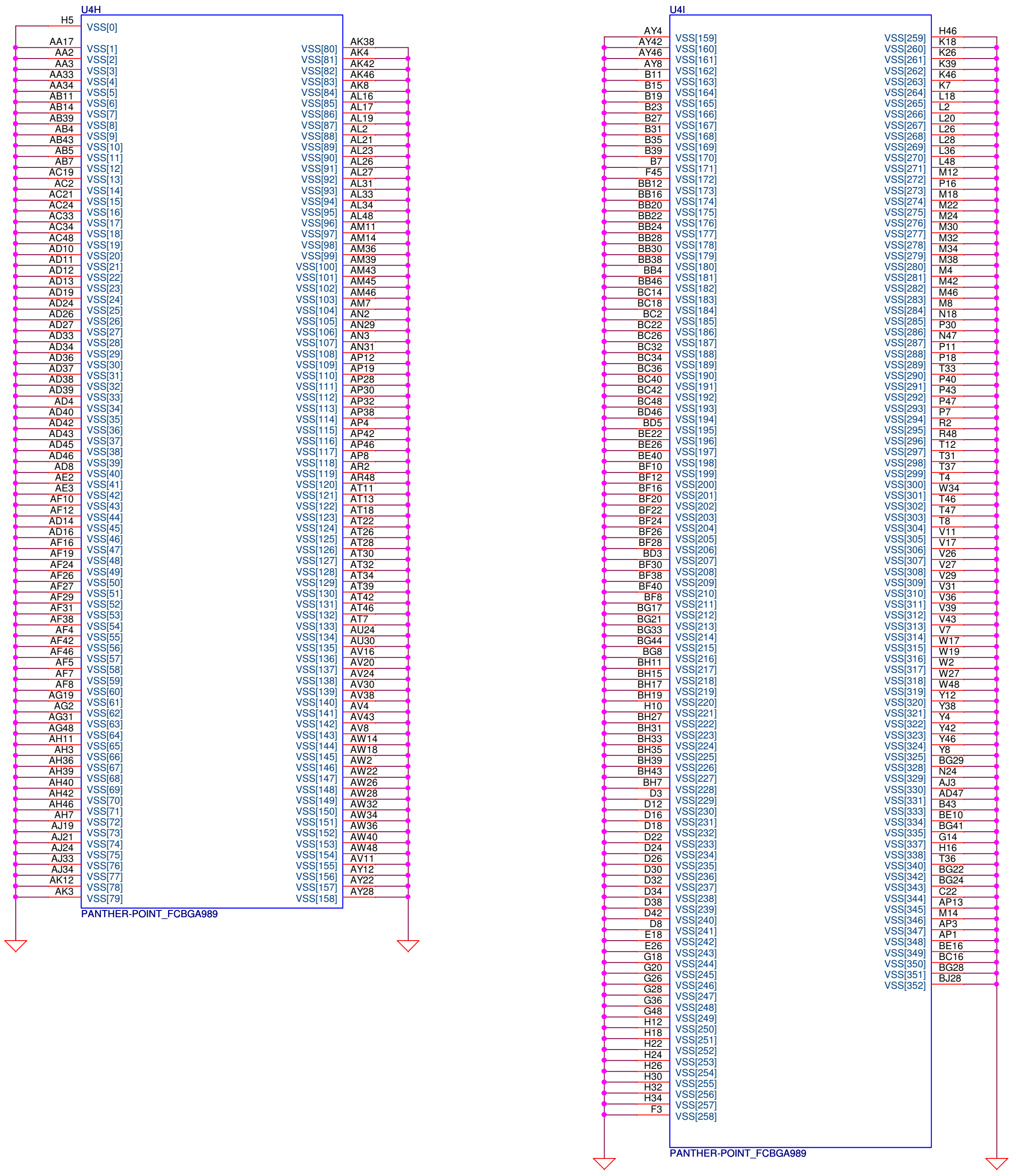

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Title: **PCH (9/9) VSS**

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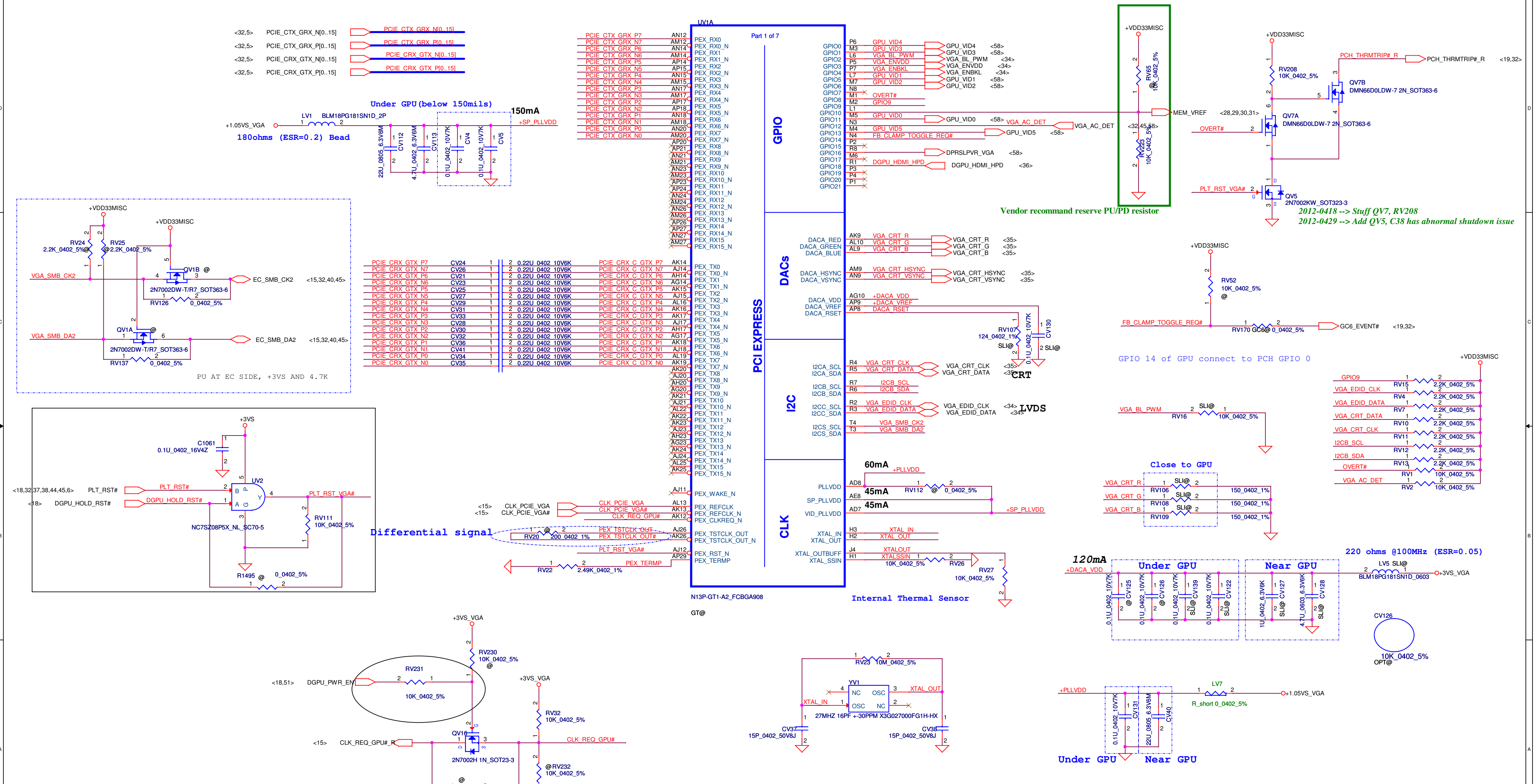

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Title: **N13P-PCIE/DAC/GPIO**

Size: | Document Number: Y490-LA8691P | Rev: 0.2

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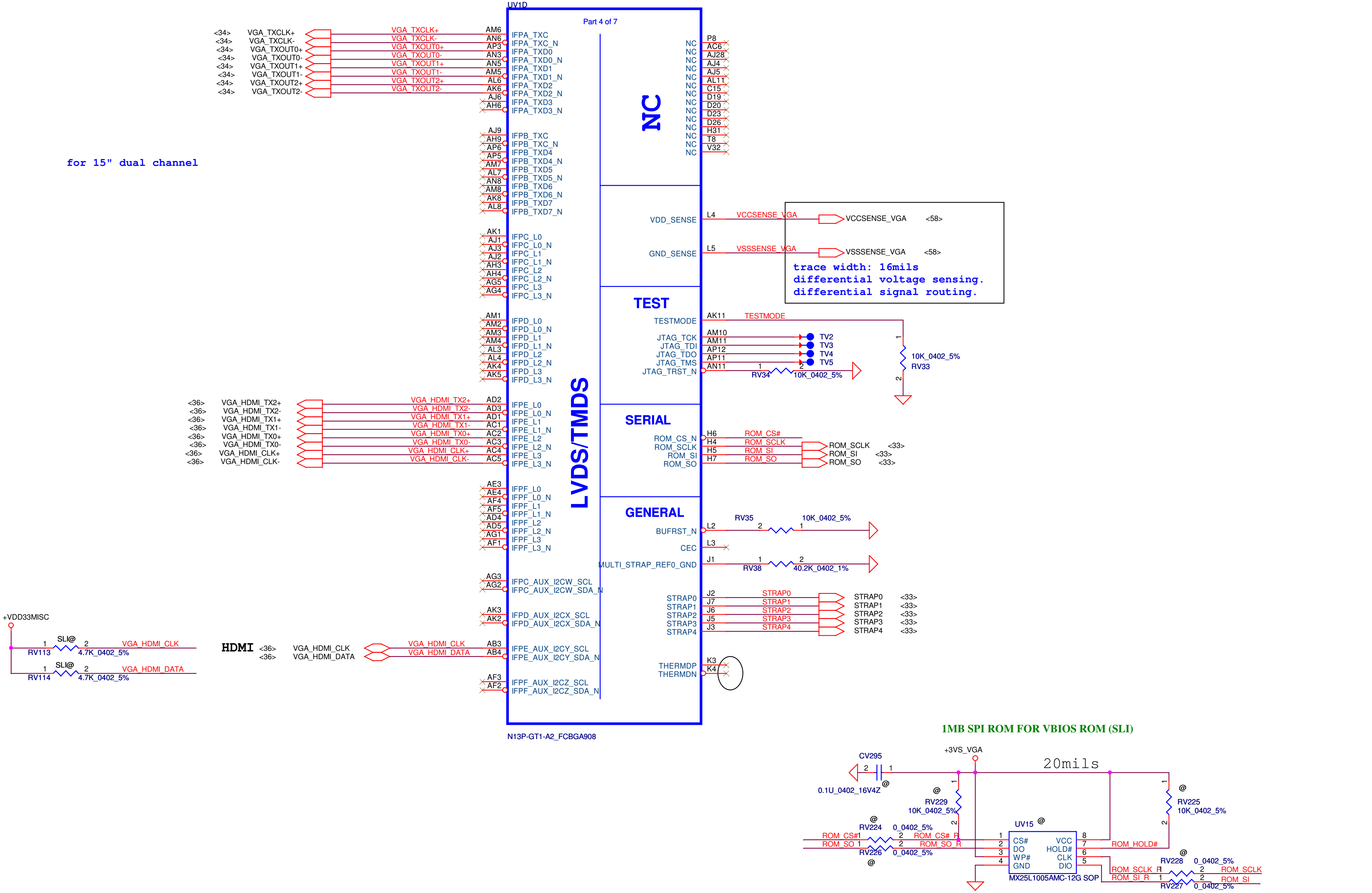
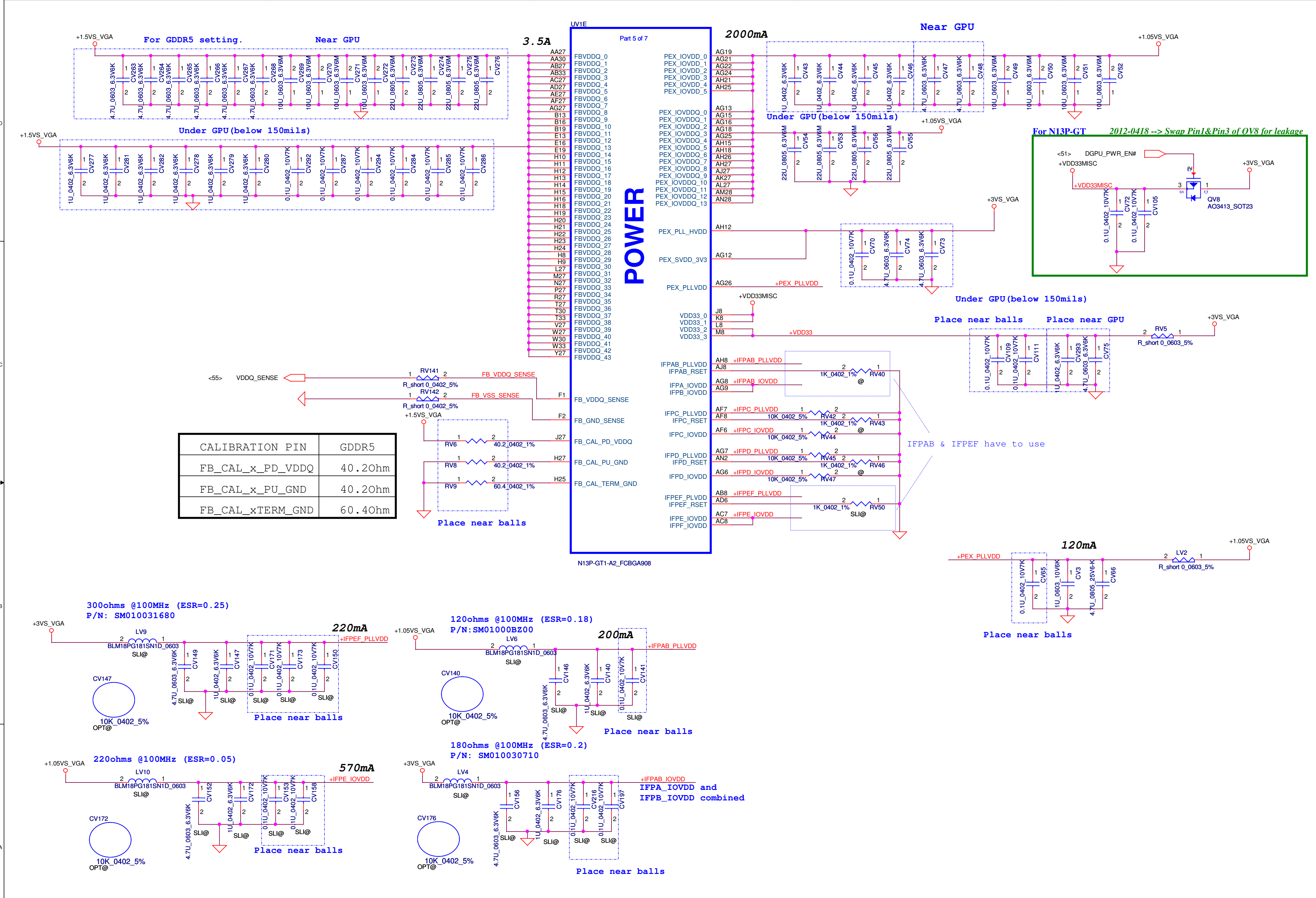


Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

LCFC Electronics, Inc.

Title: N13P-LVDS/HDMI/DP/THM

Size | Document Number: Y490-LA8691P | Rev 0.2



CALIBRATION PIN	GDDR5
FB_CAL_x_PD_VDDQ	40.2Ohm
FB_CAL_x_PU_GND	40.2Ohm
FB_CAL_xTERM_GND	60.4Ohm

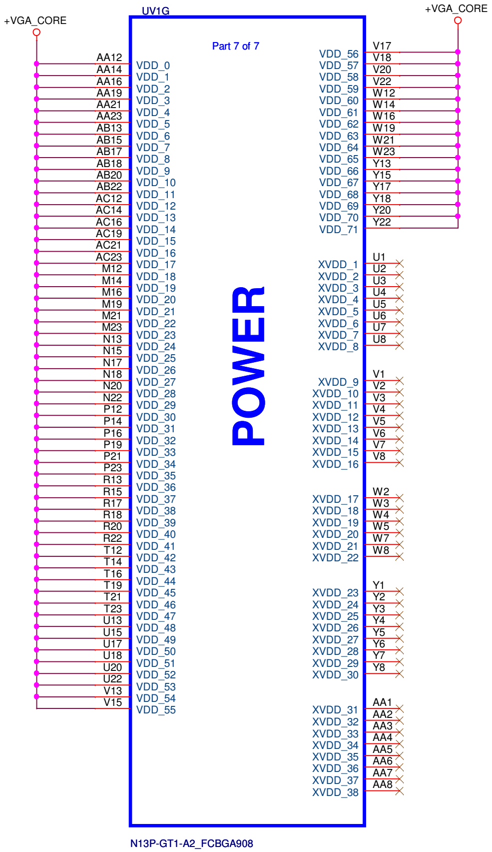
Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

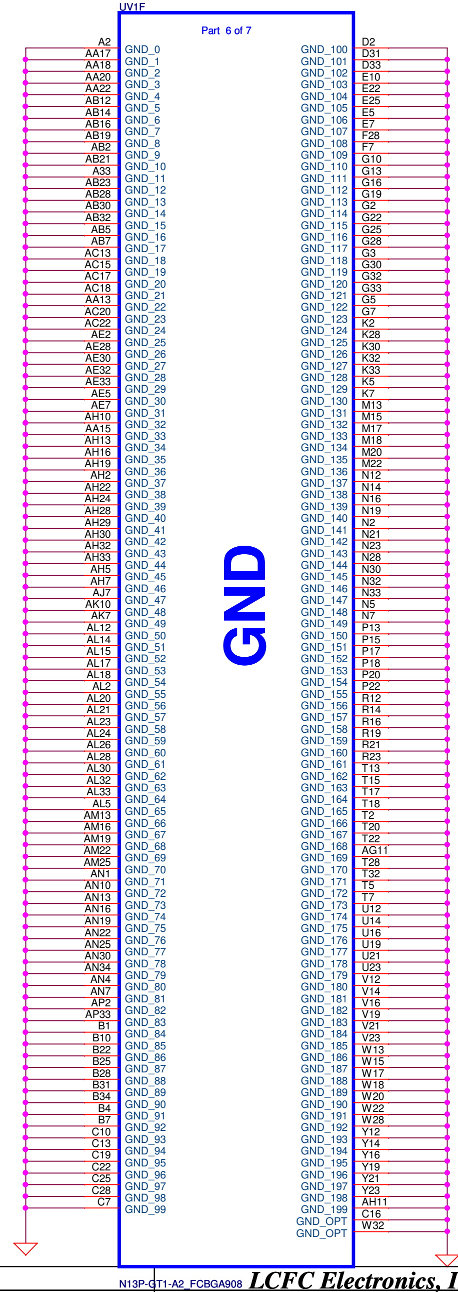
LCFC Electronics, Inc.

Title: N13P-POWER

Size | Document Number: Y490-LA8691P | Rev 0.2

Date: Thursday, May 10, 2012 |





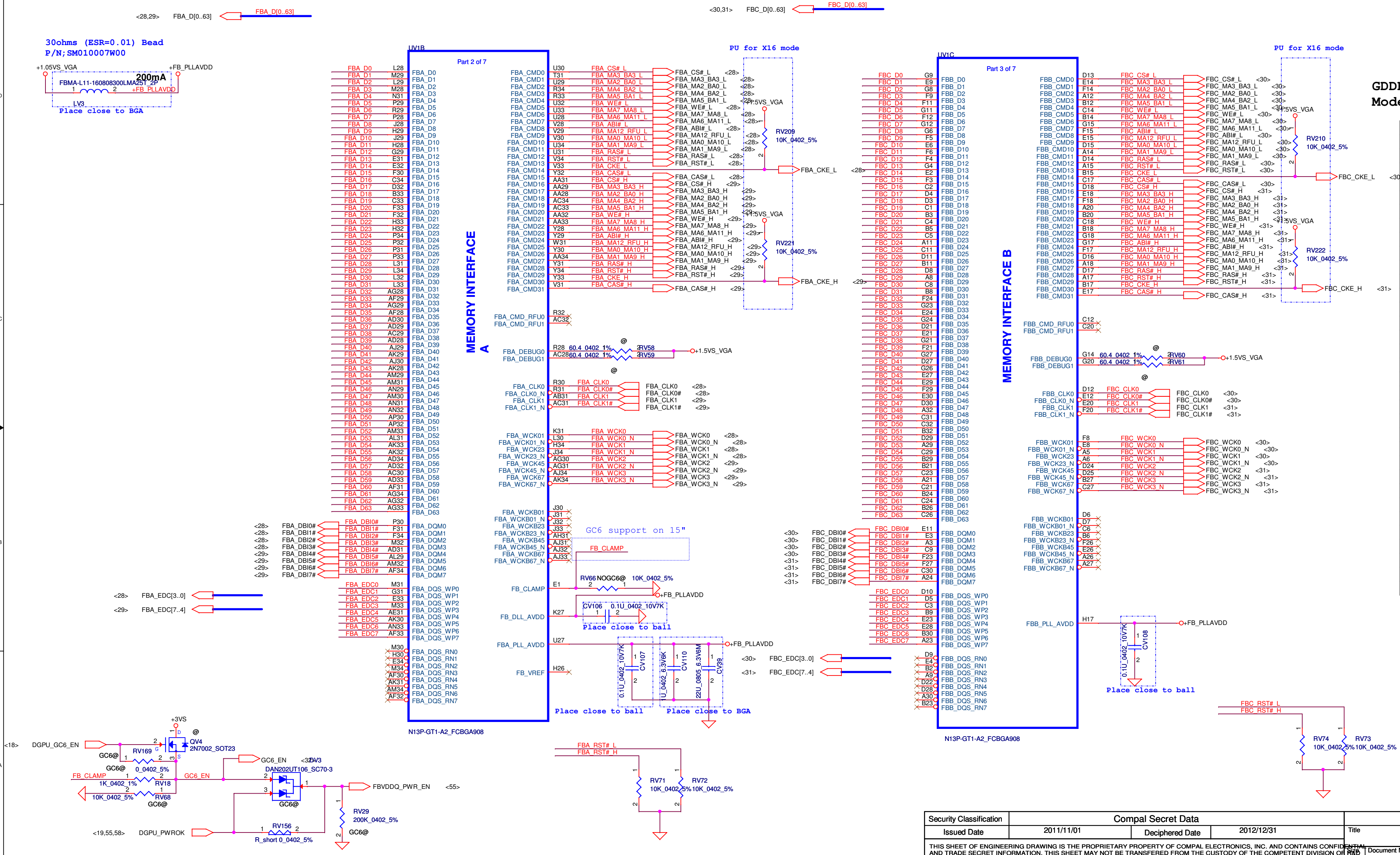
Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

LCFC Electronics, Inc.

Title: **N13P-VGA CORE, GND**

Size | Document Number: **Y490-LA8691P** | Rev 0.2

Date: Thursday, May 10, 2012 | Sheet 26 of 65



GDDR5
Mode H - Mirror Mode Mapping

	DATA Bus	
Address	0..31	32..63
FBx_CMD0	CS#	
FBx_CMD1	A3_BA3	
FBx_CMD2	A2_BA0	
FBx_CMD3	A4_BA2	
FBx_CMD4	A5_BA1	
FBx_CMD5	WE#	
FBx_CMD6	A7_A8	
FBx_CMD7	A6_A11	
FBx_CMD8	ABI#	
FBx_CMD9	A12_RFU	
FBx_CMD10	A0_A10	
FBx_CMD11	A1_A9	
FBx_CMD12	RAS#	
FBx_CMD13	RST#	
FBx_CMD14	CKE#	
FBx_CMD15	CAS#	
FBx_CMD16		CS#
FBx_CMD17		A3_BA3
FBx_CMD18		A2_BA0
FBx_CMD19		A4_BA2
FBx_CMD20		A5_BA1
FBx_CMD21		WE#
FBx_CMD22		A7_A8
FBx_CMD23		A6_A11
FBx_CMD24		ABI#
FBx_CMD25		A12_RFU
FBx_CMD26		A0_A10
FBx_CMD27		A1_A9
FBx_CMD28		RAS#
FBx_CMD29		RST#
FBx_CMD30		CKE#
FBx_CMD31		CAS#

Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

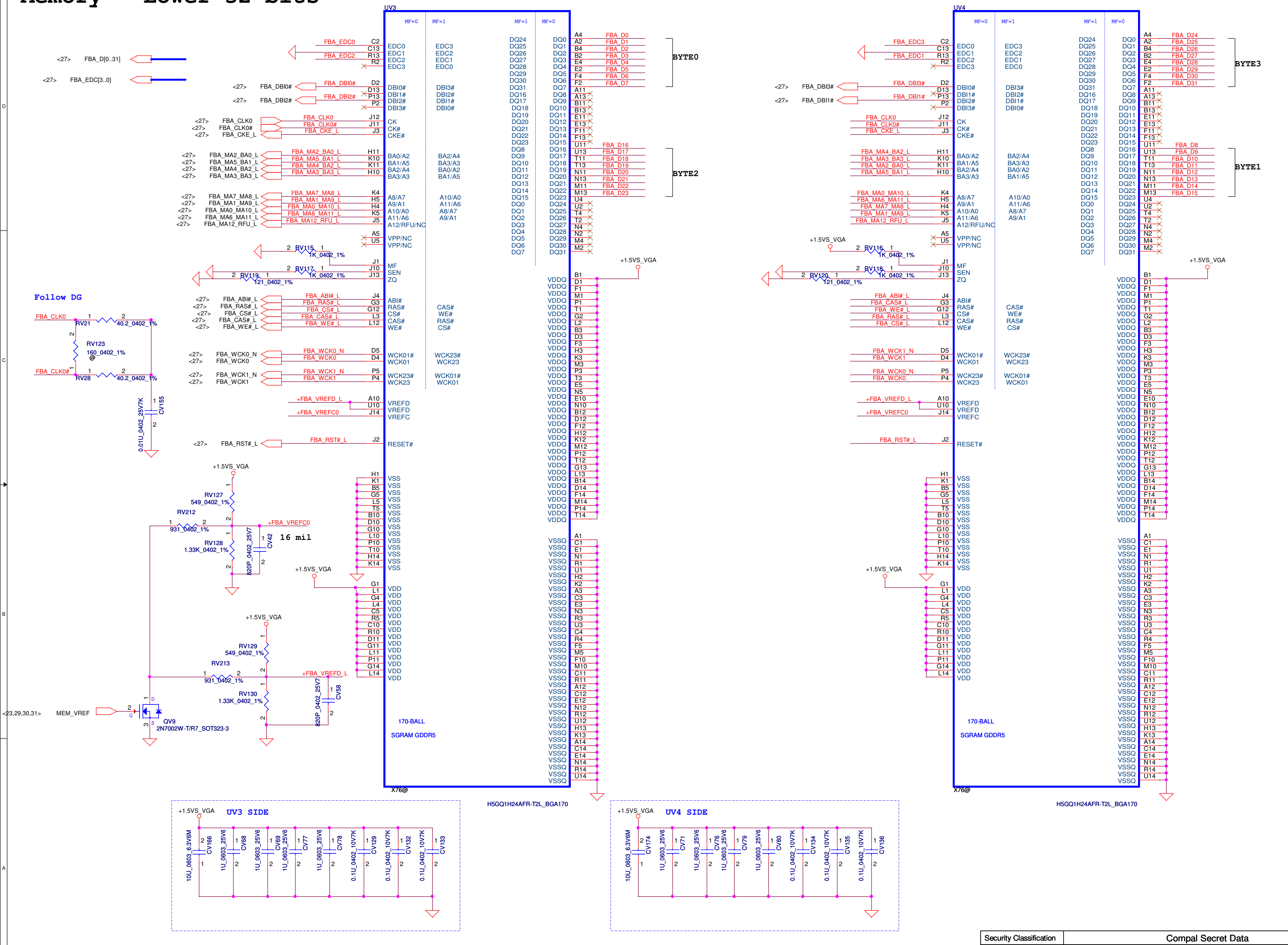
LCFC Electronics, Inc.

Title: **N13P-MEM Interface**

Size | Document Number: **Y490-LA8691P** | Rev 0.2

Date: Thursday, May 10, 2012 |

Memory - Lower 32 bits



GDDR5
Mode H - Mirror Mode Mapping

	DATA Bus	
Address	0..31	32..63
FBx_CMD0	CS#	
FBx_CMD1	A3_BA3	
FBx_CMD2	A2_BA0	
FBx_CMD3	A4_BA2	
FBx_CMD4	A5_BA1	
FBx_CMD5	WE#	
FBx_CMD6	A7_A8	
FBx_CMD7	A6_A11	
FBx_CMD8	ABI#	
FBx_CMD9	A12_RFU	
FBx_CMD10	A0_A10	
FBx_CMD11	A1_A9	
FBx_CMD12	RAS#	
FBx_CMD13	RST#	
FBx_CMD14	CKE#	
FBx_CMD15	CAS#	
FBx_CMD16		CS#
FBx_CMD17		A3_BA3
FBx_CMD18		A2_BA0
FBx_CMD19		A4_BA2
FBx_CMD20		A5_BA1
FBx_CMD21		WE#
FBx_CMD22		A7_A8
FBx_CMD23		A6_A11
FBx_CMD24		ABI#
FBx_CMD25		A12_RFU
FBx_CMD26		A0_A10
FBx_CMD27		A1_A9
FBx_CMD28		RAS#
FBx_CMD29		RST#
FBx_CMD30		CKE#
FBx_CMD31		CAS#

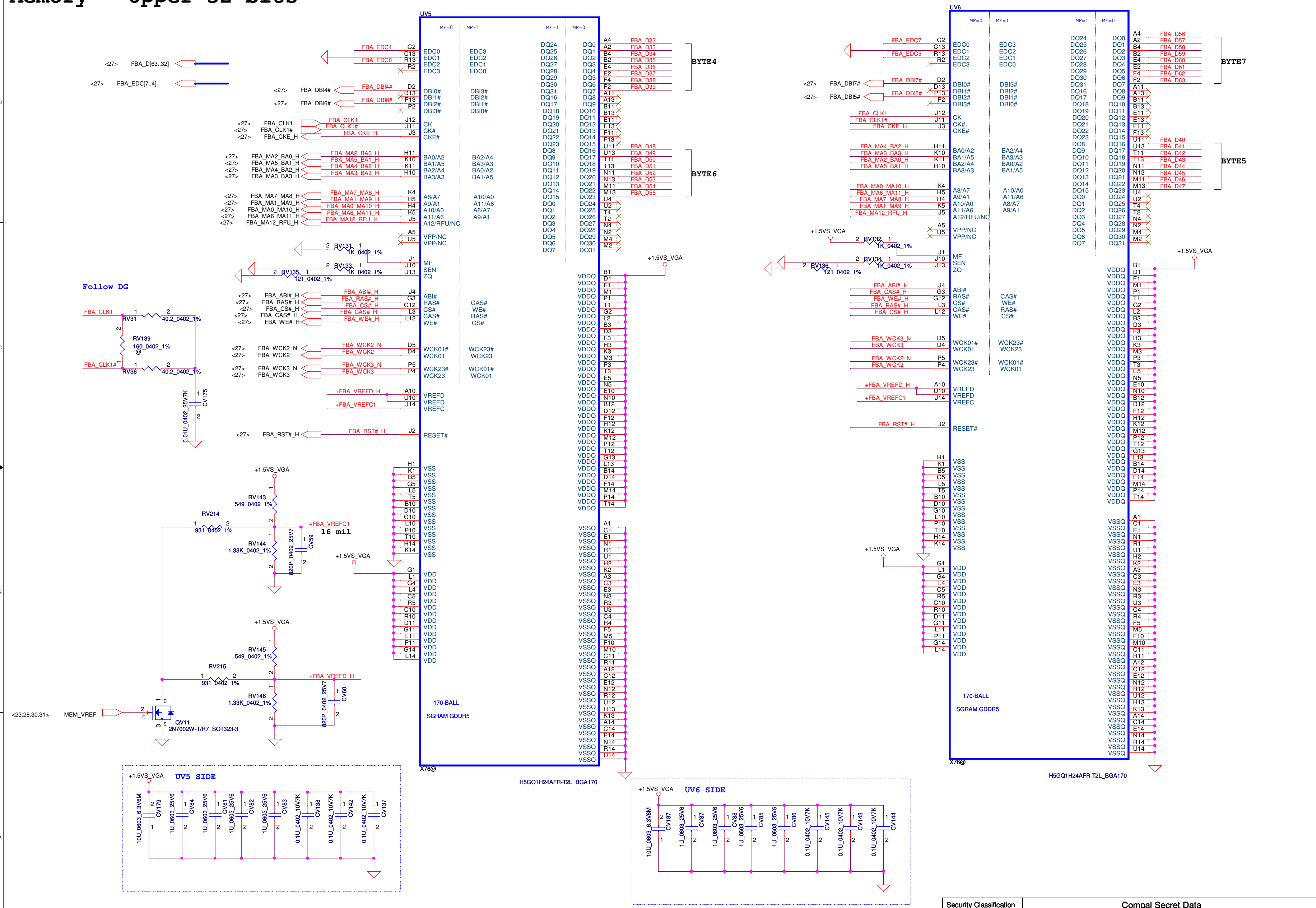
Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

LCFC Electronics, Inc.

Title: N13P-VRAM A Lower

Size | Document Number: Y490-LA8691P | Rev 0.2

Memory - Upper 32 bits



GDDR5
Mode H - Mirror Mode Mapping

	DATA Bus	
Address	0..31	32..63
FBx_CMD0	CS#	
FBx_CMD1	A3_BA3	
FBx_CMD2	A2_BA0	
FBx_CMD3	A4_BA2	
FBx_CMD4	A5_BA1	
FBx_CMD5	WE#	
FBx_CMD6	A7_A8	
FBx_CMD7	A6_A11	
FBx_CMD8	ABI#	
FBx_CMD9	A12_RFU	
FBx_CMD10	A0_A10	
FBx_CMD11	A1_A9	
FBx_CMD12	RAS#	
FBx_CMD13	RST#	
FBx_CMD14	CKE#	
FBx_CMD15	CAS#	
FBx_CMD16		CS#
FBx_CMD17		A3_BA3
FBx_CMD18		A2_BA0
FBx_CMD19		A4_BA2
FBx_CMD20		A5_BA1
FBx_CMD21		WE#
FBx_CMD22		A7_A8
FBx_CMD23		A6_A11
FBx_CMD24		ABI#
FBx_CMD25		A12_RFU
FBx_CMD26		A0_A10
FBx_CMD27		A1_A9
FBx_CMD28		RAS#
FBx_CMD29		RST#
FBx_CMD30		CKE#
FBx_CMD31		CAS#

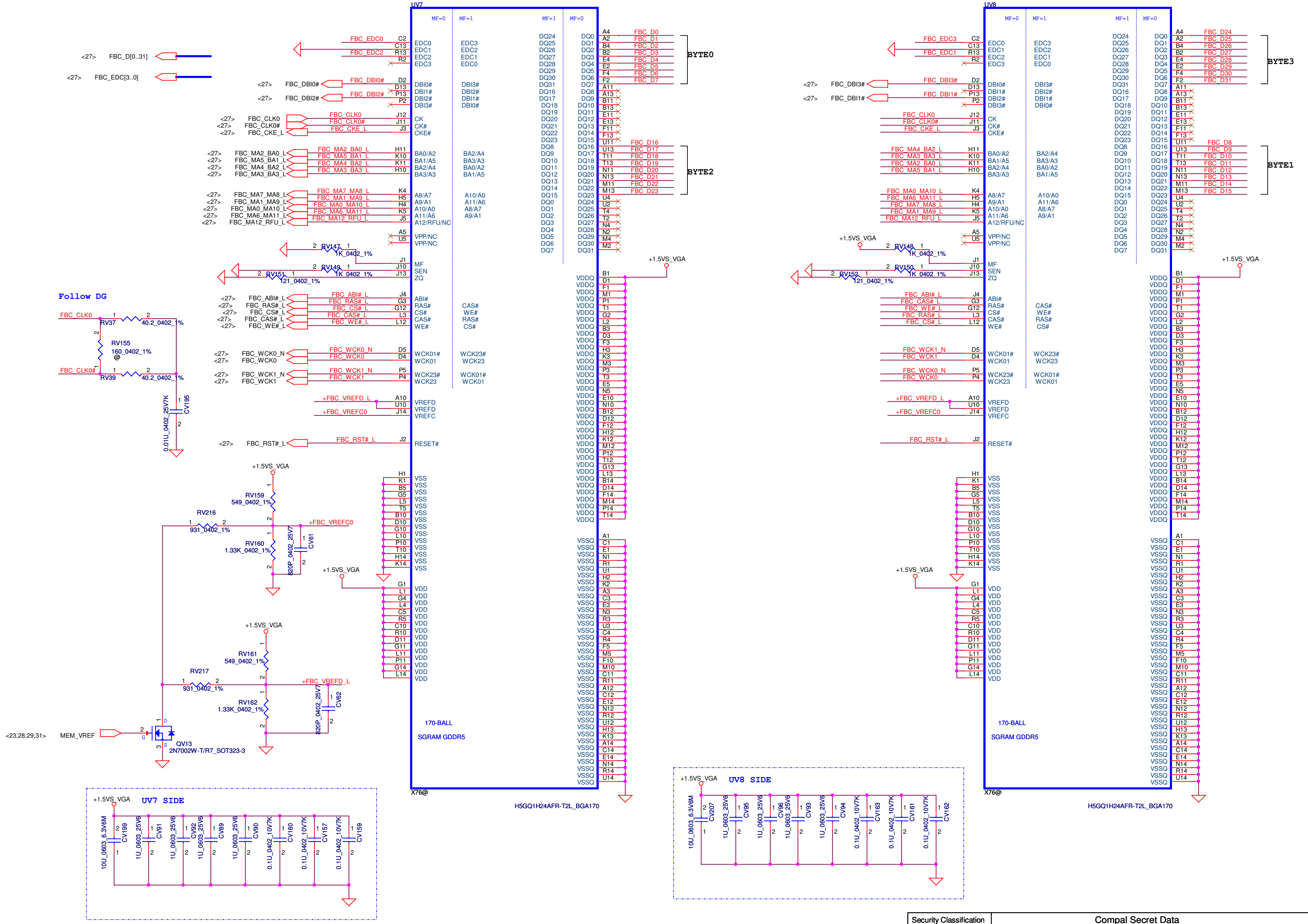
Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

LCFC Electronics, Inc.

Title: N13P-VRAM A Upper

Size | Document Number: Y490-LA8691P | Rev 0.2

Memory Partition C - Lower 32 bits



GDDR5
Mode H - Mirror Mode Mapping

	DATA Bus	
Address	0..31	32..63
FBx_CMD0	CS#	
FBx_CMD1	A3_BA3	
FBx_CMD2	A2_BA0	
FBx_CMD3	A4_BA2	
FBx_CMD4	A5_BA1	
FBx_CMD5	WE#	
FBx_CMD6	A7_A8	
FBx_CMD7	A6_A11	
FBx_CMD8	ABI#	
FBx_CMD9	A12_RFU	
FBx_CMD10	A0_A10	
FBx_CMD11	A1_A9	
FBx_CMD12	RAS#	
FBx_CMD13	RST#	
FBx_CMD14	CKE#	
FBx_CMD15	CAS#	
FBx_CMD16		CS#
FBx_CMD17		A3_BA3
FBx_CMD18		A2_BA0
FBx_CMD19		A4_BA2
FBx_CMD20		A5_BA1
FBx_CMD21		WE#
FBx_CMD22		A7_A8
FBx_CMD23		A6_A11
FBx_CMD24		ABI#
FBx_CMD25		A12_RFU
FBx_CMD26		A0_A10
FBx_CMD27		A1_A9
FBx_CMD28		RAS#
FBx_CMD29		RST#
FBx_CMD30		CKE#
FBx_CMD31		CAS#

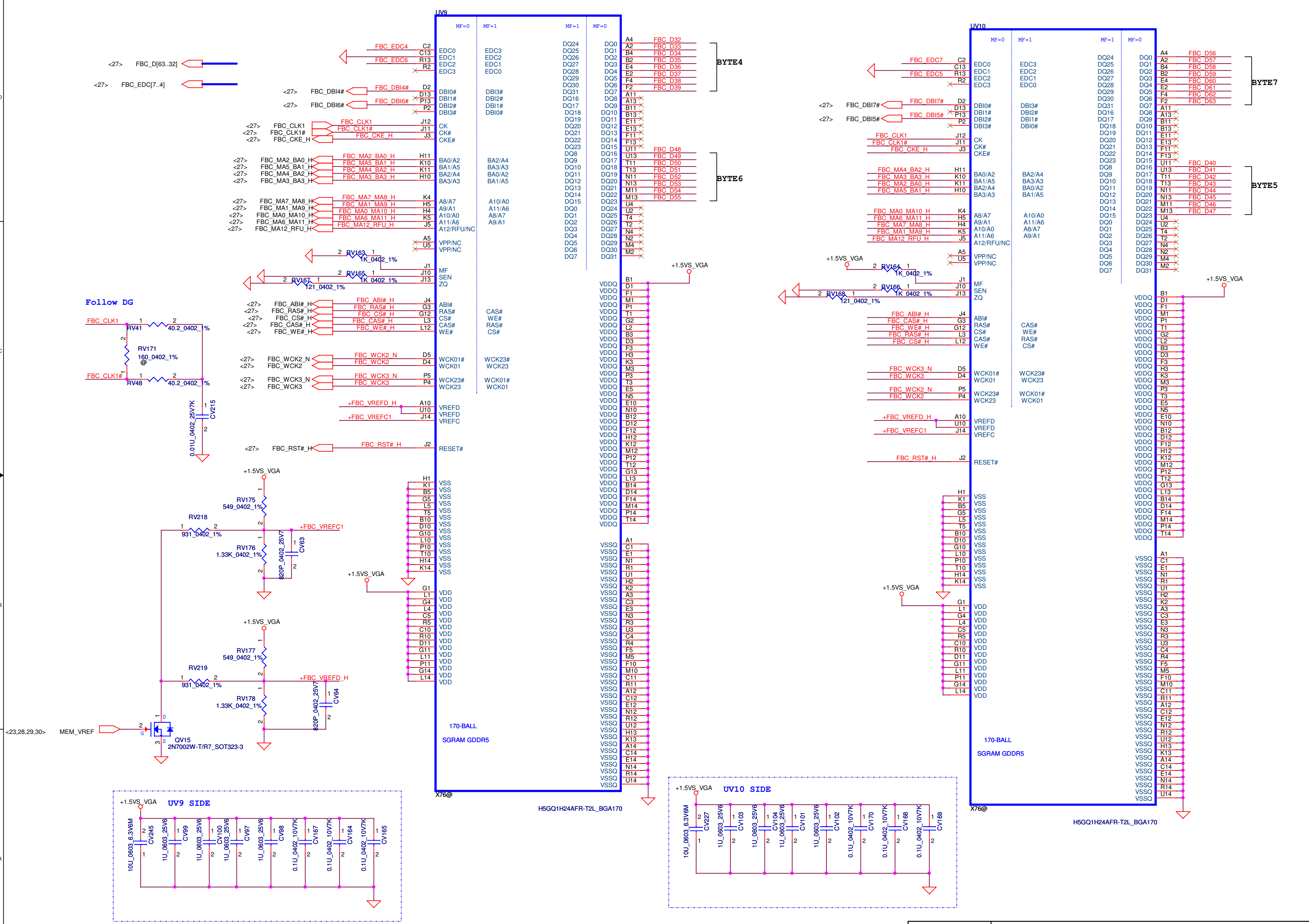
Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2011/12

LCFC Electronics, Inc.

Title: N12P-VRAM C Lower

Size: Custom | Document Number: Y490-LA8691P | Rev: 0.2

Memory Partition C - Upper 32 bits



GDDR5
Mode H - Mirror Mode Mapping

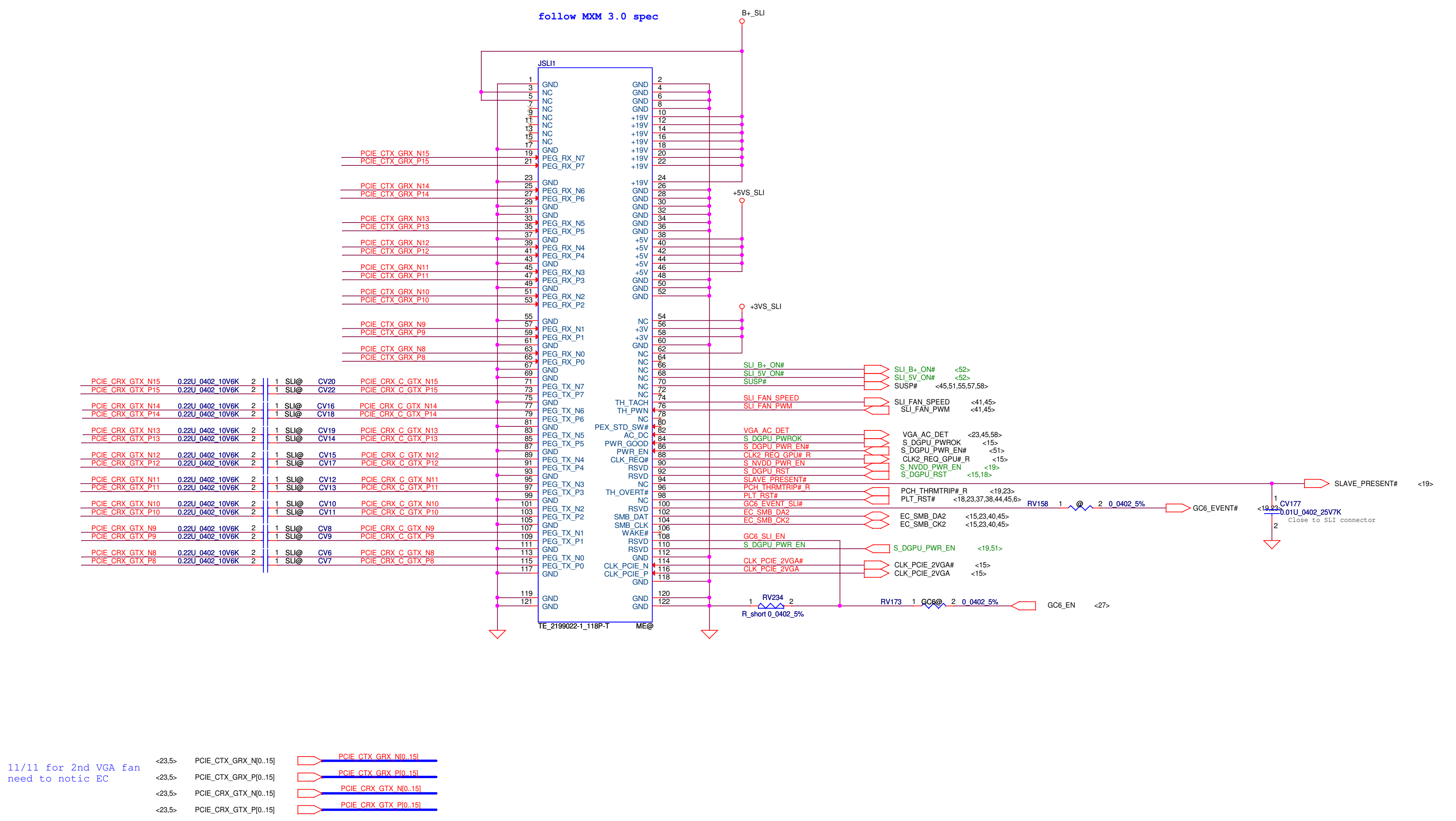
Address	DATA Bus 0..31	DATA Bus 32..63
FBx_CMD0	CS#	
FBx_CMD1	A3_BA3	
FBx_CMD2	A2_BA0	
FBx_CMD3	A4_BA2	
FBx_CMD4	A5_BA1	
FBx_CMD5	WE#	
FBx_CMD6	A7_A8	
FBx_CMD7	A6_A11	
FBx_CMD8	ABI#	
FBx_CMD9	A12_RFU	
FBx_CMD10	A0_A10	
FBx_CMD11	A1_A9	
FBx_CMD12	RAS#	
FBx_CMD13	RST#	
FBx_CMD14	CKE#	
FBx_CMD15	CAS#	
FBx_CMD16		CS#
FBx_CMD17		A3_BA3
FBx_CMD18		A2_BA0
FBx_CMD19		A4_BA2
FBx_CMD20		A5_BA1
FBx_CMD21		WE#
FBx_CMD22		A7_A8
FBx_CMD23		A6_A11
FBx_CMD24		ABI#
FBx_CMD25		A12_RFU
FBx_CMD26		A0_A10
FBx_CMD27		A1_A9
FBx_CMD28		RAS#
FBx_CMD29		RST#
FBx_CMD30		CKE#
FBx_CMD31		CAS#

Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2011/12

LCFC Electronics, Inc.

Title: N12P-VRAM C Upper

Size: Custom | Document Number: Y490-LA8691P | Rev: 0.2



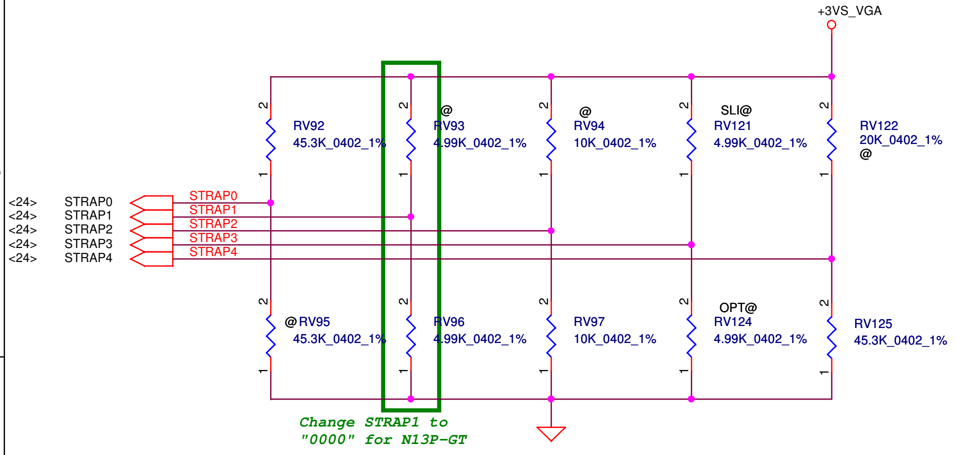
Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

LCFC Electronics, Inc.

Title: **DDRIII-SODIMM SLOT1**

Size	Document Number	Rev
Custom	Y490-LA8691P	0.2

Date: Thursday, May 10, 2012 Sheet 32 of 65

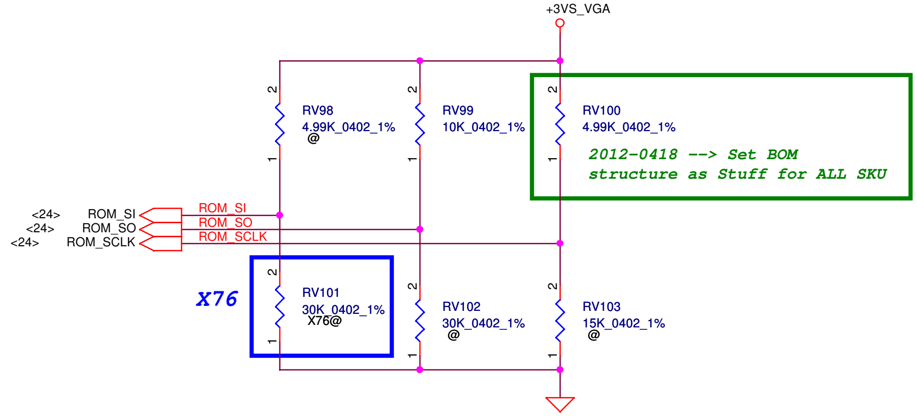


Physical Strapping pin	Power Rail	Logical Strapping Bit3	Logical Strapping Bit2	Logical Strapping Bit1	Logical Strapping Bit0
ROM_SCLK	+3VS_VGA	PCI_DEVID[4]	SUB_VENDOR	SLOT_CLK_CFG	PEX_PLL_EN_TERM
ROM_SI	+3VS_VGA	RAM_CFG[3]	RAM_CFG[2]	RAM_CFG[1]	RAM_CFG[0]
ROM_SO	+3VS_VGA	FB[1]	FB[0]	SMB_ALT_ADDR	VGA_DEVICE
STRAP0	+3VS_VGA	USER[3]	USER[2]	USER[1]	USER[0]
STRAP1	+3VS_VGA	3GIO_PAD_CFG_ADR[3]	3GIO_PAD_CFG_ADR[2]	3GIO_PAD_CFG_ADR[1]	3GIO_PAD_CFG_ADR[0]
STRAP2	+3VS_VGA	PCI_DEVID[3]	PCI_DEVID[2]	PCI_DEVID[1]	PCI_DEVID[0]
STRAP3	+3VS_VGA	SOR3_EXPOSED	SOR2_EXPOSED	SOR1_EXPOSED	SOR0_EXPOSED
STRAP4	+3VS_VGA	RESERVED	PCIE_SPEED_ CHANGE_GEN3	PCIE_MAX_SPEED	DP_PLL_VDD33V

Resistor Values	Pull-up to +3VS_VGA	Pull-down to Gnd
5K	1000	0000
10K	1001	0001
15K	1010	0010
20K	1011	0011
25K	1100	0100
30K	1101	0101
35K	1110	0110
45K	1111	0111

SLOT_CLK_CFG	
0	GPU and MCH don't share a common reference clock
1	GPU and MCH share a common reference clock (Default)

SUB_VENDOR	
0	No VBIOS ROM (Default)
1	BIOS ROM is present



3GIO_PADCFG	
3GIO_PADCFG[3:0]	
0000	Notebook Default

XCLK_417	
0	277MHz (Default)
1	Reserved

USER Straps	
User[3:0]	
1000-1100	Customer defined

PEX_PLL_EN_TERM	
0	Disable (Default)
1	Enable

PCIE_MAX_SPEED	
0	Limit to PCIE Gen1
1	PCIE Gen 2/3 Capable

FB_0_BAR_SIZE	
0	Reserved
1	Reserved
2	256MB (Default)
3	Reserved

SMBUS_ALT_ADDR	
0	0x9E (Default)
1	0x9C (Multi-GPU usage)

VGA_DEVICE	
0	3D Device (Class Code 302h)
1	VGA Device (Default)

X76

GPU	FB Memory (GDDR5)		ROM_SI	ROM_SO	ROM_SCLK	STRAP0	STRAP1	STRAP2	STRAP3	STRAP4
N13P-GT1 28nm	Samsung	K4G20325FD-FC04 2G 64Mx32	PD 30K	PU 10K	PU 5K (ALL SKU)	PU 45K	PD 5K	PD 10K	PU 5K	PD 10K
		K4G10325FG-HC04 1G 32Mx32	PD 45K							
	Hynix	H5GQ2H24MFR-T2C 2G 64Mx32 EOL	PD 25K							
		H5GQ1H24BFR-T2C 1G 32Mx32	PD 35K							
		H5GQ2H24AFR-T2C 2G 64Mx32	PD 5K							

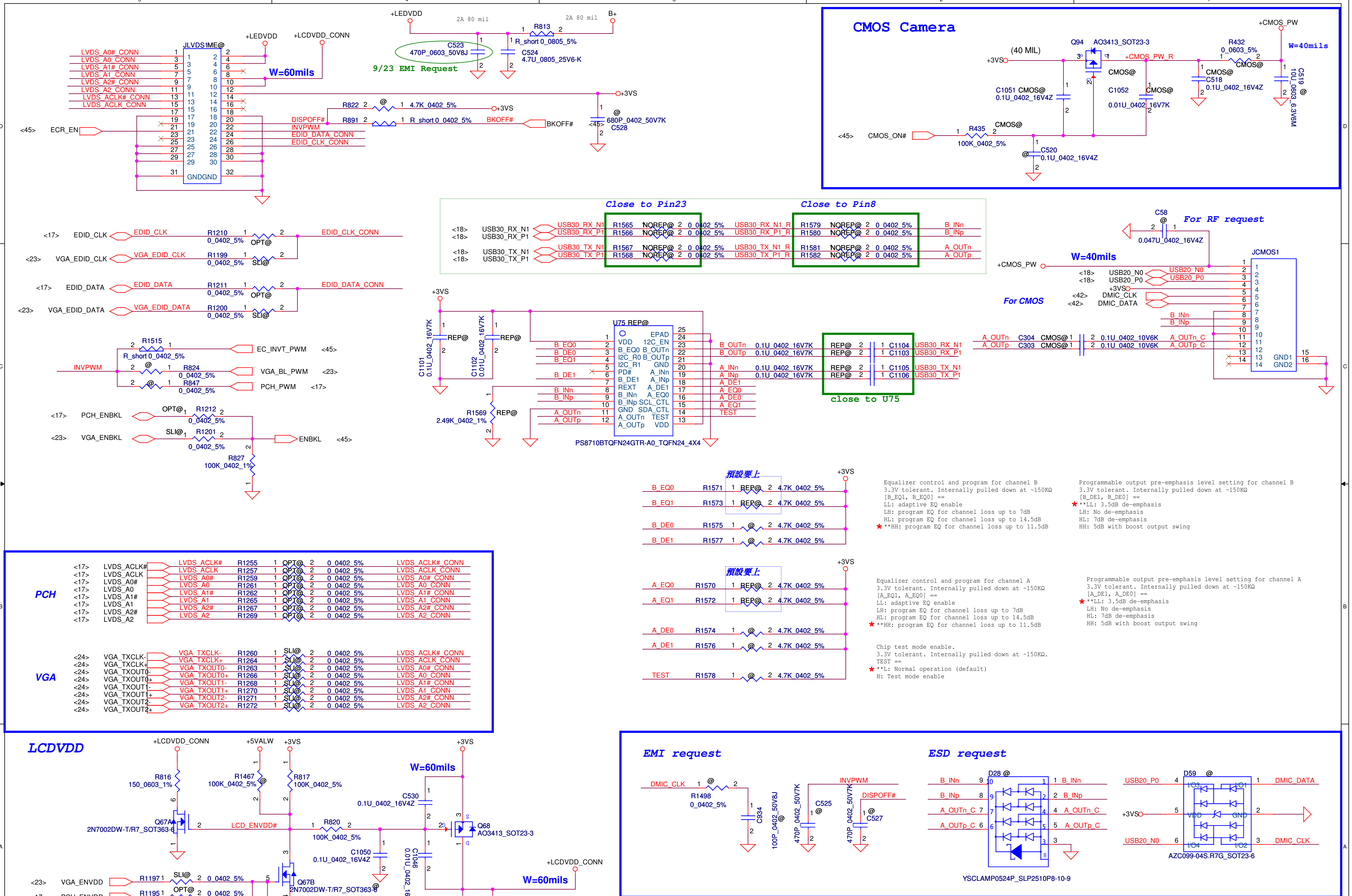
VRAM	X76	VRAM P/N
Samsung	X76409JVL01 (2G 64Mx32)	SA00005B70J
	X76409JVL51 (1G 32Mx16)	SA00003RS0J
Hynix	X76409JVL02 (2G 64Mx32)	SA00004GD0J EOL
	X76409JVL02 (2G 64Mx32)	SA00004GD1J
	X76409JVL52 (1G 32Mx16)	SA00003WL1J

Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

LCFC Electronics, Inc.

Title: **N13P_MISC**

Size: Custom | Document Number: **Y490-LA8691P** | Rev 0.2



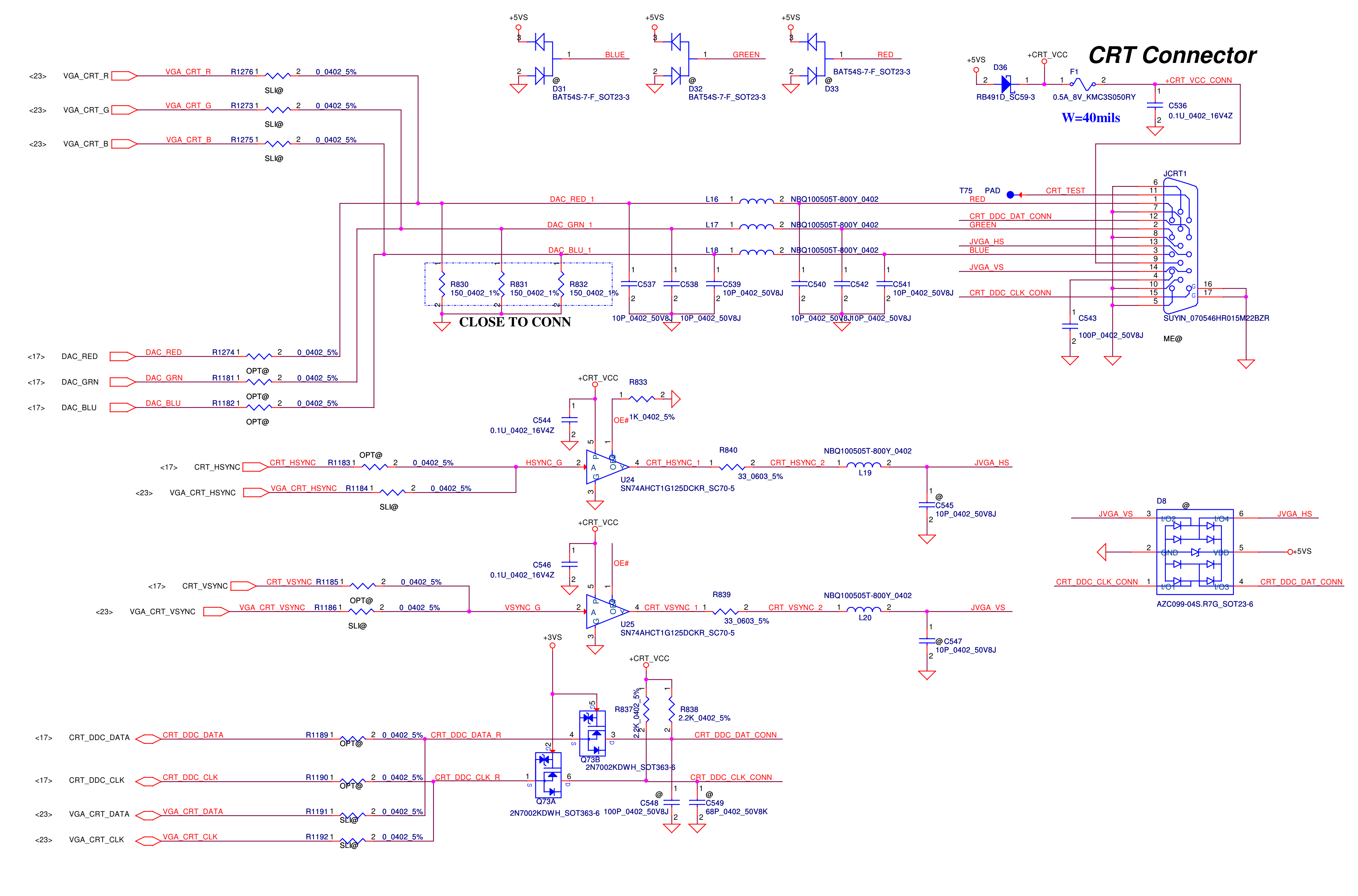
Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

LCFC Electronics, Inc.

Title: LVDS/ CMOS/ USB-ReDriver

Size: Custom | Document Number: Y490-LA8691P | Rev: 0.2

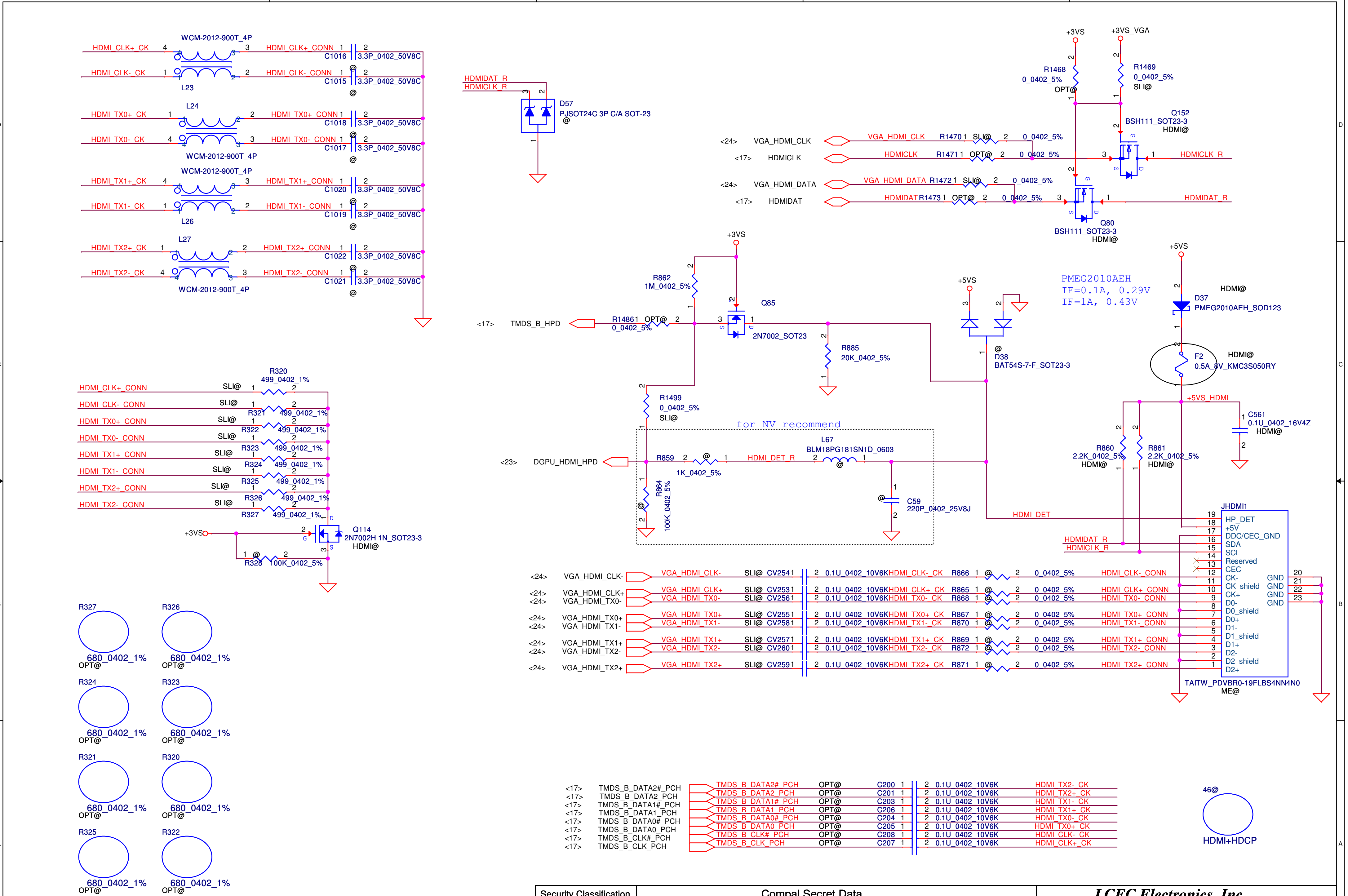
Date: Thursday, May 10, 2012 | Sheet 34 of 65



Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

LCFC Electronics, Inc.

Title	CRT Connector		
Size: Custom	Document Number: Y490-LA8691P		Rev: 0.2
Date: Thursday, May 10, 2012		Sheet 35 of 65	



Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

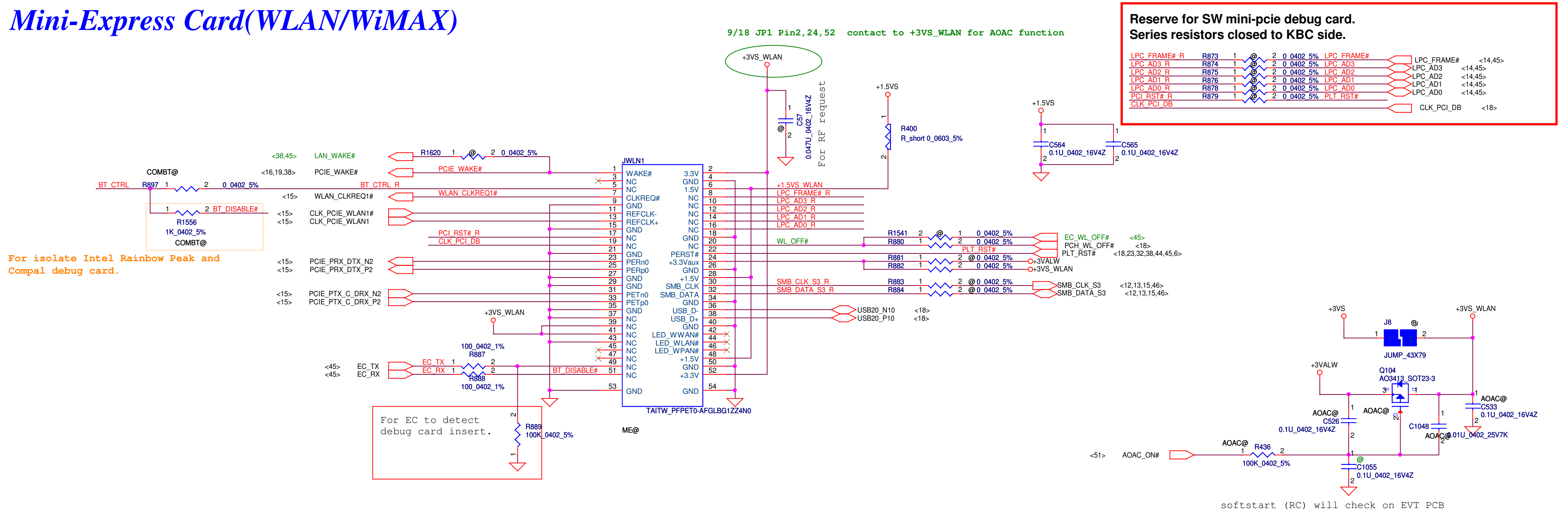
LCFC Electronics, Inc.

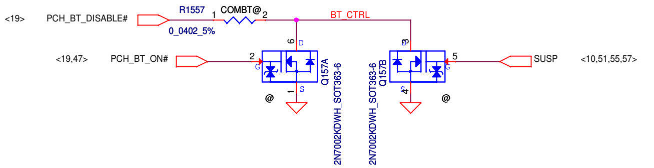
Title: **HDMI CONN**

Size	Document Number	Rev
Custom	Y490-LA8691P	0.2

Date: Thursday, May 10, 2012

Mini-Express Card(WLAN/WiMAX)



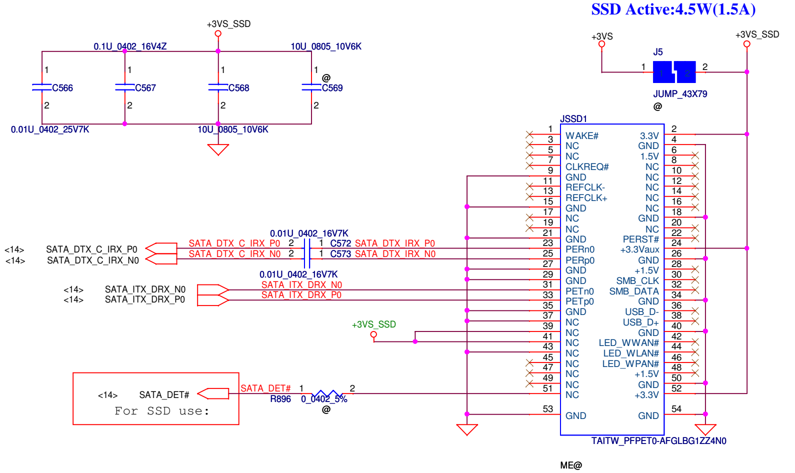


WLAN&BT Combo module circuits

	BT on module Enable	BT on module Disable
* BT_CRTL (GPIO22)	H	L
PCH_BT_ON#	L	H

Mini-Express Card(SSD)

SSD Active:4.5W(1.5A)



Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

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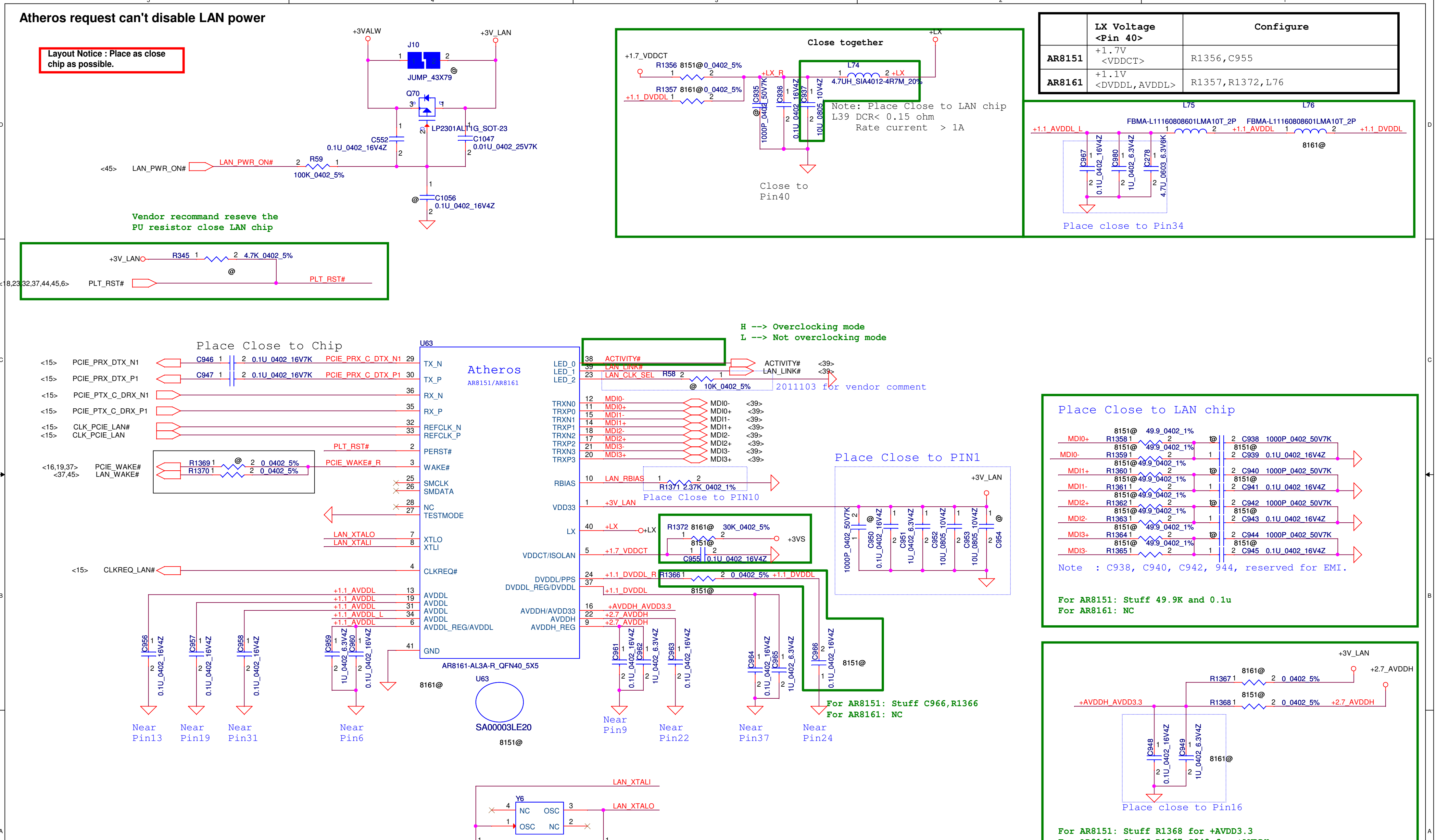
LCFC Electronics, Inc.

Title: *Mini-Card*

Size | Document Number: *Y490-LA8691P* | Rev 0.2

Date: Thursday, May 10, 2012 | Sheet 37 of 65



	LX Voltage <Pin 40>	Configure
AR8151	+1.7V <VDDCT>	R1356,C955
AR8161	+1.1V <DVDDL,AVDDL>	R1357,R1372,L76

LAN IC	X76	VRAM P/N
8161	X76409JVL04	SA000050E1J
8151	X76409JVL03	SA00003LE2J

Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

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LCFC Electronics, Inc.

Title: LAN-AR8151/8161

Size: Custom | Document Number: Y490-LA8691P | Rev: 0.2

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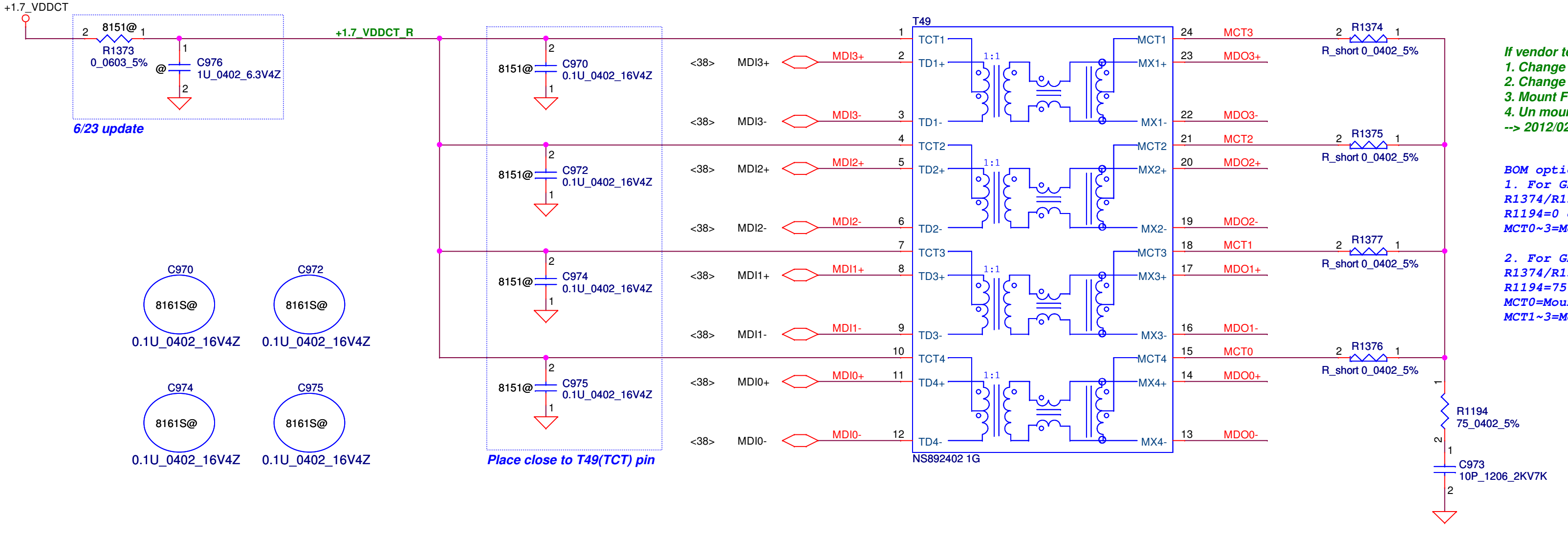
LAN Transformer



6/23 update

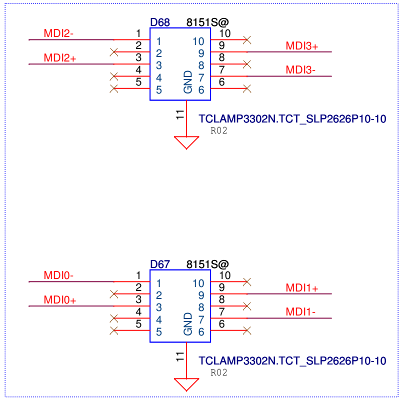
Place close to T49(TCT) pin

If vendor test result is "ok", need to change as below
1. Change R1374,R1375,R1376,R1377 to 0 ohm
2. Change R1194 to 75 ohm
3. Mount F6
4. Un mount F3,F4,F5
--> 2012/02/20 : already implement to Sch

BOM option:
1. For GDTx4
R1374/R1375/R1376/R1377=75 ohm
R1194=0 ohm
MCT0~3=Mount

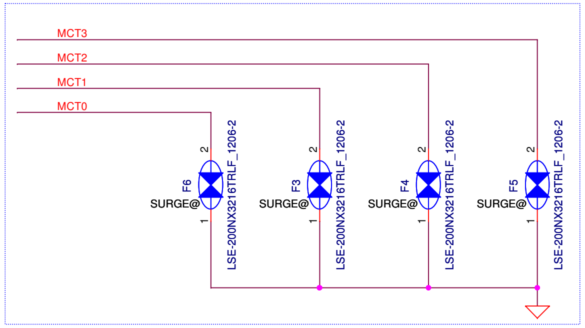
2. For GDTx1
R1374/R1375/R1376/R1377=0 ohm
R1194=75 ohm
MCT0=Mount
MCT1~3=Mount

Place Close to T49



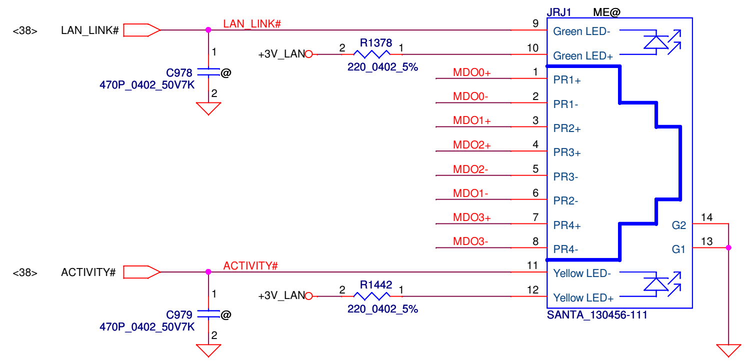
Reserve D67,D68 for EMI go rural solution

Place Close to T49



Reserve for EMI go rural solution

LAN Conn.



Security Classification	Compal Secret Data			LCFC Electronics, Inc.
Issued Date	2011/11/01	Deciphered Date	2012/12/31	Title: LAN Transformer
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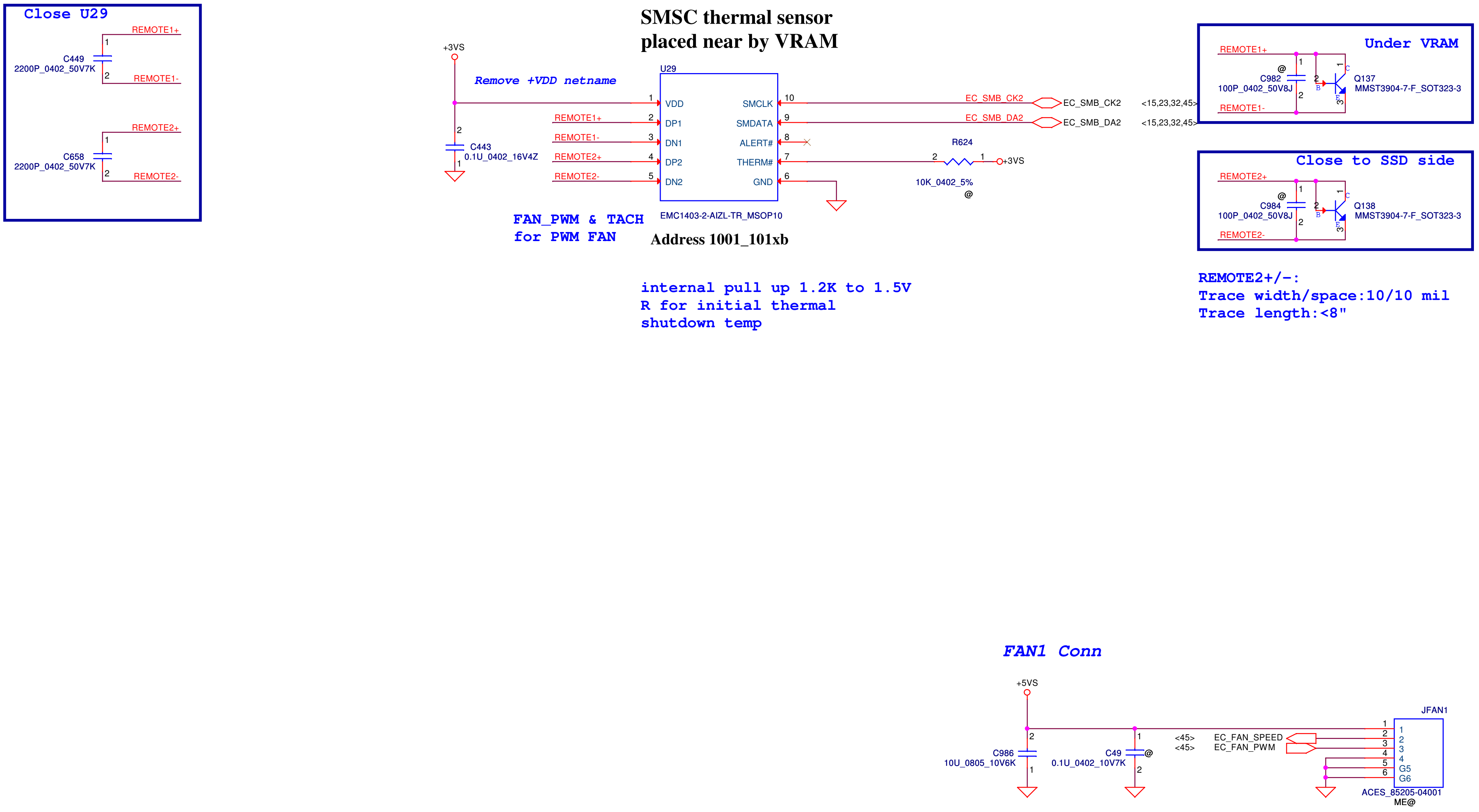
				Date: Friday, May 11, 2012; Sheet 39 of 65



Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

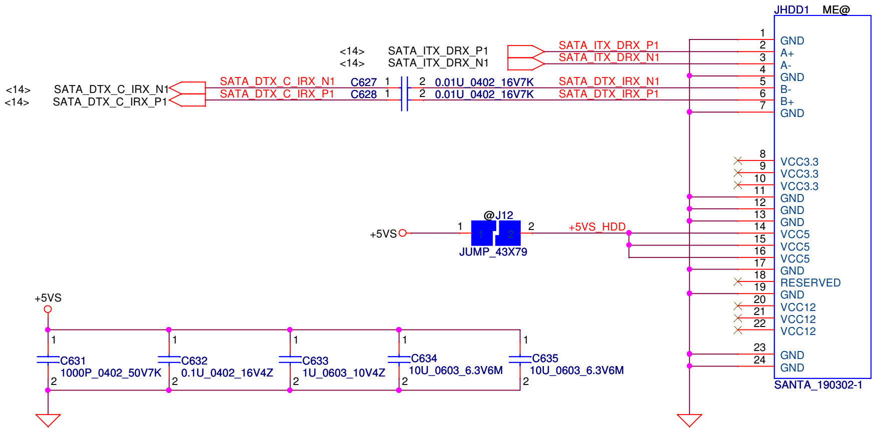
LCFC Electronics, Inc.

Title: **EMC1403/2103_Thermal sensor/FAN**

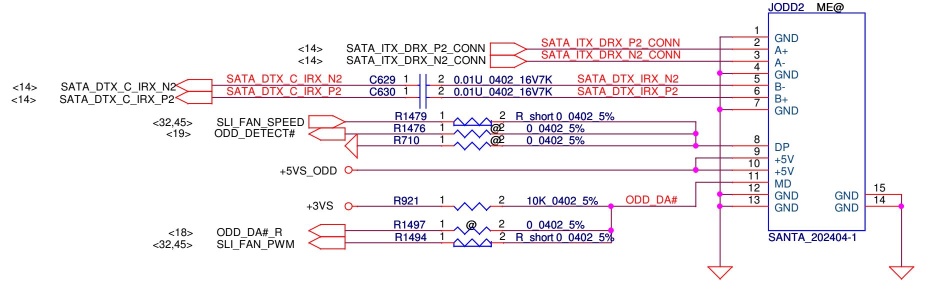
Size: Custom | Document Number: **Y490-LA8691P** | Rev: 0.2

Date: Thursday, May 10, 2012 |

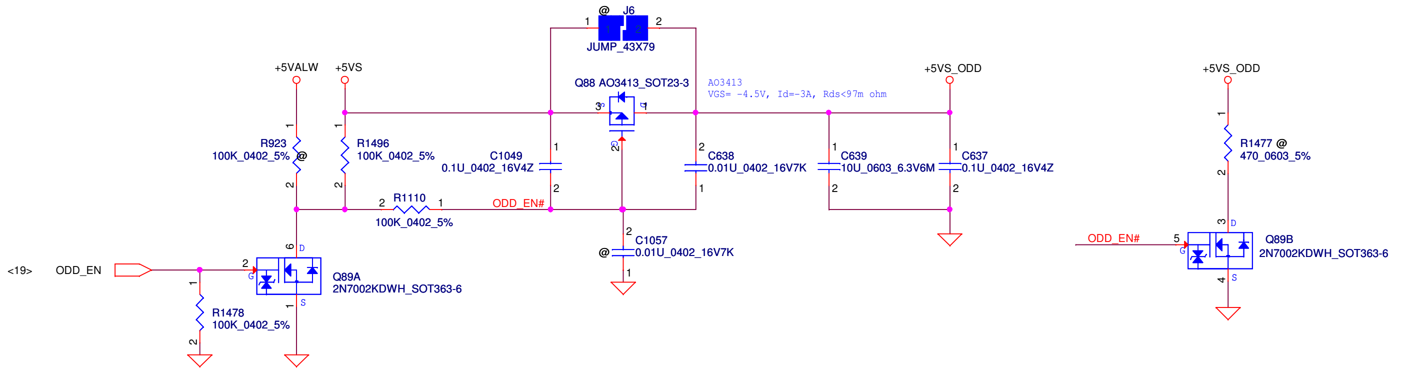
SATA HDD Conn.



SATA ODD Conn.



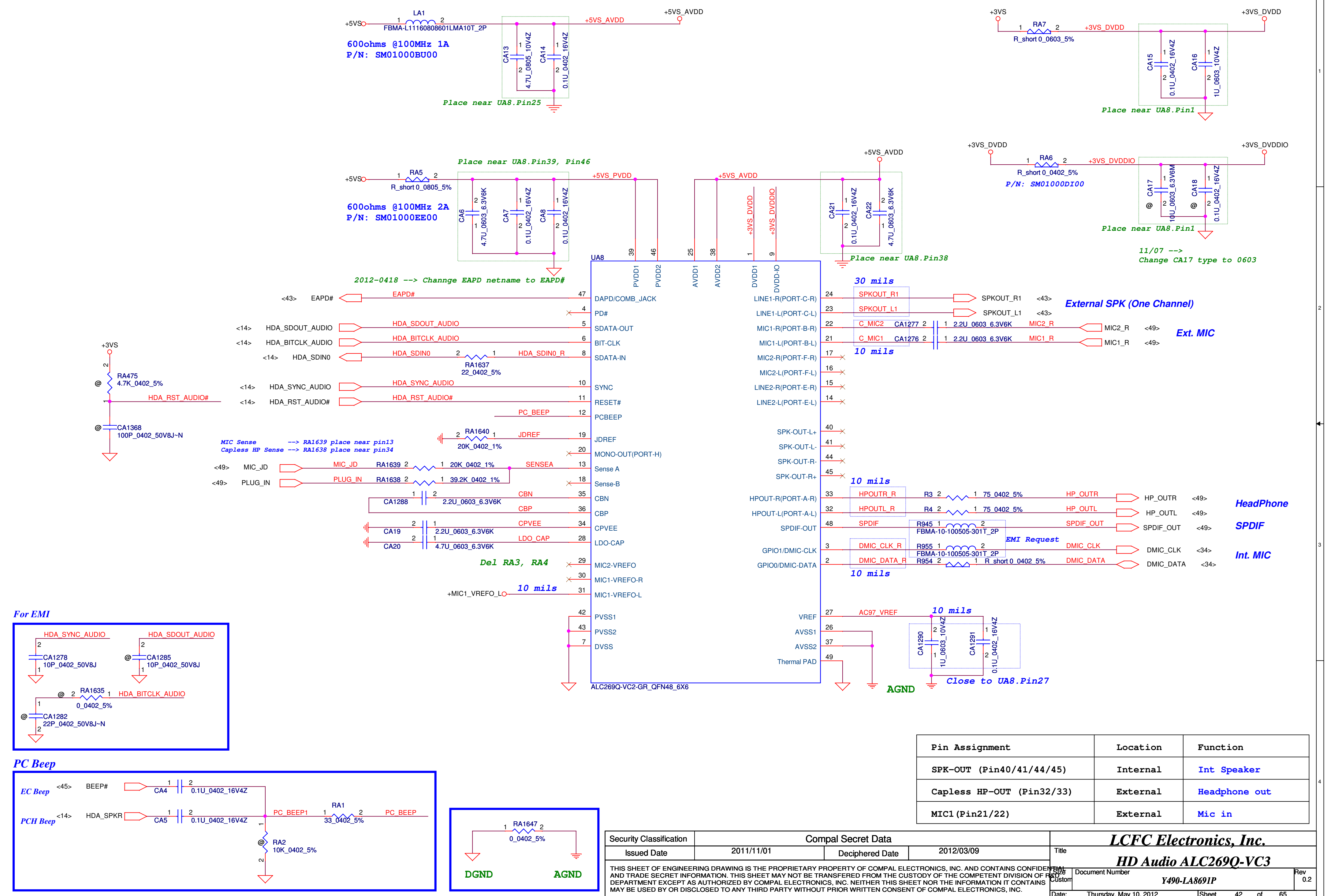
ODD Power Control



Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

LCFC Electronics, Inc.

Title	HDD/ODD Connector	
Size: Custom	Document Number: Y490-LA8691P	Rev 0.2
Date: Thursday, May 10, 2012	Sheet 41 of 65	



Pin Assignment	Location	Function
SPK-OUT (Pin40/41/44/45)	Internal	Int Speaker
Capless HP-OUT (Pin32/33)	External	Headphone out
MIC1(Pin21/22)	External	Mic in

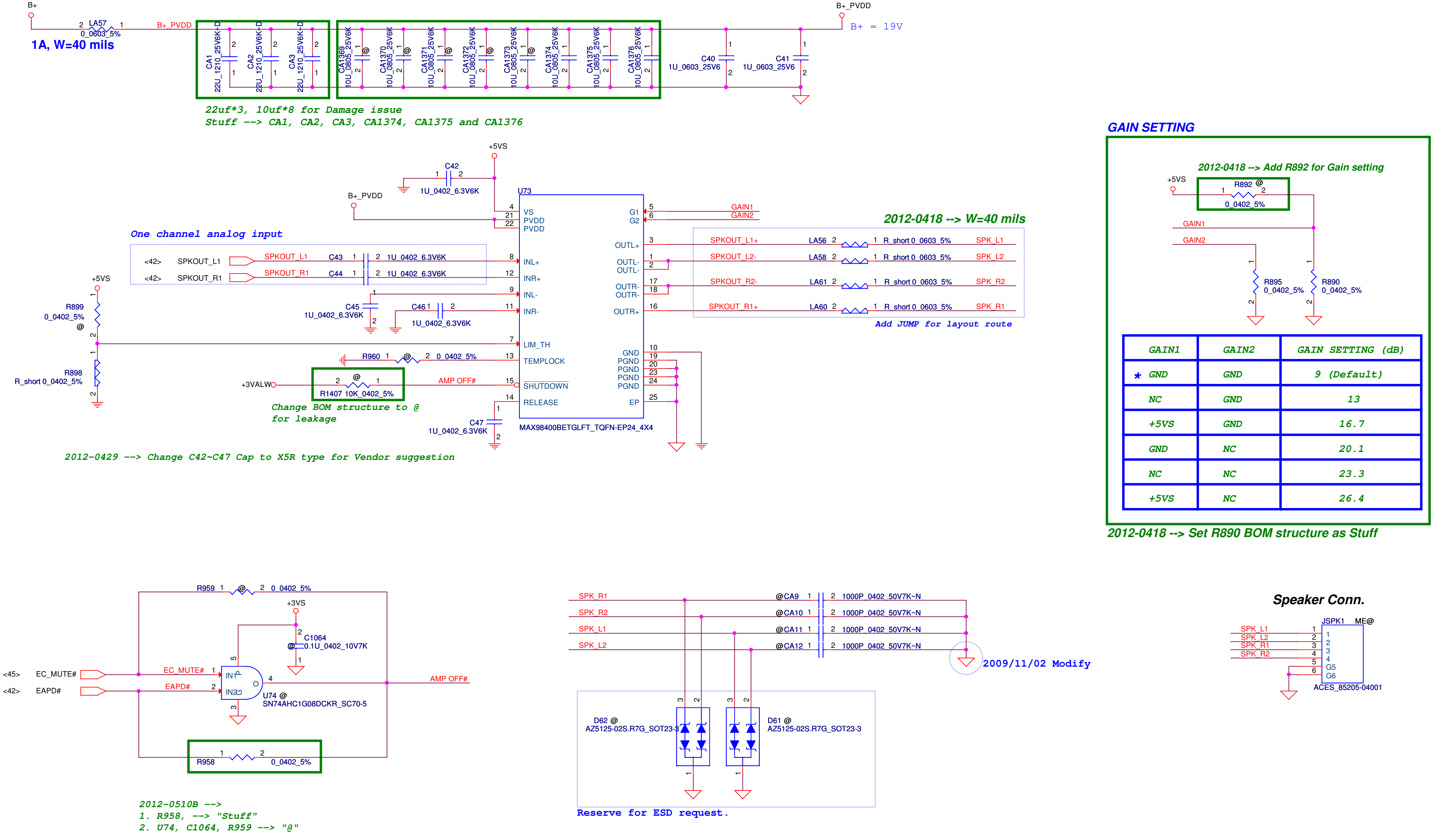
Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/03/09

LCFC Electronics, Inc.

Title: HD Audio ALC269Q-VC3

Size: Custom | Document Number: Y490-LA8691P | Rev 0.2

Date: Thursday, May 10, 2012 |



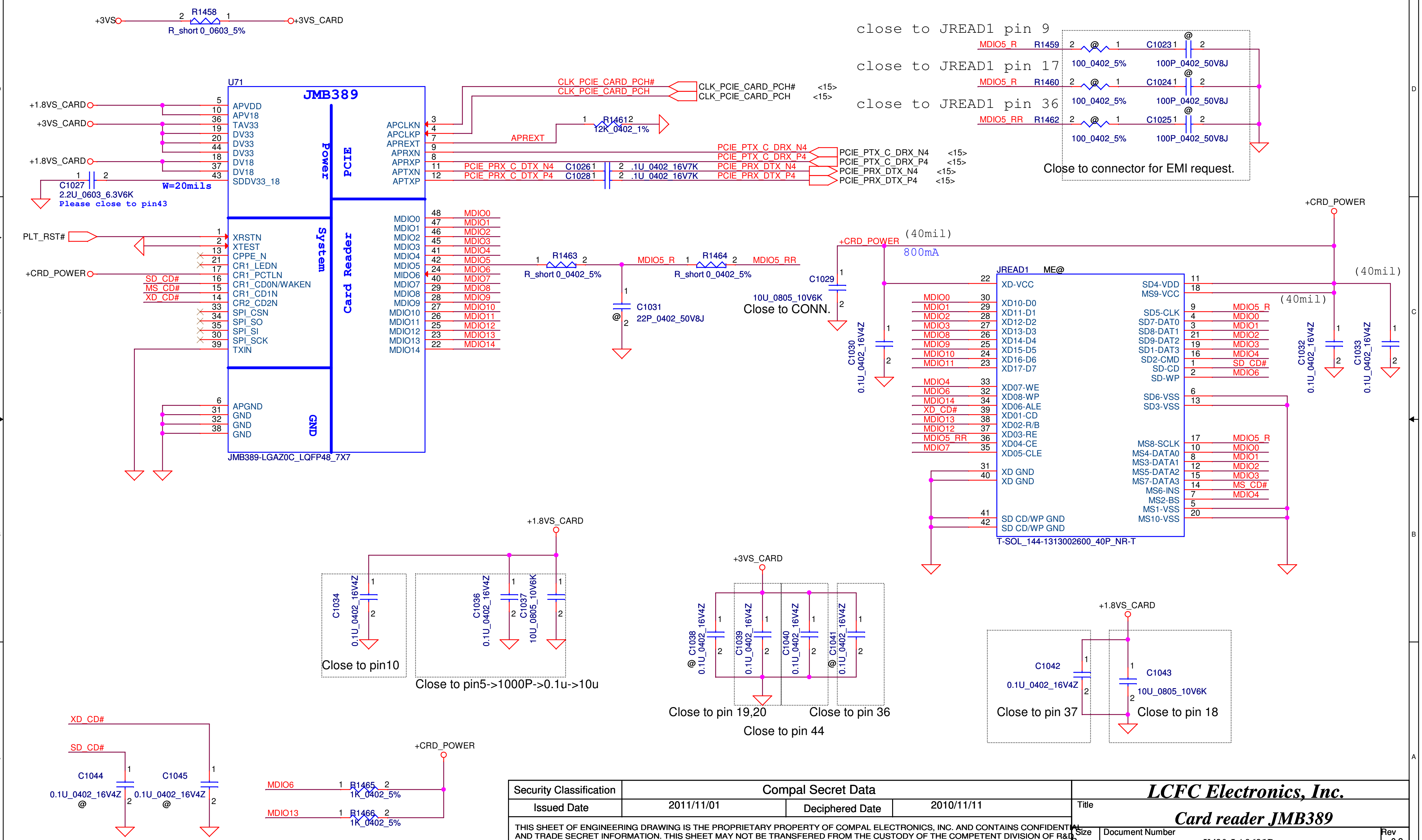
GAIN1	GAIN2	GAIN SETTING (dB)
* GND	GND	9 (Default)
NC	GND	13
+5VS	GND	16.7
GND	NC	20.1
NC	NC	23.3
+5VS	NC	26.4

Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/03/09

LCFC Electronics, Inc.

Title: **AMP-MAX98400BETG+T**

Size: Custom | Document Number: Y490-LA8691P | Rev 0.2



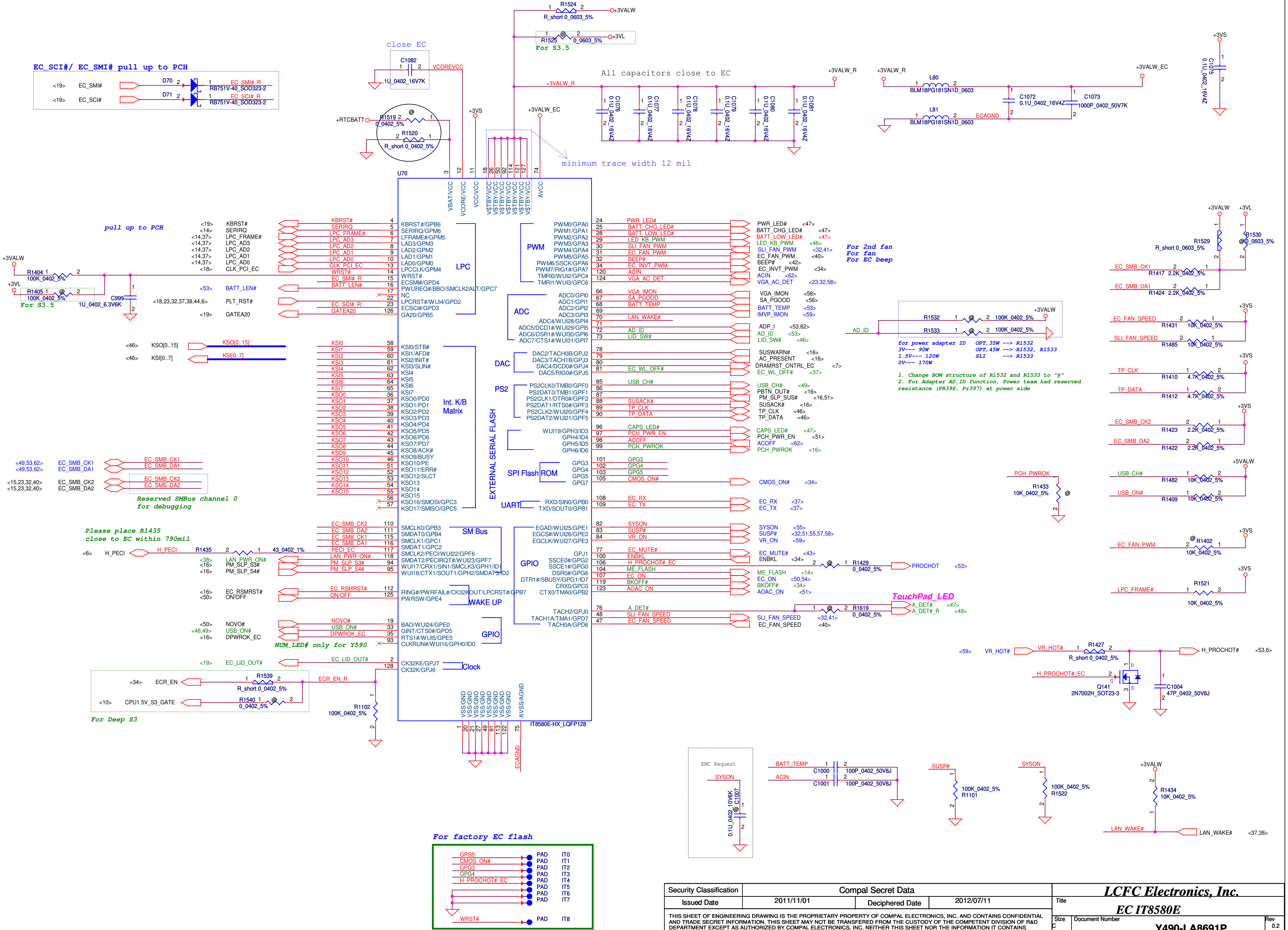
Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2010/11/11

LCFC Electronics, Inc.

Title: **Card reader JMB389**

Size	Document Number	Rev
B	Y490-LA8691P	0.2

Date: Thursday, May 10, 2012 — Sheet 44 of 65



Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/07/11

LCFC Electronics, Inc.

Title: EC IT8580E

Size: C | Document Number: Y490-LA8691P | Rev: 0.2

Date: Thursday, May 10, 2012 | Sheet 45 of 65

14" INT_KBD Conn.

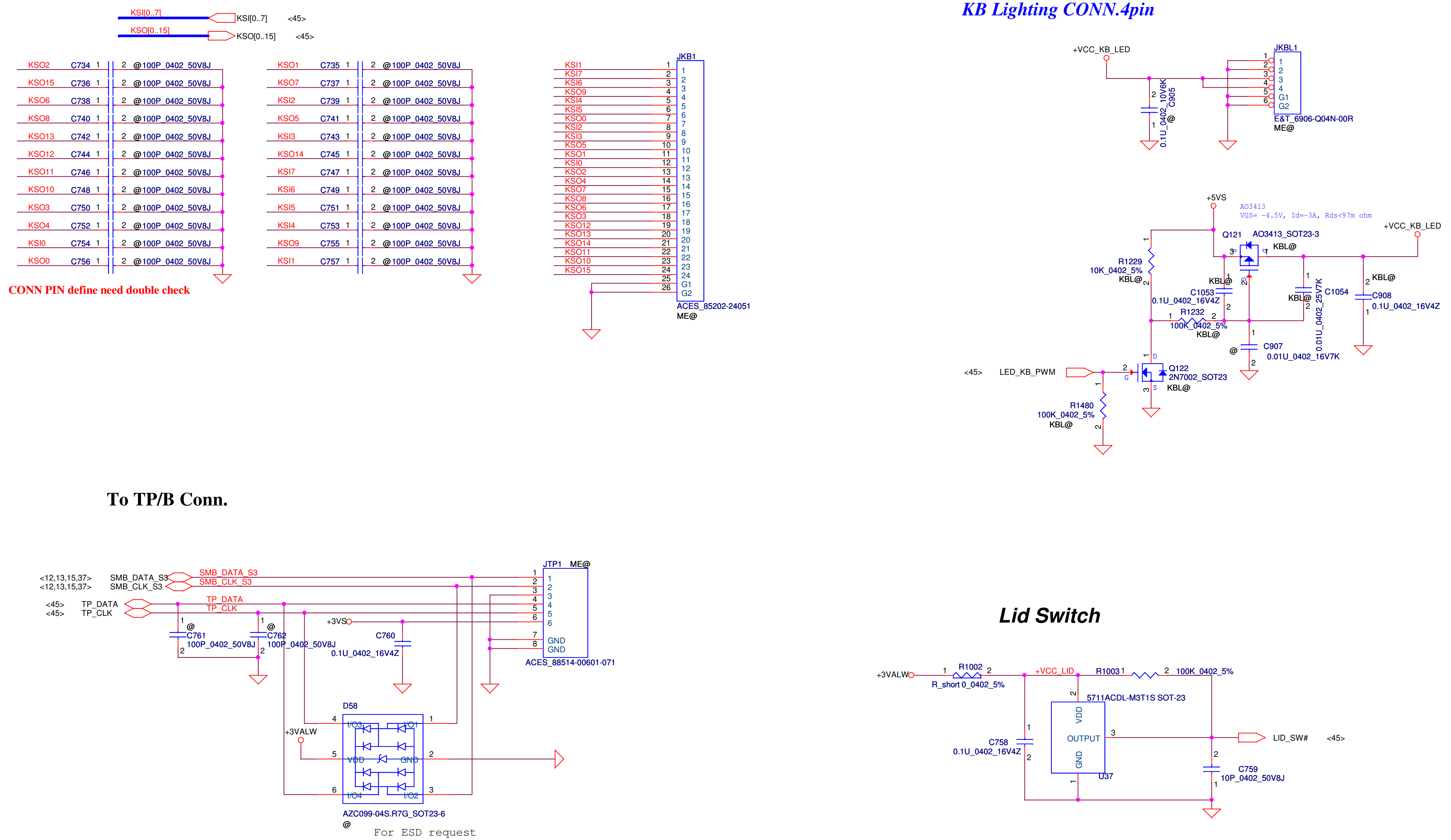
CONN PIN define need double check

KB Lighting CONN.4pin

To TP/B Conn.

For ESD request

Lid Switch



Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

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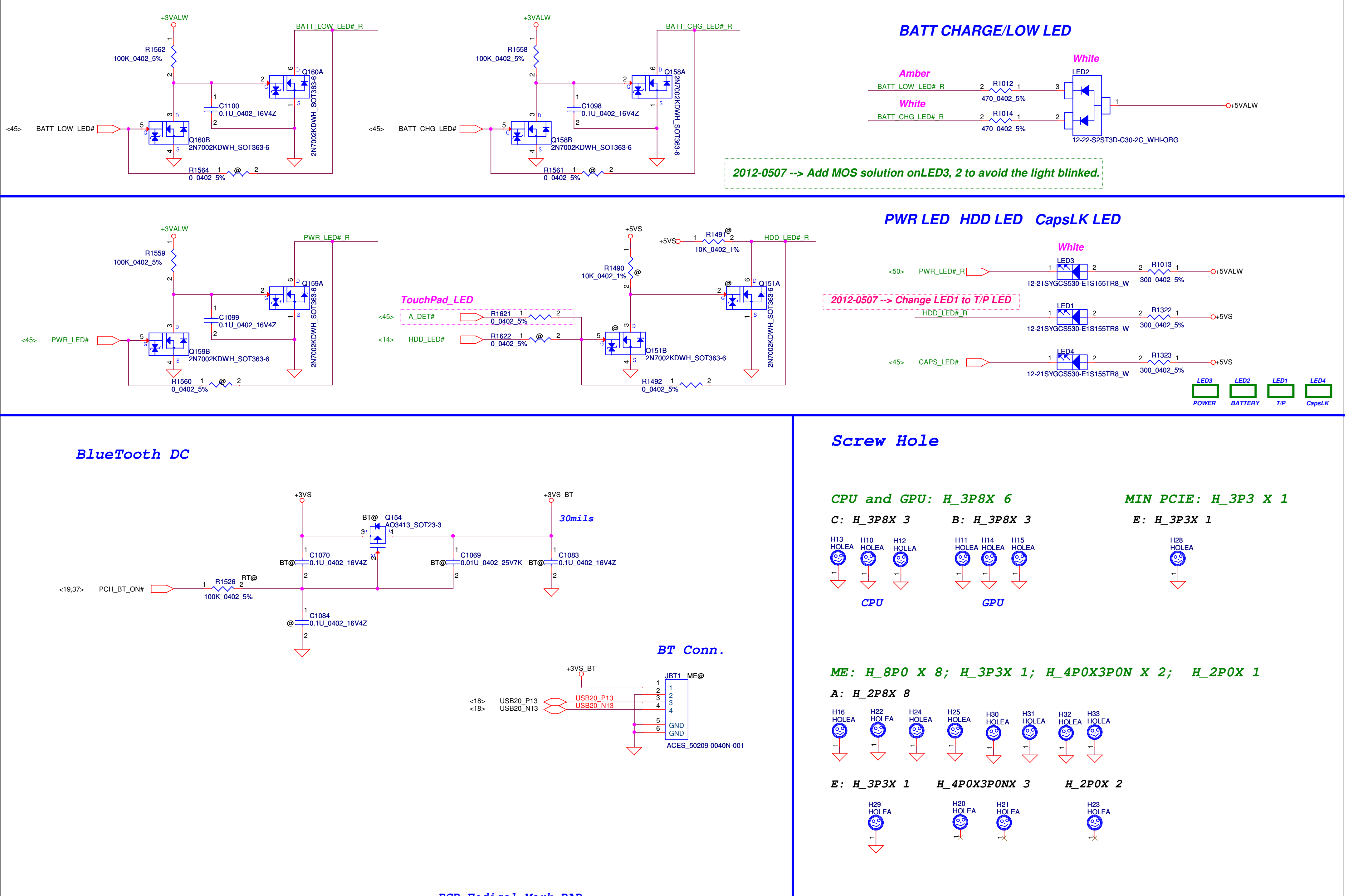
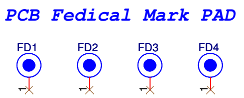
LCFC Electronics, Inc.

Title: KB /SW /LPC Debug Conn.

Size: Custom | Document Number: Y490-LA8691P | Rev: 0.2

Date: Thursday, May 10, 2012 | Sheet 46 of 65





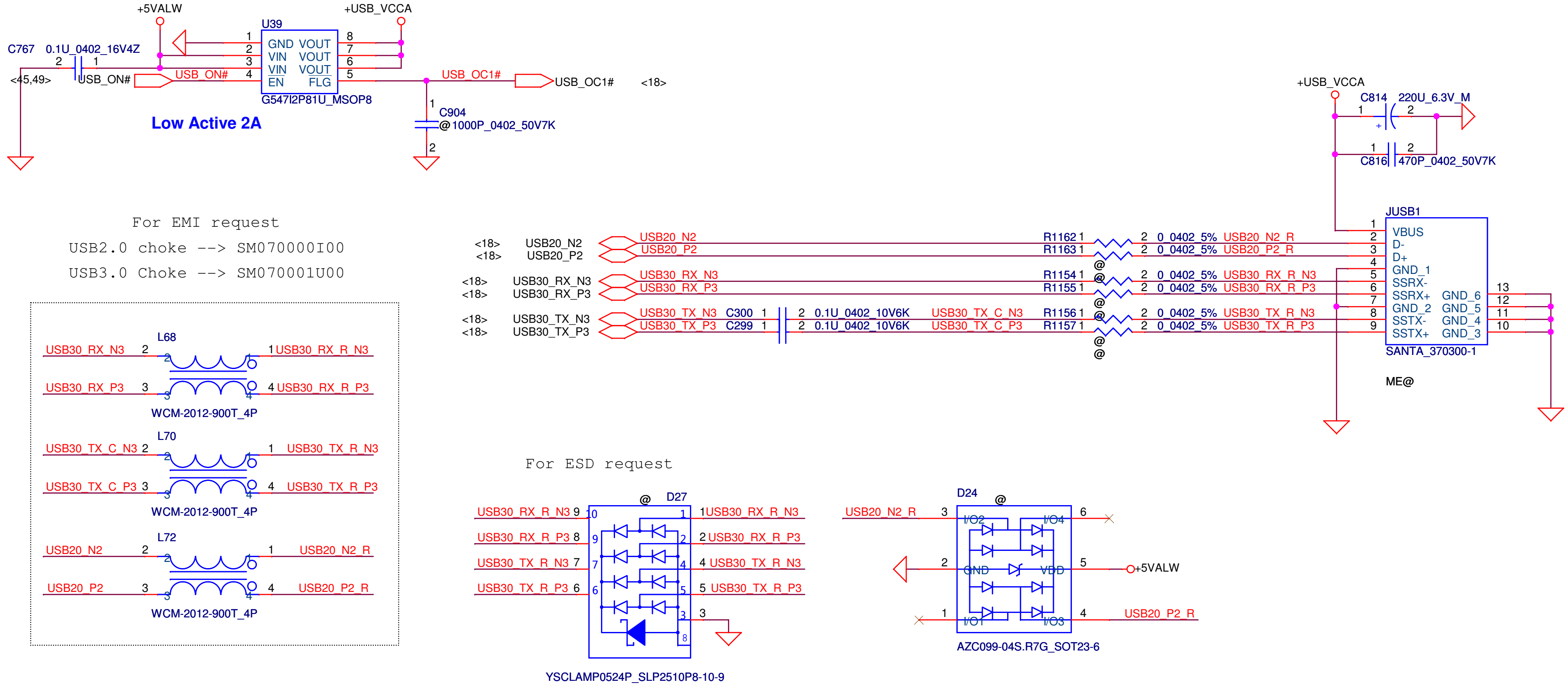
Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

LCFC Electronics, Inc.

Title: LED/EC SPI ROM/BT

Size: Custom | Document Number: Y490-LA8691P | Rev: 0.2

LEFT SIDE USB3.0 PORT X1

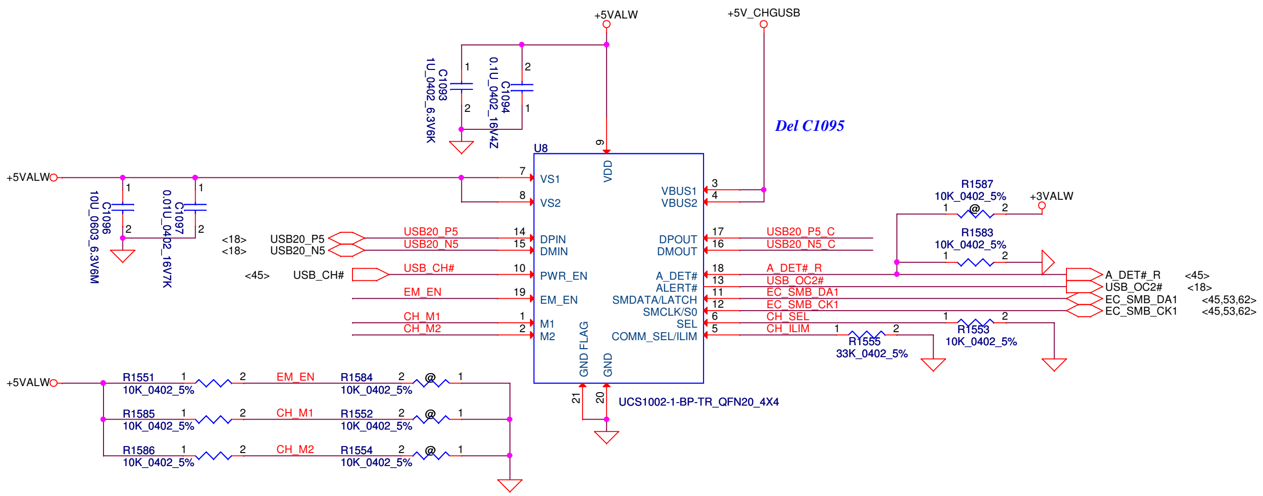


Security Classification	Compal Secret Data For EMI request	LCFC Electronics, Inc.
Issued Date 2011/11/01	Deciphered Date 2012/12/31	Title: USB3.0 ports

Size: Custom | Document Number: Y490-LA8691P | Rev 0.2

Date: Thursday, May 10, 2012 |

Sleep & Charge
Right side USB Charger Port (USB_Port5, near JMIC1)



2012-0429 --> Set default mode is "BC1.2 CDP" Mode (2.5A on S0)for USB Port5

Active Mode Selection:

M1	M2	EM_EN	ACTIVE MODE
0	0	1	Dedicated Charger Emulation Cycle
0	1	0	Date Pass-through
0	1	1	BC1.2 DCP
1	0	0	BC1.2 SDP
1	0	1	Dedicated Charger Emulation Cycle
1	1	0	Date Pass-through
* 1	1	1	BC1.2 CDP

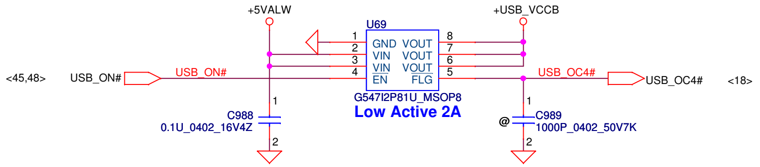
ILIM SETTING

Pull Low
0R-500mA
10K-900mA
12K-1000mA
15K-1200mA
18K-1500mA
22K-1800mA
27K-2000mA
* 33K-2500mA

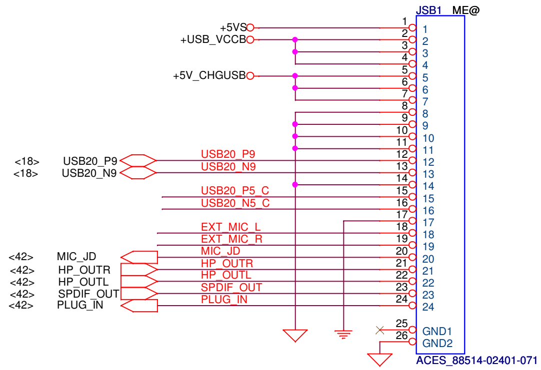
SEL Pin Decode

Pull Low
0R -1010_000
* 10K-1010_000
12K-1010_000
15K-1010_000
18K-0110_000
22K-0110_000
27K-0110_000
33K-0110_000

USB Power (USB20_P9)

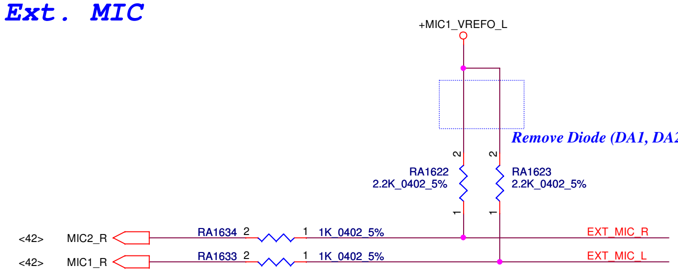


AUDIO/B Conn.



need change to 霧錫 material
COMPAL : SP010015W1J
Footprint : 88514-0240N-071

Ext. MIC



Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

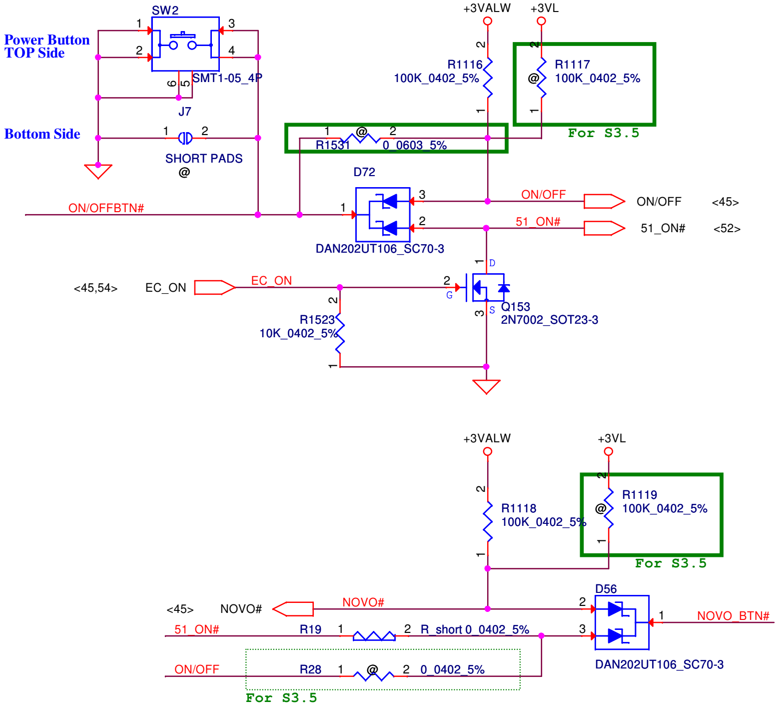
LCFC Electronics, Inc.

Title: Audio B Conn/USB charger

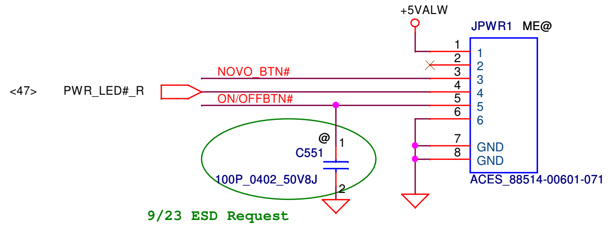
Size: Custom | Document Number: Y490-LA8691P | Rev: 0.2

Date: Thursday, May 10, 2012 |

ON/OFF switch



Power Button/B link to Function/B Conn. 10pin



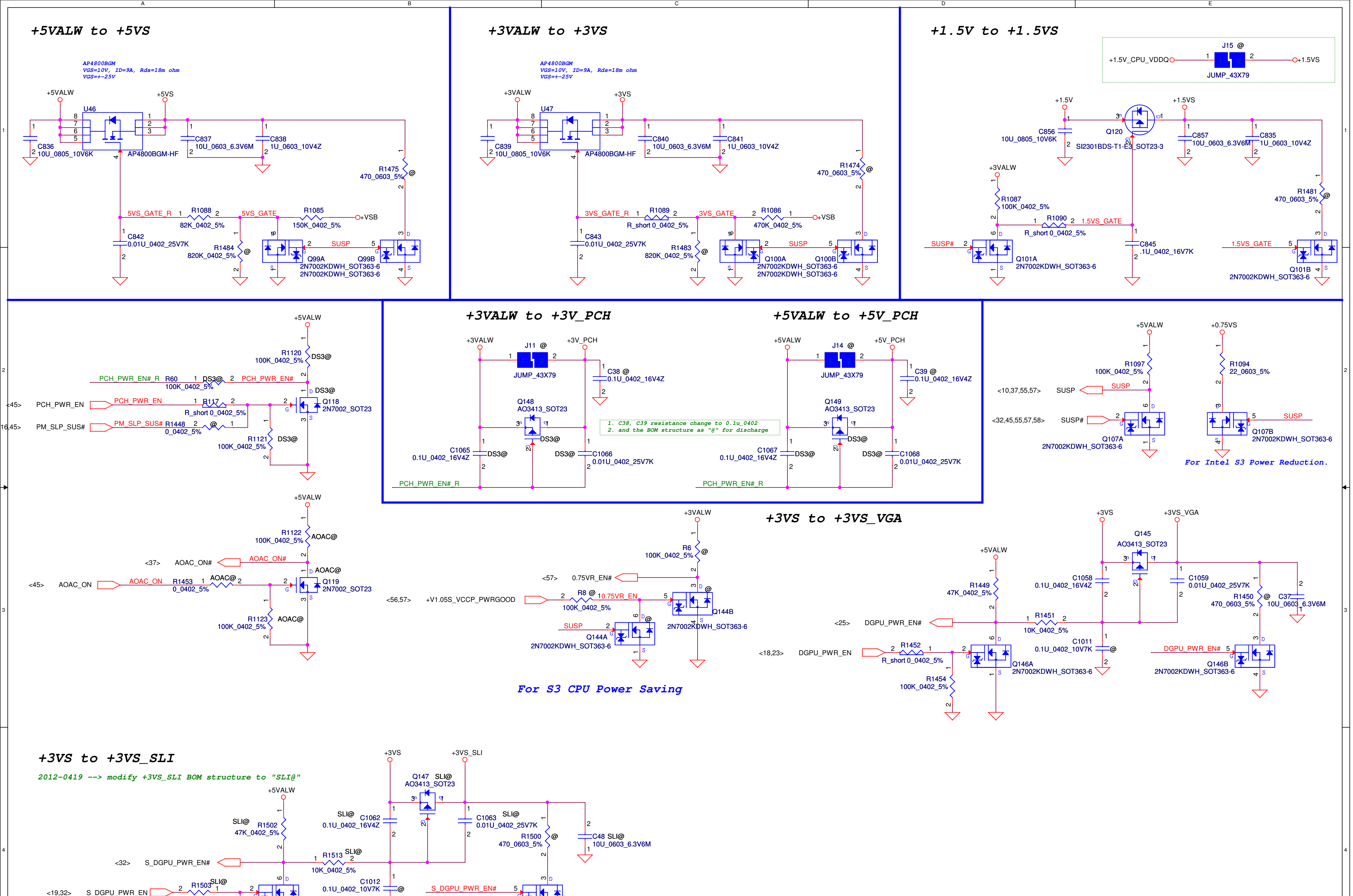
EMI REQUEST 1ST = SCA00000E00
2ST = SCA00000R00

Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

LCFC Electronics, Inc.

Title: **Other IO connector**

Size: Custom | Document Number: **Y490-LA8691P** | Rev: 0.2



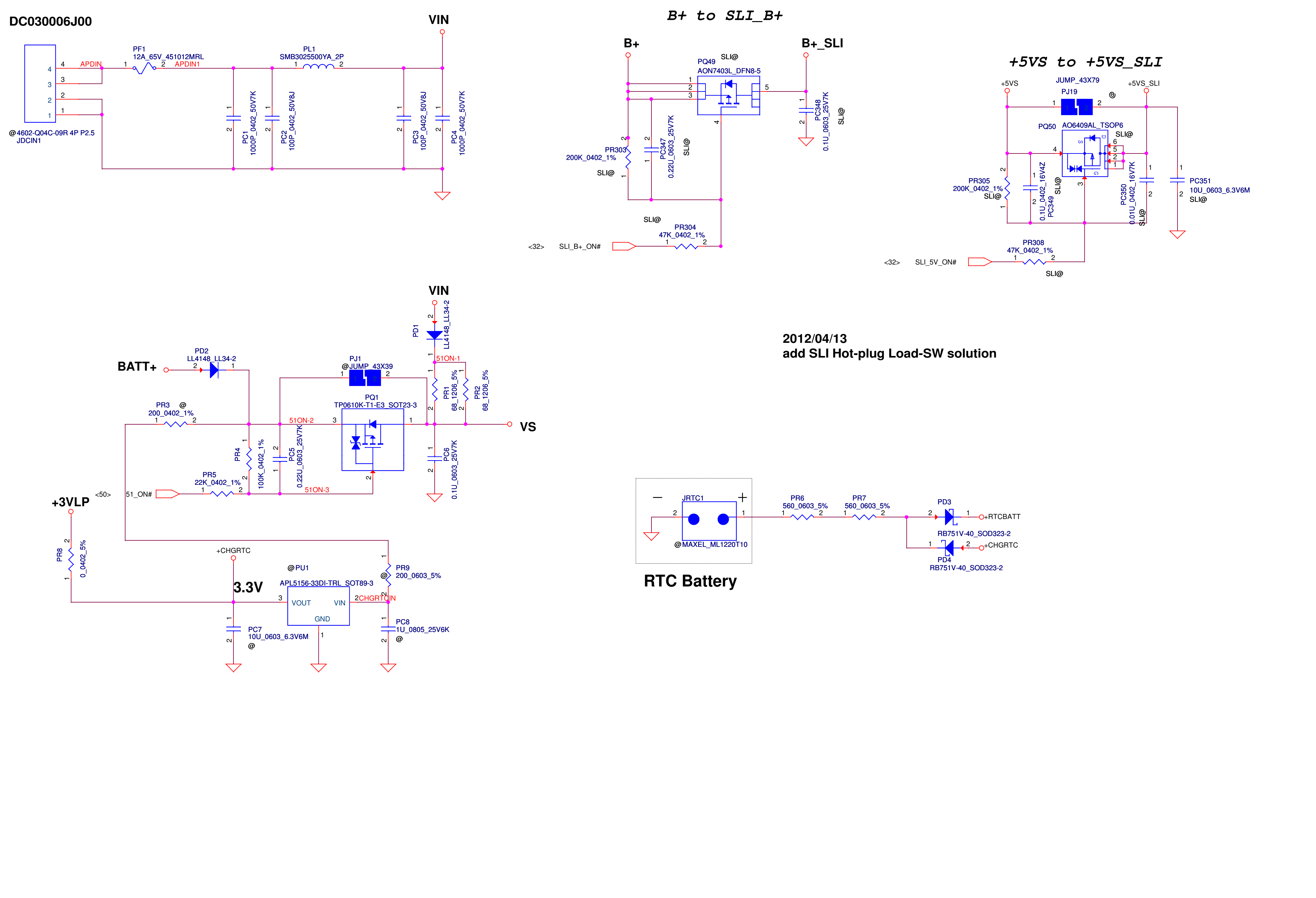
Security Classification	Compal Secret Data		
Issued Date	2011/11/01	Deciphered Date	2012/12/31

LCFC Electronics, Inc.

Title: **DC Interface**

Size	Document Number	Rev
Custom	Y490-LA8691P	0.2

Date: Thursday, May 10, 2012 Sheet 51 of 65

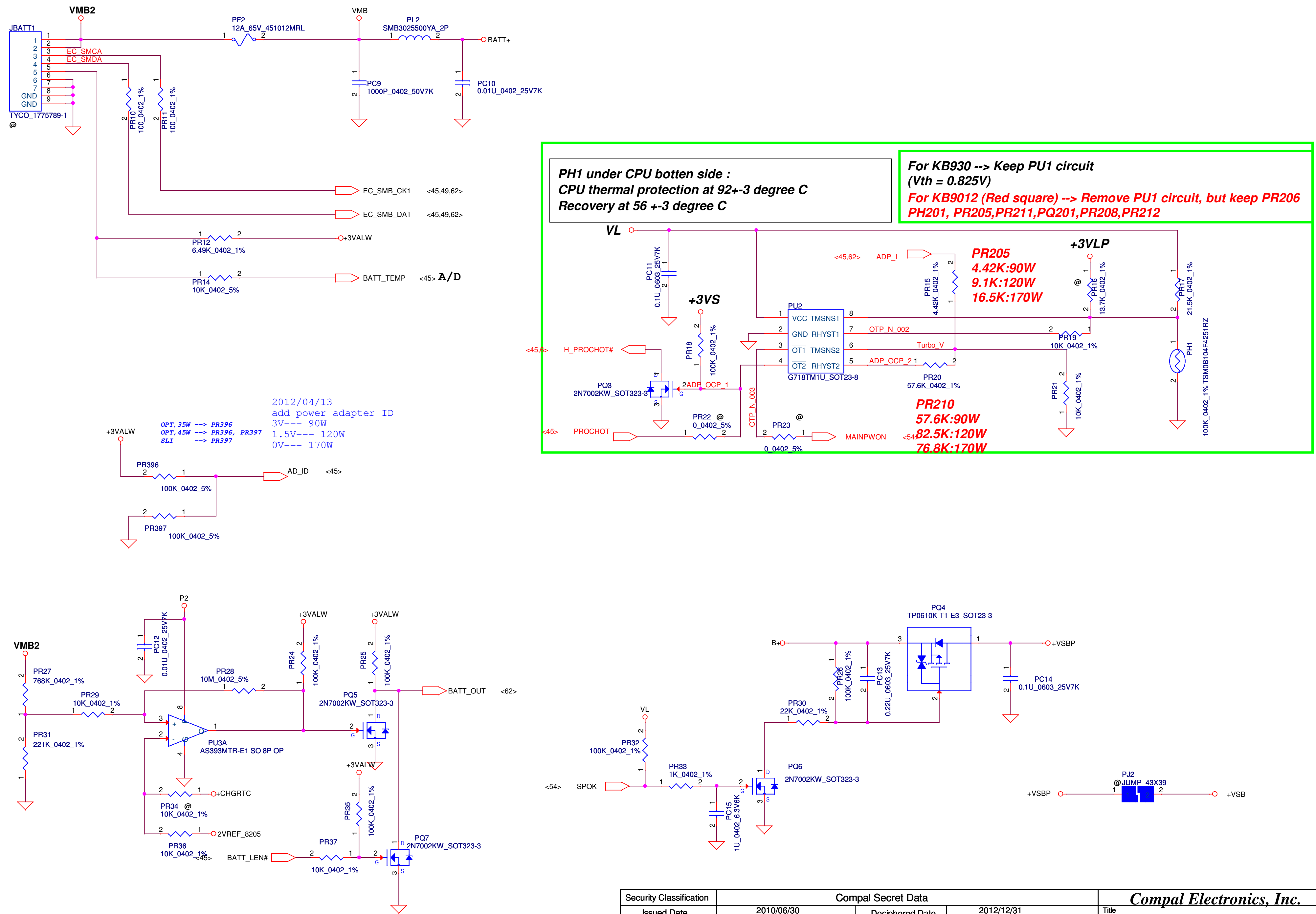


Security Classification	Compal Secret Data		
Issued Date	2011/06/30	Deciphered Date	2012/12/31

Compal Electronics, Inc.

Title: **Vin Detector**

Size	Document Number	Rev
Custom	Y490-LA8691P	0.1



Security Classification	Compal Secret Data		
Issued Date	2010/06/30	Deciphered Date	2012/12/31

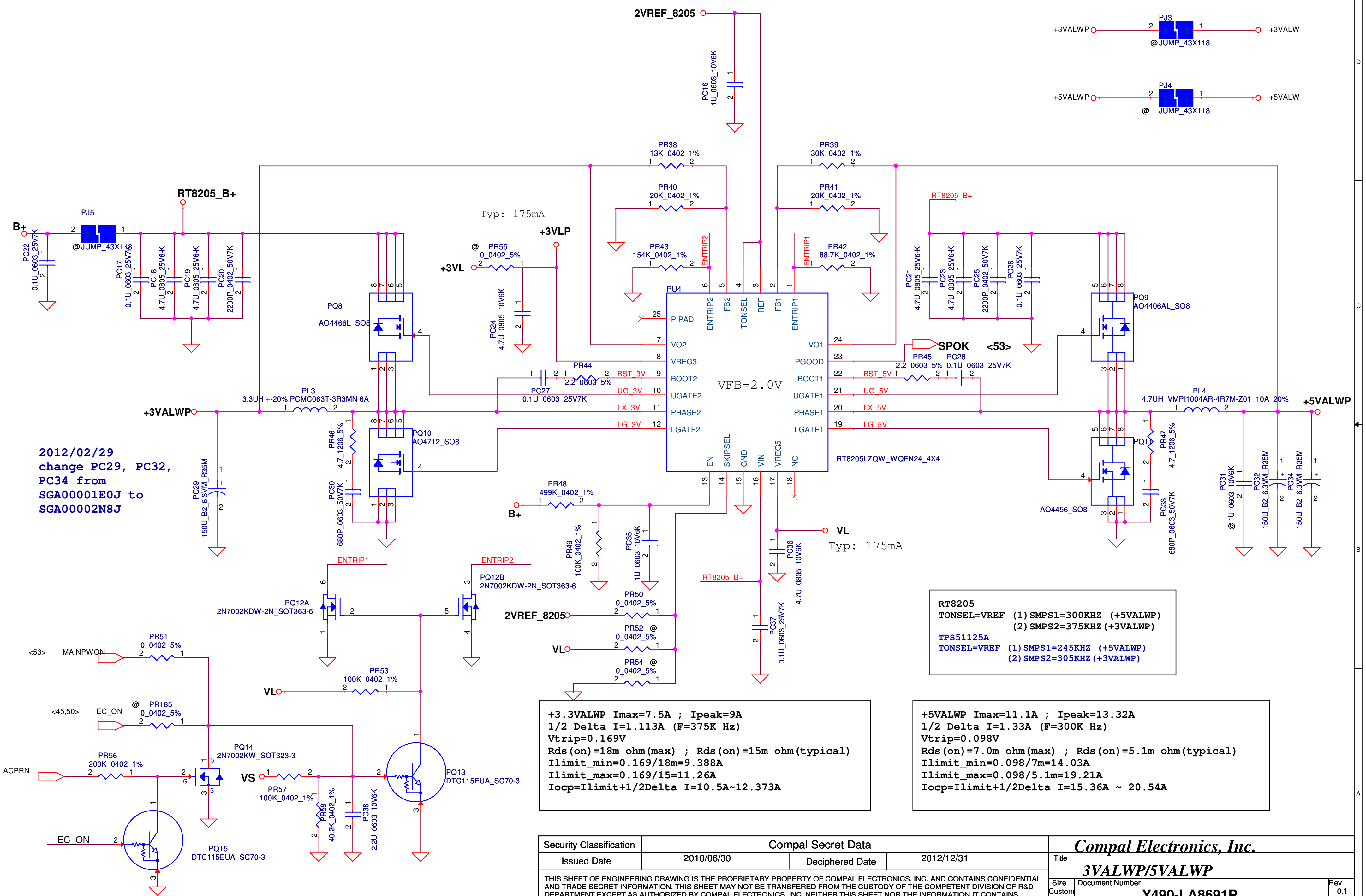
Compal Electronics, Inc.

Title: **BATTERY CONN/OTP**

Size: Custom | Document Number: Y490-LA8691P | Rev: 0.1

Date: Thursday, May 10, 2012 | Sheet 53 of 65

Note:
Use TPS51125 IC can remove RTC refernece LDO
Use TPS51427 IC must keep RTC refernece LDO



2012/02/29
change PC29, PC32, PC34 from SGA00001E0J to SGA00002N8J

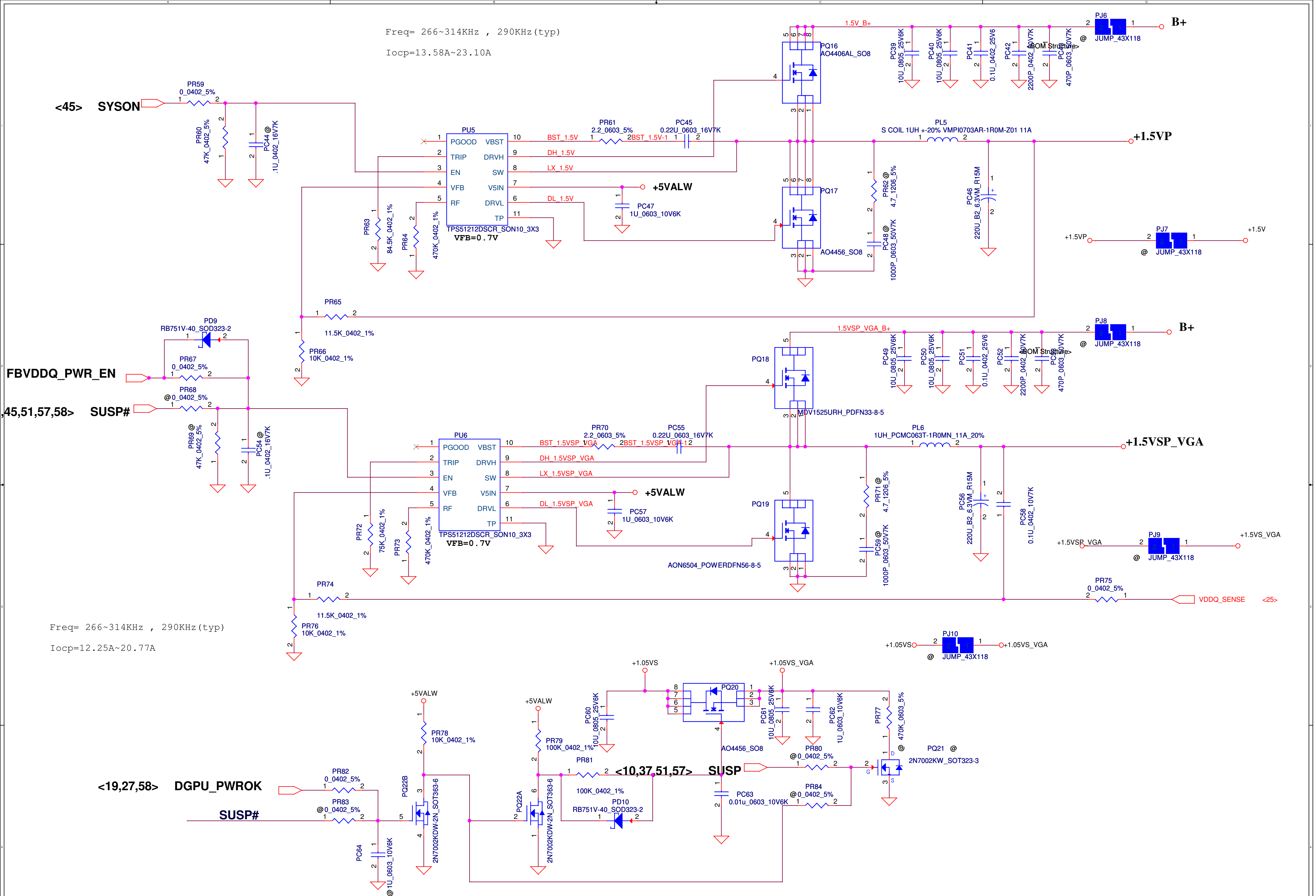
RT8205
TONSEL=VREF (1)SMPS1=300KHZ (+5VALWP)
(2)SMPS2=375KHZ (+3VALWP)
TPS51125A
TONSEL=VREF (1)SMPS1=245KHZ (+5VALWP)
(2)SMPS2=305KHZ (+3VALWP)

+3.3VALWP Imax=7.5A ; Ipeak=9A
1/2 Delta I=1.113A (F=375K Hz)
Vtrip=0.169V
Rds(on)=18m ohm(max) ; Rds(on)=15m ohm(typical)
Ilimit_min=0.169/18m=9.388A
Ilimit_max=0.169/15=11.26A
Iocp=Ilimit+1/2Delta I=10.5A~12.373A

+5VALWP Imax=11.1A ; Ipeak=13.32A
1/2 Delta I=1.33A (F=300K Hz)
Vtrip=0.098V
Rds(on)=7.0m ohm(max) ; Rds(on)=5.1m ohm(typical)
Ilimit_min=0.098/7m=14.03A
Ilimit_max=0.098/5.1m=19.21A
Iocp=Ilimit+1/2Delta I=15.36A ~ 20.54A

Security Classification	Compal Secret Data		
Issued Date	2010/06/30	Deciphered Date	2012/12/31

Compal Electronics, Inc.
Title: *3VALWP/5VALWP*
Size: Custom | Document Number: Y490-LA8691P | Rev: 0.1
Date: Thursday, May 10, 2012 |



Security Classification	Compal Secret Data		
Issued Date	2010/06/30	Deciphered Date	2012/12/31

Compal Electronics, Inc.

Title: **1.5VP/1.5VSP_VGA/1.05VSP_VGA**

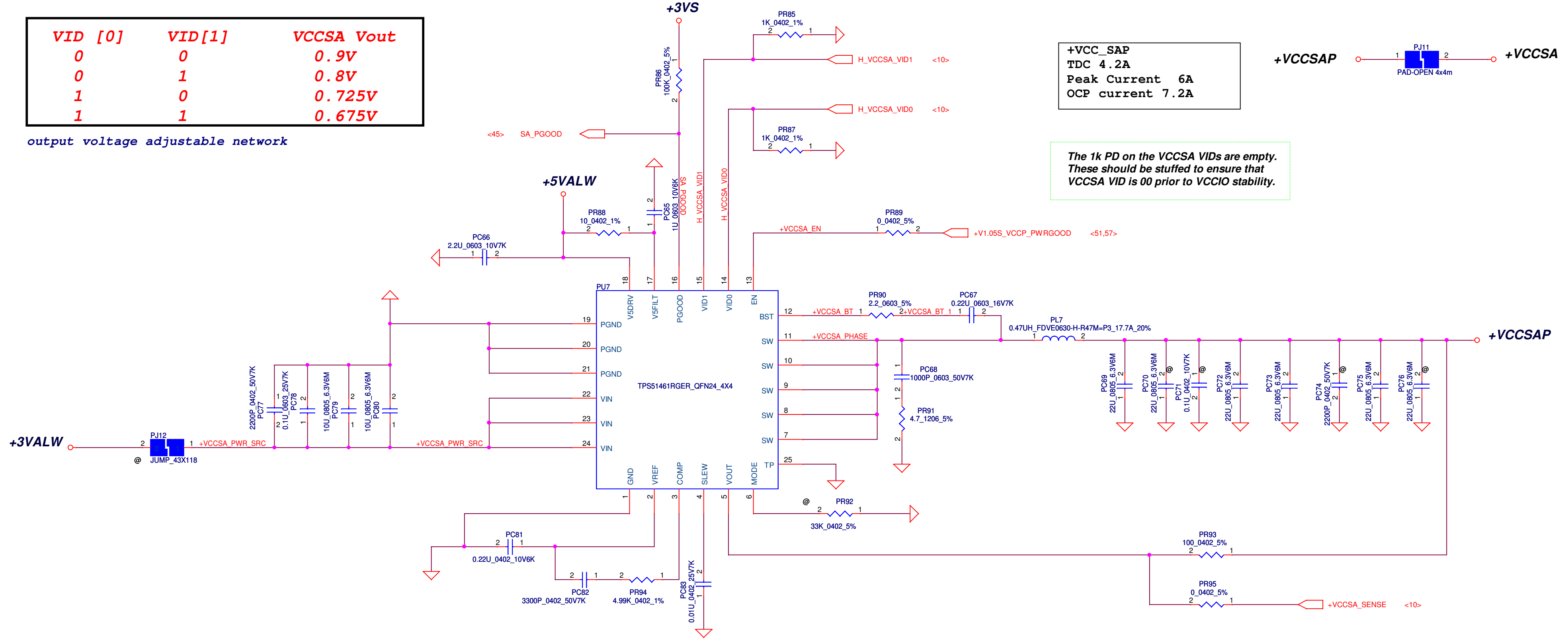
Size	Document Number	Rev
Custom	Y490-LA8691P	0.2

VID [0]	VID[1]	VCCSA Vout
0	0	0.9V
0	1	0.8V
1	0	0.725V
1	1	0.675V

output voltage adjustable network

+VCC_SAP
TDC 4.2A
Peak Current 6A
OCP current 7.2A

The 1k PD on the VCCSA VIDs are empty. These should be stuffed to ensure that VCCSA VID is 00 prior to VCCIO stability.



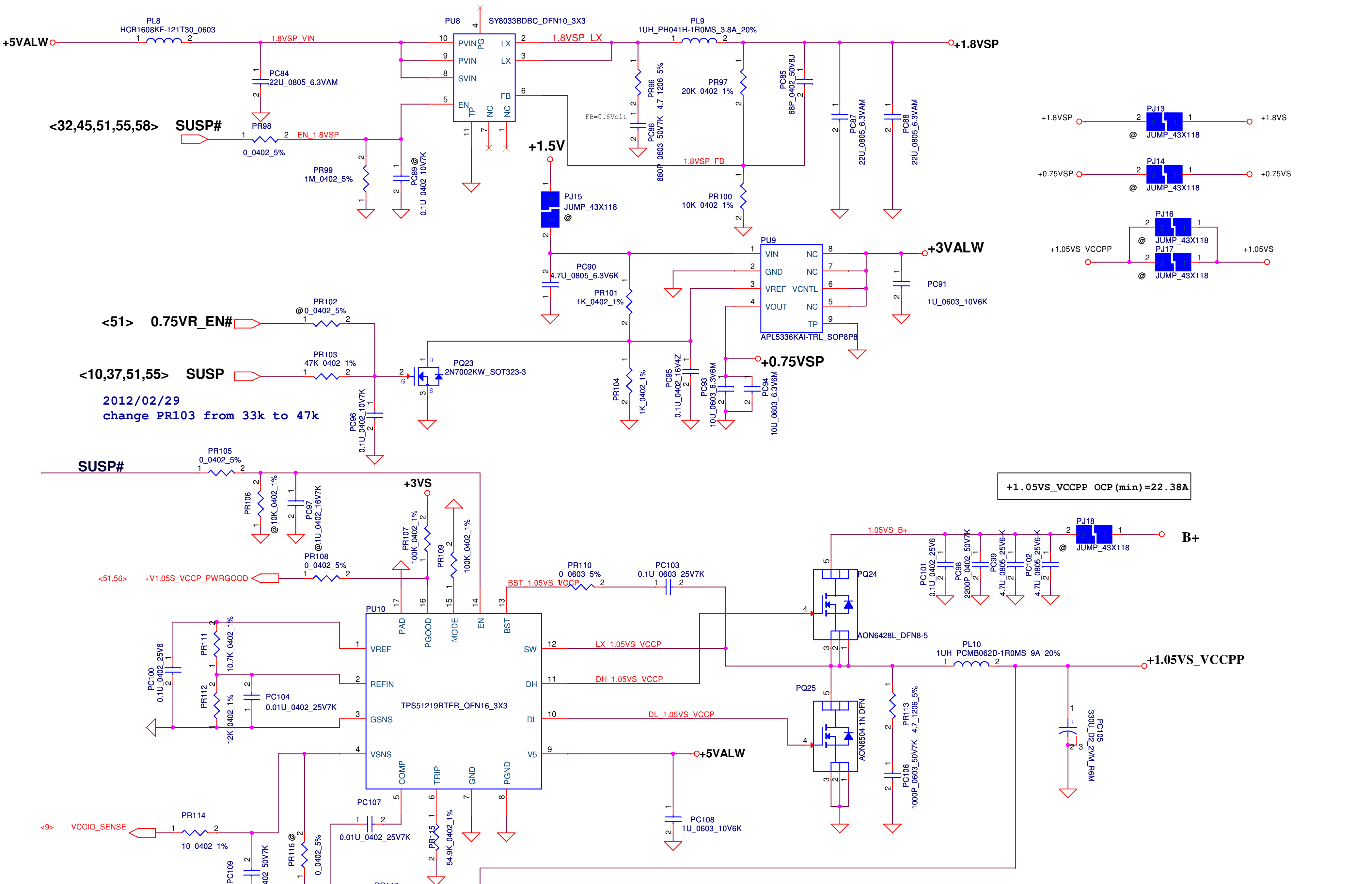
Security Classification	Compal Secret Data		
Issued Date	2011/06/30	Deciphered Date	2012/12/31

Compal Electronics, Inc.

Title: VCCSAP/1.05S_VCCPP

Size: C Document Number: Y490-LA8691P Rev: 0.1

Date: Thursday, May 10, 2012 Sheet 56 of 65



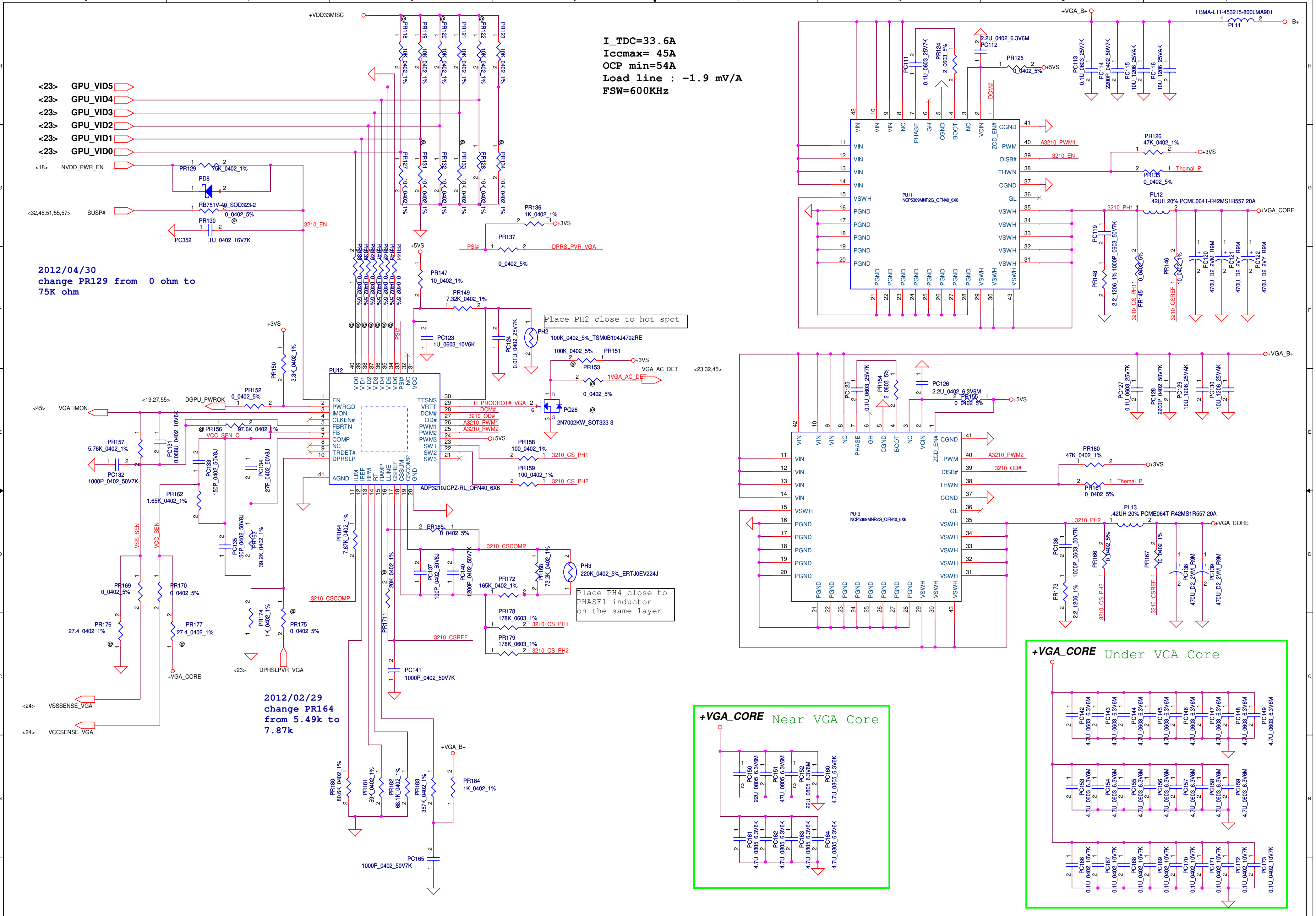
Security Classification	Compal Secret Data		
Issued Date	2010/06/30	Deciphered Date	2012/12/31

Compal Electronics, Inc.

Title: *1.8VSP/0.75VSP/1.05VS_VCCPP*

Size: Custom | Document Number: Y490-LA8691P | Rev: 0.2

Date: Thursday, May 10, 2012 |



Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2011/06/30	Deciphered Date	2012/12/31	Title: VGA_COREP
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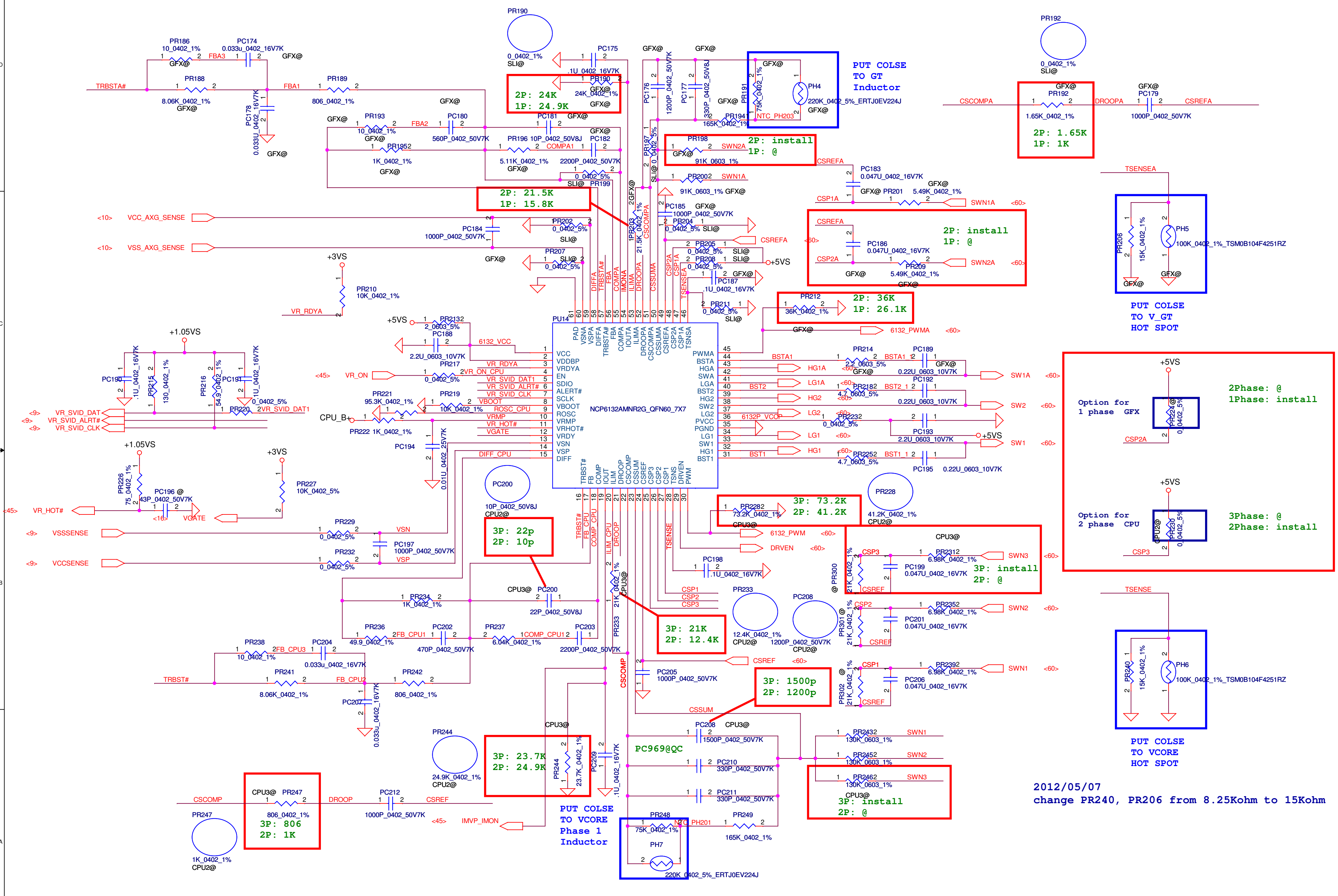
				Date: Thursday, May 10, 2012 / Sheet 58 of 65



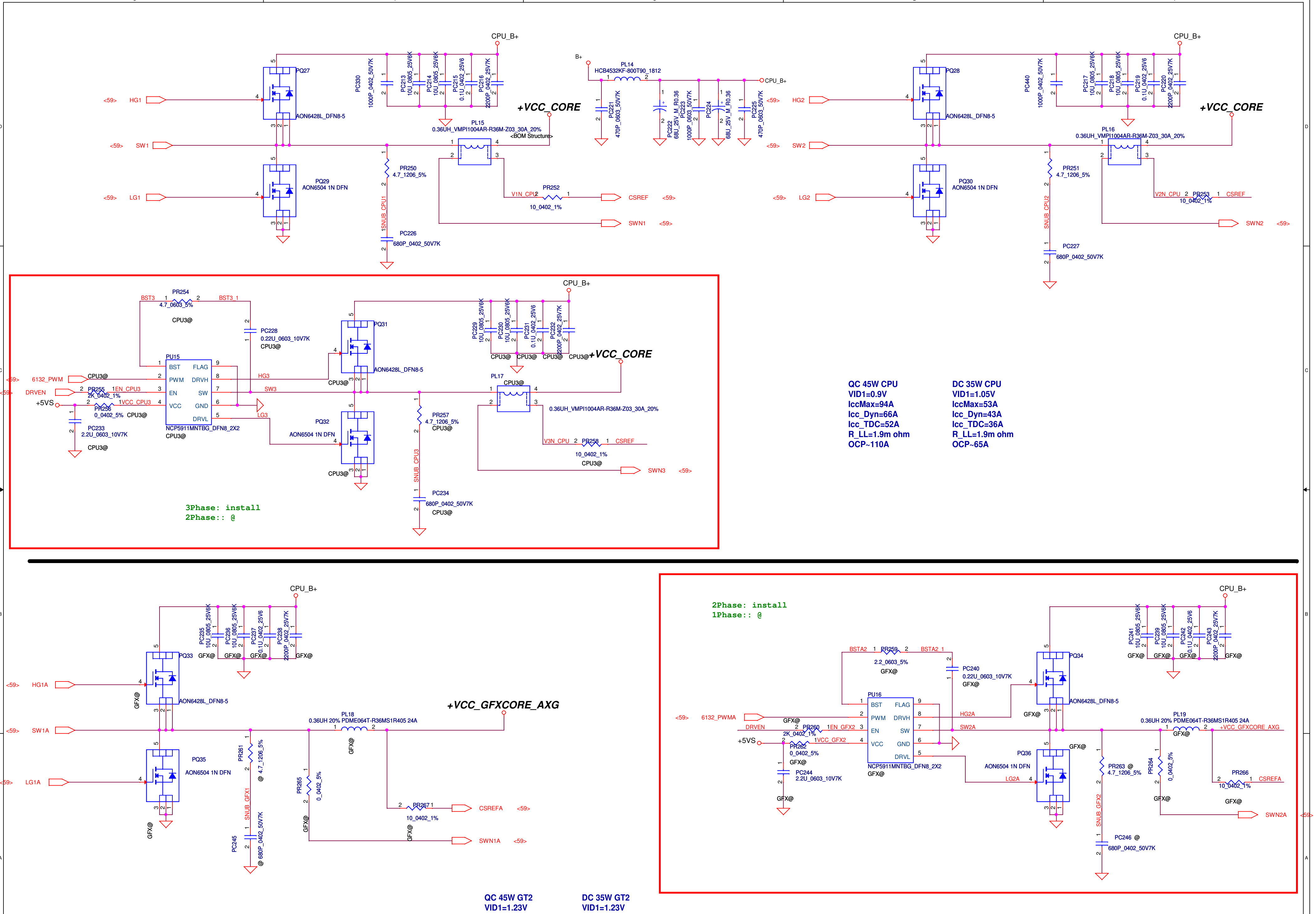
Security Classification	Compal Secret Data		
Issued Date	2011/06/30	Deciphered Date	2012/12/31

Compal Electronics, Inc.

Title: **CPU_CORE**

Size: Custom | Document Number: Y490-LA8691P | Rev: 0.3

Date: Thursday, May 10, 2012 | Sheet 59 of 65



QC 45W CPU
VID1=0.9V
IccMax=94A
Icc_Dyn=66A
Icc_TDC=52A
R_LL=1.9m ohm
OCP~110A

DC 35W CPU
VID1=1.05V
IccMax=53A
Icc_Dyn=43A
Icc_TDC=36A
R_LL=1.9m ohm
OCP~65A

QC 45W GT2
VID1=1.23V
IccMax=46A
Icc_Dyn=37A
Icc_TDC=38A
R_LL=3.9m ohm
OCP~55A

DC 35W GT2
VID1=1.23V
IccMax=33A
Icc_Dyn=20.2A
Icc_TDC=21.5A
R_LL=3.9m ohm
OCP~40A

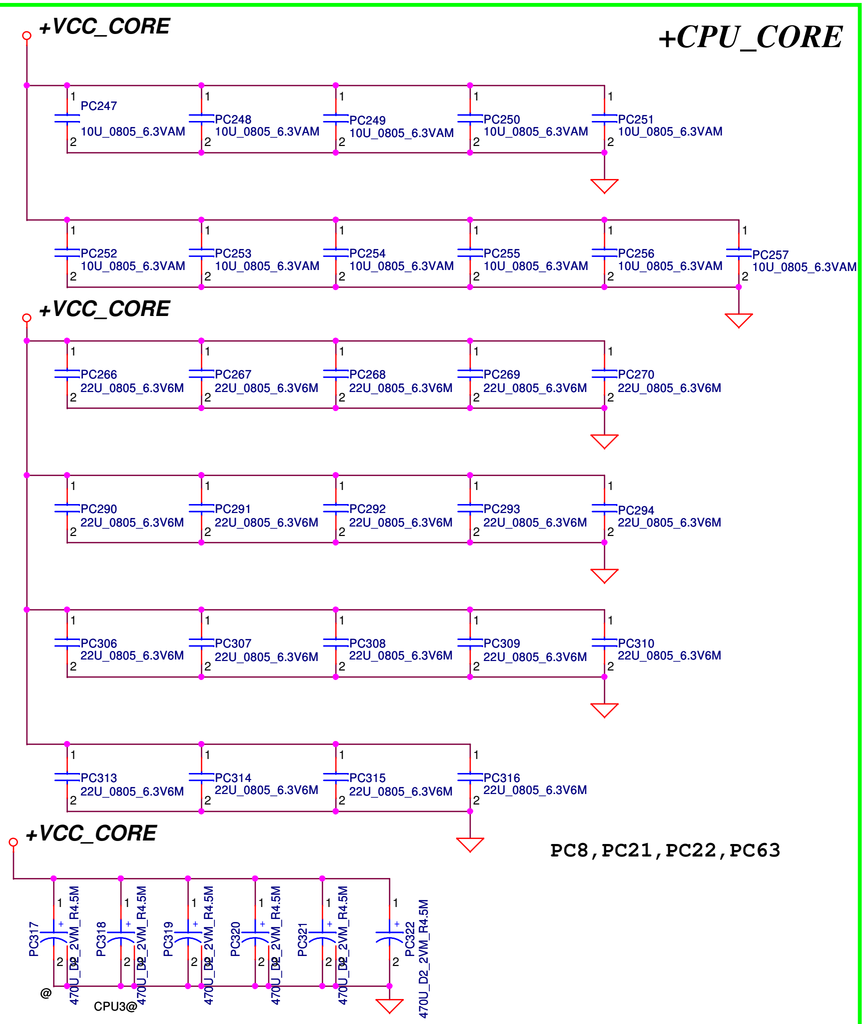
Security Classification	Compal Secret Data		
Issued Date	2011/06/30	Deciphered Date	2012/12/31

Compal Electronics, Inc.

Title: CPU_CORE

Size: C | Document Number: Y490-LA8691P | Rev: 0.3

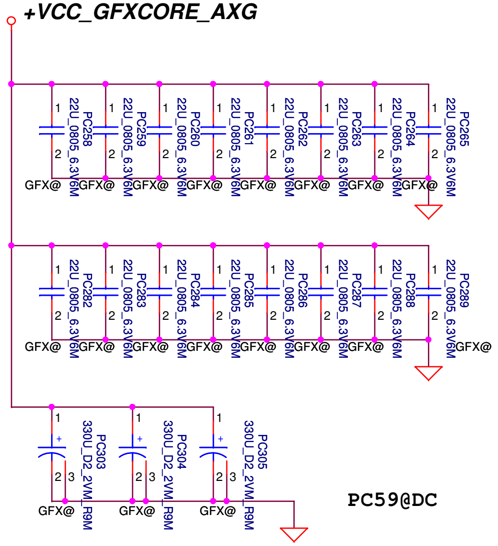
+CPU_CORE



PC8,PC21,PC22,PC63

DC:PC73,PC74,PC75,PC76,PC77,PC78(330uF/9m)
QC:PC76,PC78(470uF/4.5m),PC73,PC74,PC75(330uF/9m)

+VCC_GFXCORE_AXG

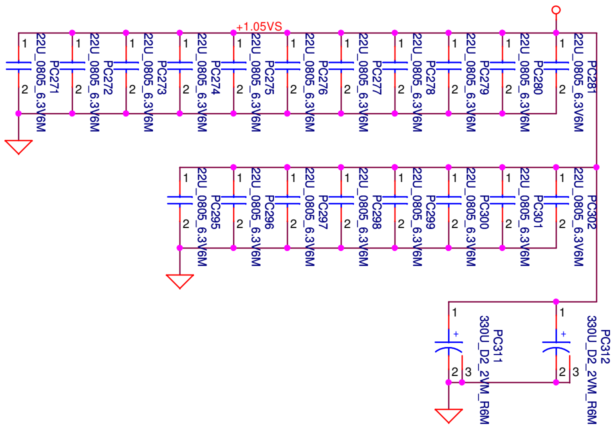


PC59@DC

PC38,PC39,PC40,PC41

Below is 458544_CRV_PDDG_0.5 Table 5-8.

Socket Bottom	5 x 22 µF (0805) 5 x (0805) no-stuff sites
Socket Top	7 x 22 µF (0805) 2 x (0805) no-stuff sites



PC32,PC49,PC54,PC55,PC56

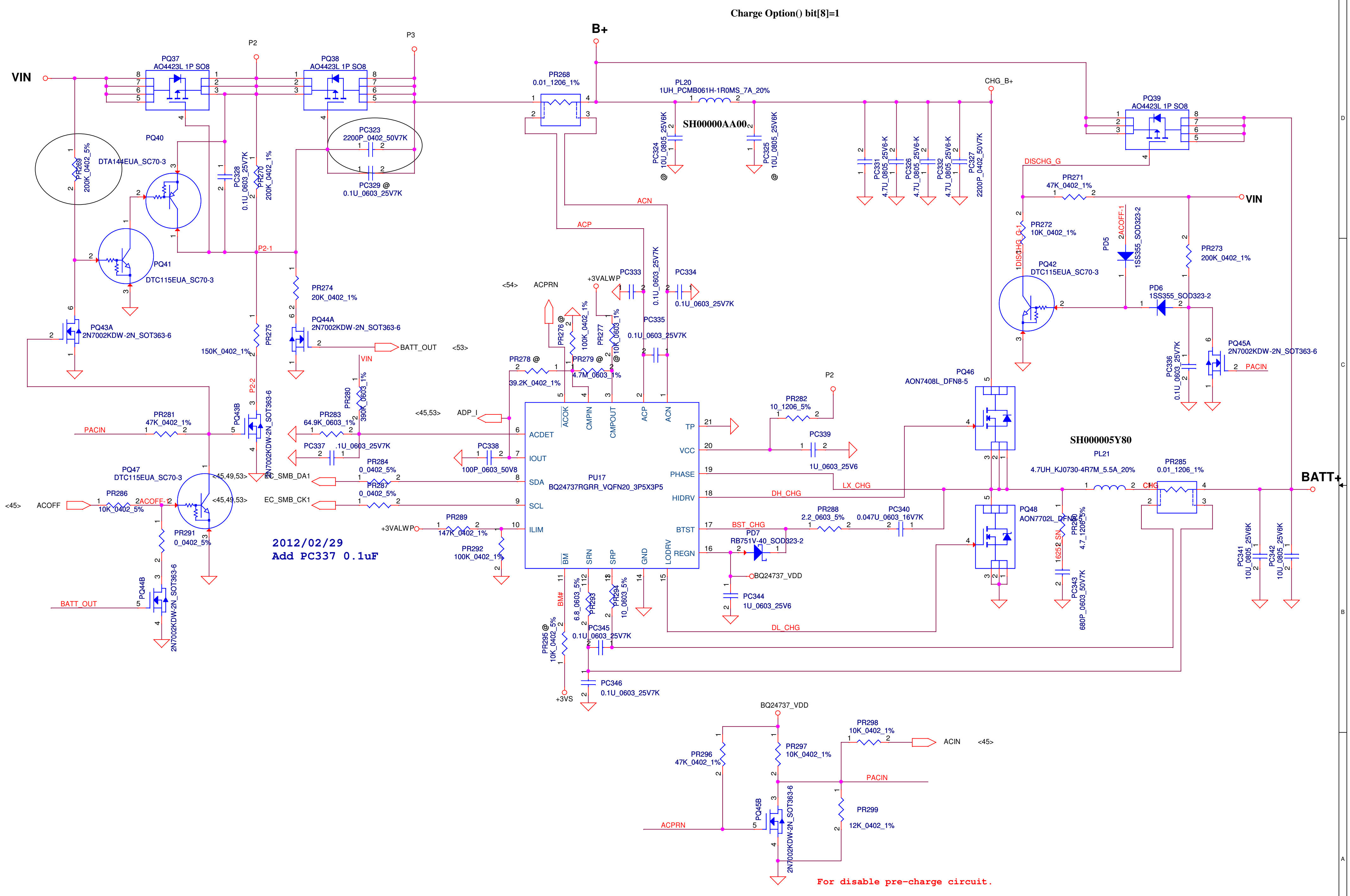
PC38,PC39,PC40,PC41

Security Classification	Compal Secret Data		
Issued Date	2011/06/30	Deciphered Date	2012/12/31

Compal Electronics, Inc.

Title: CPU_CORE1

Size | Document Number: Y490-LA8691P | Rev 0.1



Security Classification	Compal Secret Data		Compal Electronics, Inc.
Issued Date	2010/06/30	Deciphered Date	2012/12/31

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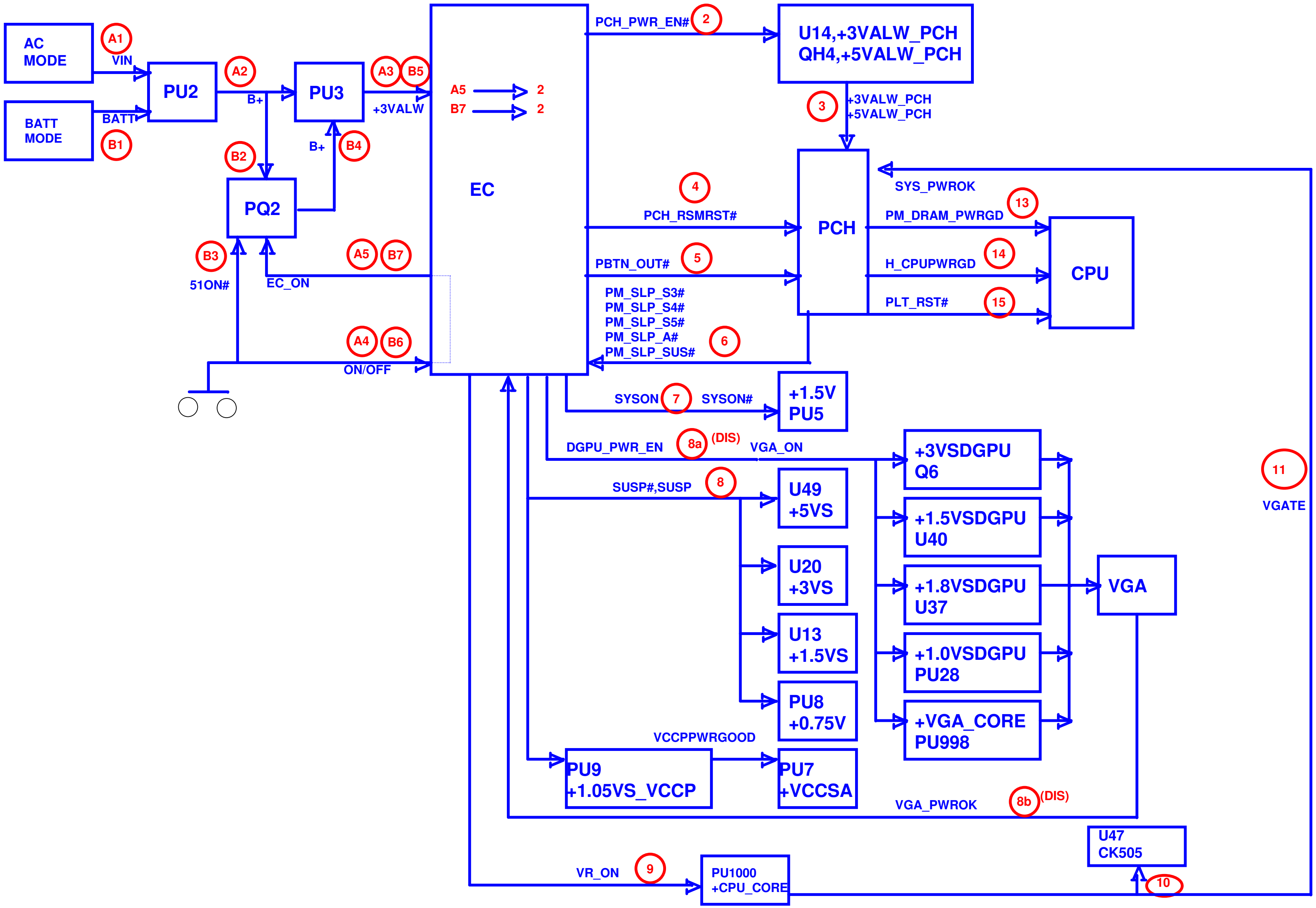
Title: CHARGER

Size | Document Number: Y490-LA8691P | Rev 0.2

Date: Thursday, May 10, 2012 | Sheet 62 of 65



Security Classification	Compal Secret Data			Compal Electronics, Inc.
Issued Date	2011/07/21	Deciphered Date	2012/12/31	Title: Power sequence

Size	Document Number	Rev
Custom	LA-8691P	0.1

Date: Thursday, May 10, 2012 | Sheet 63 of 65

Version change list (P.I.R. List)

Item	Reason for change	PG#	Modify List	Date	Phase
1	Reserve 0.1uF for Charger IC	51	Reserve PC321	201109/27	B test
2	EMI Request		change PR322,PR407,PR408,PR503,PR511,PR606,PR804,PR827 to 2.2 ohm add PC526,PC527,PC970,PC971(470uF)	201109/27	B test
3	Combine 1.05V	51	Remove one power rail +V1.05S_VCCPP Pop PR722,PR712,PR718	201109/27	B test
4	Discharge for +1.05VS_VGA by NV Request	53	Reserve PR528	201109/27	B test
5	Set VGA_CORE VBOOT voltage	56	unpop PR806 change PR813 to 147K ohm	201109/27	B test
6	For VGA_CORE power saving by NV Request	56	add PR838 0ohm	201109/27	B test
7	for CPU_CORE load line adjust	57	add PC969	201109/27	B test
8	to prevent MOS over temperature	55/58	change PQ702,PQ901,PQ902,PQ905 TPCA8065	201109/27	B test
9	for CPU_CORE test	59	Reserve PC77,PC78	201109/27	B test
10					
11					
12					
13					
14					
15					
16					
17					

HW PIR (Product Improve Record)

QIQY5 LA-8691P SCHEMATIC CHANGE LIST
REVISION CHANGE: 0.2
GERBER-OUT DATE: 2012/03/09

NO	DATE	PAGE		MODIFICATION LIST	PURPOSE
01)	03/14	10	R64	Change R64 BOM structure from "@" to "DS3@"	For Deep S3 Function

Security Classification	Compal Secret Data		
Issued Date	2011/07/21	Deciphered Date	2012/12/31

Compal Electronics, Inc.

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